CLASSIFIED

TOP SECRET

CLASSIFIED



1. **Revised Cardiac Risk Index**
   a. **Includes IHD & preop Cr >2
   b. The principal theme of guidelines: preop intervention is rarely necessary simply to lower risk of Surgery. An intervention is indicated/not indicated irrespective of need for Surgery. Preop testing should be performed only if result is likely to influence periop management.

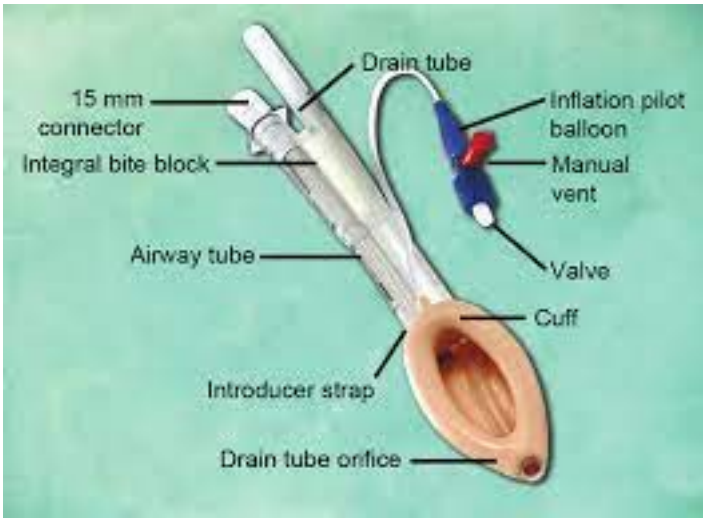
(Table 1-9) Revised Cardiac Index	Cardiac Risk Stratification based on surgical procedures in pts with CAD
High-risk Surgery • Abdominal aortic aneurysm **(AAA)** • **Peripheral vascular operation** • **Thoracotomy** • **Major abdominal** operation **Ischemic heart disease** • History of **myocardial infarction** • History of a **+ exercise testing** • **Current** complaints of **angina** pectoris • Use of **nitrate therapy** • Presence of **Q waves** on electrocardiogram **Congestive heart failure** • History of **CHF** • History of **pulmonary edema** • History of **paroxysmal nocturnal dyspnea** • Physical examination showing **rales** or **S3 gallop** • **pulmonary vascular redistribution** on CXR **Cerebrovascular disease History of stroke** • History of transient ischemic attack **Insulin-dependent diabetes mellitus** **Preoperative serum creatinine concentration mg/dL**	**High (Reported cardiac risk often >5%)** • **Emergent** major operations, particularly in the elderly • **Aortic** & other **major vascular** open surgeries • **peripheral vascular** surgery • **Prolonged** surgical procedures w/ **large fluid shifts** &/or **blood loss** **Intermediate (Reported cardiac risk generally (1-5%)** • **Carotid** endarterectomy • **Head** & **neck** surgery • **Intraperitoneal** & **intrathoracic** surgery • **Orthopedic** surgery • **Prostate** surgery **Low (Reported cardiac risk generally <1%)** • **Endoscopic** procedures • **Superficial** procedures • **Cataract** surgery • **Breast** Ambulatory procedures

2. **Goldman Cardiac Risk Index**:

Risk Factor:	Points:
Signs of CHF	**11**
MI ≤ 6 months	**10**
Arrhythmia	**7**
≥ 5 PVCs/min	**7**
Age > 70yrs	5
Emergency Surgery	4
Significant aortic stenosis	3
Chronic liver or kidney disease or bedridden	3
Intraperitoneal, intrathoracic, or aortic operation	3

Risk Index	Risk of Death & Major Cardiac Complications
Class I: 0-5 pts	**1.3%**
Class II: 6-12 pts	**4.7%**
Class III: 13-25 pts	**15.3%**
Class IV: 25-53 pts	**56%**

3. **Antitrust Laws**
   a. The antitrust laws are designed to ensure competition in the marketplace, including the marketplace for health services.
   b. Under the antitrust laws, all CRNAs & all MDAs are considered to be competitors, both within the specialties & b/w them.
   c. Exists to protect competition, not competitors
4. **What is an Exemplary damage??? trust???**
   a. Exemplary damage is another term for punitive damage; awarded to punish defendant
5. **Hepatorenal Syndrome Type 2 (definition)**
   a. HRS → Criteria: **ascites**, **not in shock**, **NOT exposed to nephrotoxic drugs**, has **no parenchymal renal disease**, has a **Cr level >1.5 mg/dL**, & **does not improve after 2 days of diuretic withdrawal & albumin therapy** (1g/kg/day up to 100 g/day).
      i. **Type 1** → (acute SEVERE decompensation, Cr >2.5 mg/dL), which is **usually fatal**,
      ii. **Type 2** → (chronic, MODERATE renal failure, Cr >1.5 & GFR <40 mL/min).
   b. Functional renal failure → **prognosis very poor** even though there is no intrinsic renal abnormality; exact pathophysiology is unknown,but appears that **dehydration** & **diminished RBF** often precede the ↓in **GFR** & **development of azotemia**.
      i. RRT is mainstay of supportive Tx until underlying liver dysfunction normalized or a donor organ becomes available for transplant.
   c. What procedure**:** parcentesis
6. **LMA**
   a. **Cuff volume** →→→→


Drain tube
15 mm
connector
Integral bite block
Inflation pilot
balloon
Manual
vent
Airway tube
Valve
Cuff
Introducer strap
Drain tube orifice

Chiari malformation: brain tissue extends into the spinal canal

FOI Notes from Apex:
FBO in the *awake*, *spontaneously ventilating* patient is the **gold standard** for managing difficult airway.

Other indications include:
C-spine limitation:
Severe cervical stenosis
Cervical fracture
Chiari malformation
Vertebral artery insufficiency

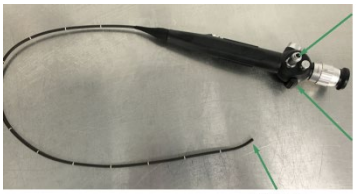
Limited mouth opening:
TMJ disease,
Facial burns
Mandibular-maxillary fixation

NO absolute contraindications to fiberoptic bronchoscopy
Relative Contraindications:
Hypoxia (lack of time)
Secretions not relieved by suction or antisialagogue
Hemorrhage that impairs visualization
Uncooperative patient (for awake)
Local anesthetic allergy (for awake)

7. Nerve fiber chart

Fiber Type	Subtype	Myelination	Function	Diameter (pm)	Velocity	Block Onset
A	α (alpha)	**Heavy**	Skeletal muscle - motor **Proprioception**	12-20	+++++	4th
	ß (beta)	**Heavy**	Touch Pressure	5- 12	++++	4th
	y (gamma)	Medium	Skeletal muscle - **tone**	3-6	+++	3rd
	δ (delta)	Medium	Fast pain Temperature Touch	2-5	+++	3rd
B		Light	**Pre**ganglionic ANS fibers	3	++	1st
C	Sympathetic	0	**Post**ganglionic ANS fibers	0.3- 1 3	+	2nd
	Dorsal root	0	Slow pain Temperature Touch	0.4- 1.2	+	2nd

a. **Order of nerve size**
   i. (C, B, **Aδ**, **Ag**, **Aß**, **Aα**) ← Smallest to largest
b. **Order of block Onset/Offset**
   i. **Onset: (B, C, Ay, Aß, Aα)**
   ii. **Offset: (Aα, Aß, Ay, C, B)**
c. **Match fibers w/ function (3 questions on this)**
   i. **Preganglionic, Pain, Touch/Pressure, Proprioception**



8. Method for awake FOI
a. Anesthetize upper airway (topical LA, SLN block, transtracheal block).
b. Secretions should be suctioned before inserting the FOB thru the ETT.
c. It is essential that the FOB exit the tip of ETT & not the Murphy eye.
d. The FOB & the ETT are manipulated to bring the larynx into view, & the FOB is advanced into the trachea.
e. Inflation of the ETT cuff during advancement of the FOB in the pharynx serves to create an enlarged pharyngeal space.
f. Because secretions tend to adhere to the pharyngeal walls, ETT cuff inflation also helps keep the optics of the FOB from being obscured.
g. The inflated cuff further aims the tip of the ETT anteriorly.
h. The target should always be kept in center of the provider's field of vision by flexion & rotation as the FOB is slowly advanced.
i. As the FOB passes through the vocal cords, the tracheal rings will become visible.
j. The scope is advanced to just above the carina, & then the ETT is threaded over the scope.
k. If resistance is encountered when advancing the ETT, force should not be exerted as the FOB can become kinked & result in diversion of the ETT into the esophagus & damage the FOB.
l. Resistance to advancement often means that the ETT is impacted on a vocal cord.
m. This can be relieved by rotating the ETT as it is gently advanced.
n. The appropriate depth of ETT placement can be verified by observing distance b/w the carina & tip of the ETT as FOB is withdrawn.
o. If there is any resistance when removing the FOB, it is either through the Murphy eye or kinked in the pharynx.
p. In both instances, the ETT & the scope must be withdrawn together to avoid damaging the FOB bronchoscope. (Miller)

9. Methergine (when NOT to give)
Ergot alkaloids are potent α-blockers that cause direct smooth muscle contraction
a. Ergot alkaloid; **NOT** given IV (results in arterial & venous constriction, severe HTN, cerebral bleeding, HA, N/V); usually 0.2 mg IM; cause ↑BP, CVP, & PAOP; metabolized & eliminated chiefly by liver; plasma ½ life = 2 hrs but uterine effects last much longer; ↑uterine contractility; used if oxytocin fails
b. CONTRAINDICATED in Pregnancy-induced HTN & Preeclampsia

10. Prostaglandin used for uterine atony, what is a side effect?
a. Bronchospasm

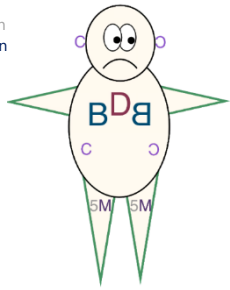
11. Rapid IV push protamine
a. Hypotension (Nagelhout)

12. RAPID IV push HIGH dose oxytocin (select multiple)
a. Preservative may cause systolic & ***especially diastolic*** hypotension, flushing, & tachycardia when infused at high doses.

13. Hematologic level most unchanged in full term parturient
a. Platelet count remains stable or is slightly ↓in 3rd trimester

14. Chemo drugs (4 questions)
a. Cisplatin (alkylating agent) → **acoustic nerve injury + nephrotoxicity**
b. Vincristine & vinblastine (tubulin-binding drug) → **peripheral neuropathy**
c. Bleomycin (antitumor Abx) = pulm fibrosis (keep FiO2 ≤ 30)
d. Doxorubicin (antitumor Abx) = cardiotoxic *AKA adriamycin*
e. 5-fluorouracil & methotrexate (antimetabolite) = **bone marrow suppression**
f. **Which one needs **preop echo** besides Doxorubicin? Adriamycin

C: Cisplatin (Alkylating agent) = Acoustic n. injury + Nephrotoxicity
V: Vincristine & vinblastine (Tubulin-binding drug) = Peripheral neuropathy
B: Bleomycin (Antitumor antibiotic) = Pulmonary fibrosis (keep Fi02 ≤ 30%)
D: Doxorubicin (Antitumor antibiotic) = Cardiotoxic
5: 5-fluorouracil (Antimetabolite) = Bone marrow suppression
M: Methotrexate (Antimetabolite) = Bone marrow suppression



15. MOST COMMON CAUSE intraop cardiac arrest
a. Preop complications (65%), surgical procedures (24%), intraop pathologic events (9%), & caused by anesthetic management (2%)
b. Intraop hemorrhage (33%) (NIH via Google)

16. IVF for HYPOchloremic metabolic alkalosis
a. NS (Barash & M&M)

Prime **M**inister	**P**ropriception **M**otor
Pressurized **T**o	**P**ressure **T**ouch
Sign	Muscle **S**pindle
The **C**ontract **F**ast	**T**ouch **C**old (temperature) **F**ast Pain
	Autonomic
	Autonomic, Slow pain.

Fiber Type	Subtype	Myelination	Function	Diameter (pm)	Velocity	Block Onset
A	α (alpha)	**Heavy**	Skeletal muscle - motor **Proprioception**	12-20	+++++	4th
	ß (beta)	**Heavy**	Touch Pressure	5- 12	++++	4th
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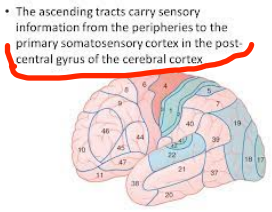
Erlanger and Gasser Classification of Nerve Fiber - Mnemonic

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Fibre Type	MNEMONIC	Function	Diameter (µm)	Conduction Velocity (m/s)
A alpha	**P**rime **M**inister	**P**ropriception **M**otor	12-20	70-120
A beta	**P**ressurized **T**o	**P**ressure **T**ouch	5-12	30-70
A gamma	**S**ign	Muscle **S**pindle	3-6	15-30
A delta	**T**he **C**ontract **F**ast	**T**ouch **C**old (temperature) **F**ast Pain	2-5	12-30
B		Autonomic	<3	3-15
C		Autonomic, Slow pain.	0.3-1.3	0.7-2.3



17. **Pyloric stenosis**
    a. **NOT** a Surgical emergency; Occurs when hypertrophy of the pyloric muscle creates a mechanical obstruction at the gastric outlet (b/w stomach & duodenum). An olive shaped mass can be palpated just below the xiphoid process. The infant presents w/ non-bilious projectile vomiting. It occurs in 1st 2-12 wks of life. It is MOST COMMON in males.
    b. **Pathophysiology**: Vomiting depletes water, hydrogen, chloride, Na+, & K+. Excessive vomiting → hypoNa, ↓K+, hypochloremic, metabolic alkalosis + compensatory respiratory acidosis. Metabolic alkalosis shifts the O2-hemoglobin dissociation curve to the left (left = love), & this reduces O2 release at the tissue level. Kidneys compensate for metabolic alkalosis by increasing bicarb excretion. The infant excretes alkalotic urine. As dehydration continues, the kidneys retain Na+ & water under the influence of ↑aldosterone. To maintain electroneutrality, kidneys lose H to the urine. This causes a paradoxical acidification of the urine. If dehydration is not corrected, impaired tissue perfusion ↑lactate production & produces metabolic acidosis. This is a late complication!
    c. **Anesthetic Management**: Pyloric stenosis is a **medical** (not surgical) emergency. Surgery (pyloromyotomy) should be postponed until the fluid, e-lyte, & acid-base status are optimized. Severe dehydration should be corrected w/ 20 mL/kg of 0.9% NS. Maintenance fluids consist of D5 0.45% NS at 1.5x the calculated maintenance rate. Anticipate a full stomach, so empty stomach before induction, intubate either awake or w/ RSI, & extubate awake. Postop apnea is common. This is possibly d/t fact that CSF pH remains alkalotic even after serum acid-base status is normalized.
18. **Renal tubular secretion near adult level by how many month old**
    a. Renal tubular function → full concentrating ability @ **~2 yrs** of age
    b. GFR: reaches adult level at **8-24 months**
19. **1st line treatment neuropathic pain**
    a. Antidepressant? (*Gabapentin* wasn't an option); amitriptyline
20. **Obesity Hypoventilation Syndrome S/S (select 4)**
    a. Nocturnal episodes of central apnea (apnea w/o resp efforts) reflecting progressive desensitization of the resp center to nocturnal hypercarbia; culminates as Pickwickian syndrome: daytime hypersomnolence, arterial hypoxemia, polycythemia, hypercarbia, resp acidosis, pulm HTN, & RV failure, (Stoelting)
    b. Triad of obesity, daytime hypoventilation, & sleep-disordered breathing w/o alternative neuromuscular, mechanical, or metabolic cause. Combo of obesity & chronic hypoventilation that ultimately results in pulmonary hypertension & cor pulmonale. The presence of both obesity (BMI >30 kg/m2) & awake arterial hypercapnia (PaCO2 >45 mm Hg) in the absence of known causes of hypoventilation supports Diagnosis. Prolonged OSA also alters the control of breathing → CNS-mediated apneic events. This ↑reliance on hypoxic drive for ventilation. The main ventilatory impairment of OHS is alveolar hypoventilation independent of intrinsic lung disease. Other characteristics of OHS include daytime hypersomnolence, hypercapnia, hypoxemia, & polycythemia. RV failure eventually ensues. These pts also have an ↑sensitivity to resp depressant effects of GAs. (Barash)
    c. Obesity, Hypersomnolence during the day, Hypoxemia, Hypercarbia, Resp acidosis, Compensatory metabolic alkalosis, Polycythemia, Pulm HTN, Right HF (Apex)
21. **Ve % ↑from Vt 700 RR 10 to Vt 700 RR 12**
    a. 7,000 ↑to 8,400 (1,400 difference)
    b. ? % of 7000 = 1400
    c. = 20% ↑
22. **DM Diagnosis criteria**
    a. FPG level of 126 mg/dL or greater or a random glucose level > 200 mg/dL, 2 hr GTT >200, HgbA1c >6.5.
23. **DM1 Pathophysiology**:
    a. beta cell destruction
24. **DM anesthetic management**:
    a. delayed gastric emptying → ↑risk aspiration; impaired thermoregulation → ↑risk of hypothermia; lactate in LR can be converted to glucose & may contribute to hyperglycemia
25. **Who sets standards within a profession**
    a. CRNAs are professionals who set their own standard & when the court needs to know what this standard is, it seeks expert testimony.
    b. While experts may sometimes refer to statements by a professional association, only the members of the profession can establish the standard of care & they do it by their conduct. (AANA website)
26. **Mechanoreceptors end where (medulla, pons, 2 other choices)**
    a. Cerebral cortex in postcentral gyrus (Apex)
27. **Laminae 7-9 reside**:
    a. in ventral grey matter & are motor
28. **Airway assessment**
    a. **Select 2** (move mandible posterior to bite lip, opening mouth, neck ROM, & ?)
29. **Slow IV push in critically ill patient best because**:
    a. avoid hypotension?
30. **S/S hemolytic transfusion reaction under GA**
    a. Hypotension, free Hgb in plasma or urine, ARF reflects precipitation of stromal & lipid contents (not free Hgb of hemolyzed erythrocytes in distal renal tubules. DIC causing a coagulopathy is initiated by material released from hemolyzed erythrocytes. hemoglobinuria
31. **What is the highest mortality w/ hemolytic reaction? Or What transfusion reaction has highest morbidity?**
    a. TRALI
32. **Which blood product causes a protein reaction?**
    a. FFP (Apex) **Whole blood* wasn't a choice***
33. **Disease MOST COMMON w/ blood transfusion**
    a. CMV > Hep B > Hep C > HIV (Apex) **CMV wasn't a choice***
34. **Jet ventilation**
    a. **mode high frequency, small vol =**
        i. Need 50 psi w/ a metered & adjustable hand-controlled valve & a leur-lock connector is down-regulated to 15-30 psi of O2 w/ insufflations of 1-1.5 sec @ 12 insufflations/min; 14G cath provides 400-700 Vt (Barash)
        ii. Inspiratory pressures should not exceed 50 psi on the regulator, & in most instances, 25 psi is sufficient. A 1-sec inspiration at 25 psi w/ a rate of 20 bpm delivers a 285-mL Vt or 5.7 L/min ventilation. Exhalation occurs passively through the upper airway. (Nagelhout)
35. **At what PSI will the jet ventilator cause barotrauma?**
    a. Inspiratory pressures should **NOT** exceed 50 psi



36. Identify parts of a jet ventilation attachment thingy:

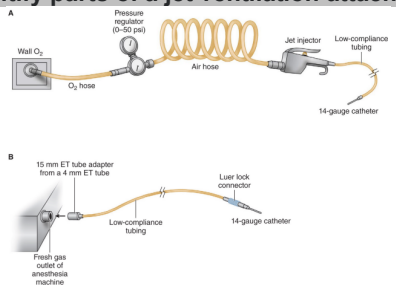


FIGURE 55-6 A,B: Two systems for transtracheal jet ventilation after cricothyrotomy (see Figure 55–5). A jet ventilator and pressure regulator (as shown in A) provide better control of the inspiratory cycle. Both systems use low-compliance tubing and a high-pressure source of oxygen.

a.

37. Clonidine

- **a.** selective *partial* α2 agonist. Neuraxial clonidine: effective analgesic for chronic CA & nonCA pain, as well as for postop pain. Clonidine has antihypertensive effects & has been shown to potentiate postop analgesia induced by LAs.
- **b.** Spinal: causes a 30% prolongation of sensory & motor block of LAs; dose ranges 10-50 µg.. Epidural: doses ranging from 75-150 µg.
- **c.** Caudal: 1 µg/kg dose. Intrathecal administration of clonidine 37.5-150 µg w/ bupivacaine results in a dose-dependent ↑in sensory blockade & more pain-free intervals in the postop period.
- **d.** Addition of clonidine intrathecally or epidurally was associated w/ significant reduction of HR & BP.
- **e.** Black Box warning concerning neuraxial use in OB: related maternal hemodynamic instability.
- **f.** The warning states, "OB, PP, or Periop Use: weigh risk/benefit; epidural clonidine generally not recommended for OB, PP, or periop pain management d/t risk of hemodynamic instability, esp. hypotension & bradycardia."
- **g.** Clonidine ↓ risk of intraop pain & ↑risk of arterial hypotension, w/o evidence of dose-responsiveness.
- **h.** Finally, clonidine has no relevant impact on the time to achieve complete sensory or motor block, on the extent of the cephalad spread of the sensory block, or on the risk of bradycardia.
- **i.** *SE of IV clonidine: vasodilation, hypotension & tachycardia

GLOMERULOSA → MINERALOCORTICOIDS → SALT
FASICULATA → → GLUCOCORTICOIDS → SUGAR
RETICULARIS → → → ANDROGENS → SEX

38. Strongest anti-inflammatory corticosteroid

	Glucocorticoid Effect (Anti-inflammatory potency)	Mineralocorticoid Effect (Sodium retaining potency)	Equivalent Dose (mg)	Duration
Endogenous Steroids				
• *Cortisol* → NEED CHOLESTEROL FOR CORTISOL CREATION	1	1	20	8-12
Cortisone	0.8	0.8	25	8-36
• *Aldosterone*	0	3000	N/A	24
Synthetic Steroids				
Prednisone	4	0.8	5	12-36
Prednisolone	4	0.8	5	12-36
Methylprednisolone	5	0.5	4	12-36
Dexamethasone •	25	0	0.75	36-54
Betamethasone •	25	0	0.75	36-54
Triamcinolone	5	0	4	12-36
Fludrocortisone	10	250	2	24

- **a.** Aldosterone does **NOT** have glucocorticoid effects
- **b.** Dexamethasone, triamcinolone, & betamethasone do **NOT** have mineralocorticoid effects;
- **c.** **Triamcinolone** is more likely to cause sedation (not euphoria) & anorexia (not ↑appetite)
- **d.** **One about corticosteroids → most common multipurpose coanalgesic used for cancer pain**:
  - **i.** ↓pain perception, have a sparing effect on opioid requirements, improve mood, ↑appetite, & lead to weight gain

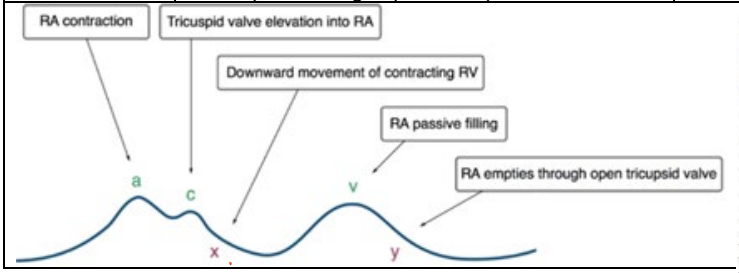
39. CVP waveform:

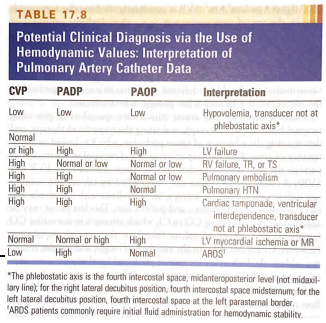
- a. **No a wave & large c wave =** Afib or V-pacing if underlying rhythm is asystole
- b. **Large a wave**: tricuspid stenosis, diastolic dysfunction, MI, chronic lung disease → LVH, AV dissociation, junctional rhythm, V-pacing asynchronous, PVCs
- c. **v wave**: adding volume to R atria
- d. • **Large v wave → tricuspid regurg**• acute ↑in intravascular volume, RV papillary muscle ischemia
- e. **Match a (atrial kick), c (closure of valve), x, y, & v**

40. Which point correlates w/ closure of tricuspid valve? C- Wave

a.

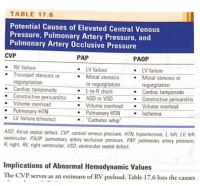
CVP Waveform	Mechanical Event	Electrical Event
A wave	Right atrial contraction (atrial kick 20%)	Just after P wave (atrial depolarization)
C wave	Right ventricular contraction • bulging of tricuspid valve into RA (closure of TV)	Just after QRS complex (ventricular depolarization)
X descent	RA relaxation	ST segment
V wave	Passive filling of RA	Just after T wave begins (ventricular repolarization)
Y descent	RA empties through open tricuspid valve	After T wave ends




TABLE 17.8
Potential Clinical Diagnosis via the Use of Hemodynamic Values: Interpretation of Pulmonary Artery Catheter Data

CVP	PADP	PAOP	Interpretation
Low	Low	Low	Hypovolemia, transducer not at phlebostatic axis*
Normal or high	High	High	LV failure
High	Normal or low	Normal or low	RV failure, TR, or TS
High	High	Normal or low	Pulmonary embolism
High	High	Normal	Pulmonary HTN
High	High	High	Cardiac tamponade, ventricular interdependence, transducer not at phlebostatic axis*
Normal	Normal or high	High	LV myocardial ischemia or MR
Low	High	Normal	ARDS†

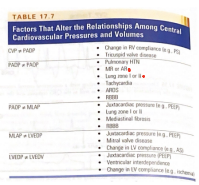
*The phlebostatic axis is the fourth intercostal space, midanteroposterior level (not midaxillary line); for the right lateral decubitus position, fourth intercostal space midsternum; for the left lateral decubitus position, fourth intercostal space at the left parasternal border.
†ARDS patients commonly require initial fluid administration for hemodynamic stability.


TABLE 17.6
Potential Causes of Elevated Central Venous Pressure, Pulmonary Artery Pressure, and Pulmonary Artery Occlusive Pressure

CVP	PAP	PADP
• RV failure • Tricuspid stenosis or regurgitation • Cardiac tamponade • Constrictive pericarditis • Volume overload • Pulmonary HTN • LV failure (chronic)	• LV failure • Mitral stenosis or regurgitation • L-to-R shunt • ASD or VSD • Volume overload • Pulmonary HTN • "Catheter whip"	• LV failure • Mitral stenosis or regurgitation • Cardiac tamponade • Constrictive pericarditis • Volume overload • Ischemia

ASD, Atrial septal defect; CVP, central venous pressure; HTN, hypertension; L, left; LV, left ventricular; PADP, pulmonary artery occlusive pressure; PAP, pulmonary artery pressure; R, right; RV, right ventricular; VSD, ventricular septal defect.

Implications of Abnormal Hemodynamic Values
The CVP serves as an estimate of RV preload. Table 17.6 lists the causes


TABLE 17.7
Factors That Alter the Relationships Among Central Cardiovascular Pressures and Volumes

CVP ≠ PADP	• Change in RV compliance (e.g., PE) • Tricuspid valve disease • Pulmonary HTN
PADP ≠ PAOP	• MR or AR • Lung zone I or II • Tachycardia • ARDS • PEEP
PAOP ≠ MLAP	• Juxtacardiac pressure (e.g., PE/LP) • Lung zone I or II • Mediastinal fibrosis • PEEP
MLAP ≠ LVEDP	• Juxtacardiac pressure (e.g., AS) • Mitral valve disease • Change in LV compliance (e.g., ischemia)
LVEDP ≠ LVEDV	• Juxtacardiac pressure (PEEP) • Ventricular interdependence • Change in LV compliance (e.g., ischemia)

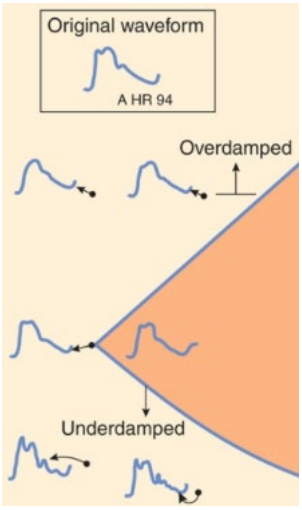
41. **Art line Rapid upstroke followed by widened pulse pressure & low diastolic pressure?**
    a. **aortic regurg**
42. **Art line compliance questions → Arterial BP transducer: high** → ABOVE PHLEBOSTATIC ACCESS ↳ LOW BP
    a. **Hypotension** BELOW → HIGH BP
43. **Orbicularis oculi Is controlled by what cranial nerve?**
    a. Facial nerve (Barash) **7th or VII**
44. **Which nerve is more associated w/ diaphragm?**
    a. Orbicularis oculi
45. **Hotspot to choose placement of TOF leads**
    a. (medial portion to get ulnar nerve)
46. **TOF → Ulnar nerve hotspot** → ULNAR NERVE *
    a. contraction of adductor pollicis (adducts thumb); **Best** for **recovery** from NMB
47. **TOF →Facial nerve**:
    a. orbicularis oculi (closes eyelid) or corrugator supercilia (eyebrow twitch) **Best** for **onset** of NMB:
48. **TOF →Posterior tibial nerve/Ulnar nerve**:
    a. Flexor hallicus (big toe flexion) or adductor pollicis (thumb adduction) **Best** for **recovery** of blockade
49. **Order of Muscle Paralyzation**:
    a. Eyes → Extremities → Trunk → Abdomen → Diaphragm (restored in opposite order)
50. **Factors that affect NMB**:
    a. Hypothermia (↑DOA), Resp acidosis (↑block & ↓reversal), ↑Mg+ (↑block), ↓K++, ↓Ca2++2, Age (neonate vs geriatric)
51. **Gastroparesis in DM is d/t which nerve?**
    a. vagus ↑ R/F ASPIRATION ***
        i. Reflects ANS dysfunction; diabetic autonomic neuropathy, also ↓ gastric secretion & motility → gastroparesis. (Stoelting)
        ii. Autonomic insufficiency as a result of microvascular changes (Nagelhout)
52. **Anest mgmt patient w/ acromegaly having pituitary tumor removed**
    a. Hypertrophy occurs in skeletal, connective & soft tissues. Tongue & epiglottis are enlarged, making patient susceptible to upper airway obstruction. Incidence of difficult intubation is 20-30% & may be clinically unpredictable. Hoarseness may reflect thickening of vocal cords or paralysis of a RLN d/t stretching. Dyspnea or stridor is associated w/ subglottic narrowing. Peripheral nerve or artery entrapment, HTN, & DM are common. Anest mgmt is complicated by distortion of facial anatomy & upper airway. Induction of GA may put patient at ↑risk if mask fit is improper or vocal cord visualization is impaired. When preop Hx suggests upper airway or vocal cord involvement, it is prudent to consider awake intubation. (Barash)
    b. Pts w/ acromegaly may have HTN, cardiomyopathy, DM, & OP as well as cartilaginous & soft-tissue hypertrophy of the larynx & enlargement of tongue, complicating intubation. For transsphenoidal Surgery, HOB is elevated 15°. Monitor for VAE. Use of submucosal injection of epi-containing solutions or topical vasoconstrictors to assist in hemostasis may result in large BP ↑s. An anesthetic technique that incorporates muscle relaxation & allows for smooth extubation & rapid neuro assessment is desirable. N2O should be omitted if air is injected Surgery to aid w/ tumor visualization. Preop: prep patient for awakening w/ nasal packing. May need smaller diameter ETT. Use nasotracheal tubes w/ caution because of possible turbinate enlargement. Intubate awake w/ optical guidance or FOI. Have equip for trach available. ETT should remain in place until patient fully awake & has total return of reflexes. Monitor blood glucose & e-lytes. May need stress dose steroids. Perform Allen test before inserting A-lines. (Nagelhout)
    c. **PACU management**: Mannitol? Hyperventilate? Possibly Desmopressin d/t risk for DI?
    d. Growth hormone (somatotropin): facilitates growth of all tissues in body
53. **Drugs that exacerbate/precipitate s/s of Parkinsons:**
    a. **Phenothiazines** & **Butyrophenones:**
        i. Drugs in these category→ droperidol, Haldol, & Reglan (inhibits plasma cholinesterase)
54. **Which would you not give to a patient with Parkinson's?**
    a. Alfentanil & Reglan → ALFENTANIL → DYSTONIC REACTIONS; REGLAN → DOPAMINE ANTAGONIST
55. **Parkinson's is a result of?**
        i. A degeneration of dopaminergic neurons in the substantia nigra (where dopamine is released)
56. **Endogenous opioids & receptors**
    a. Inhibitory: enkephalin/dynorphin
    b. The endogenous opioid system is composed of a family of structurally related endogenous peptides that act at a 4-member opioid receptor family consisting of the μ-opioid receptor (MOR), κ-opioid receptor (KOR), δ-opioid receptor (DOR), & orphanin FQ/nociception (NOP) receptor. At least 3 MOR subtypes have been described: μ1 is predominantly involved in opioid analgesia, μ2 is involved in opioid-induced resp depression, & μ3 is involved in opioid-induced immune suppression. The endogenous opioid peptides include endorphins, enkephalins, & dynorphins, each of which display different affinities for the μ-, κ-, & δ-opioid receptors. β-endorphins have a high affinity for the MOR, met- & leu-enkephalins for the KOR, & Dynorphin A for the DOR. (Barash)

Classification of Opioid Receptors

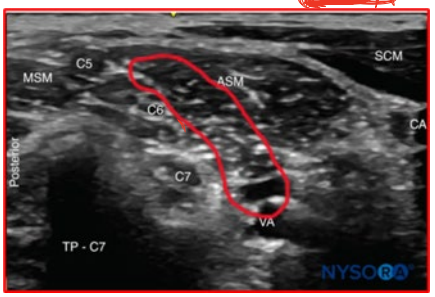
	Mu_1[a]	Mu_2[a]	Kappa	Delta
Effect	Analgesia (supraspinal, spinal) Euphoria Low abuse potential Miosis Bradycardia Hypothermia Urinary retention	Analgesia (spinal) Depression of ventilation Physical dependence Constipation (marked)	Analgesia (supraspinal, spinal) Dysphoria, sedation Low abuse potential Miosis Diuresis	Analgesia (supraspinal, spinal) Depression of ventilation Physical dependence Constipation (minimal) Urinary retention
Agonists	Endorphins[b] Morphine Synthetic opioids	Endorphins[b] Morphine Synthetic opioids	Dynorphins	Enkephalins
Antagonists	Naloxone Naltrexone Nalmefene	Naloxone Naltrexone Nalmefene	Naloxone Naltrexone Nalmefene	Naloxone Naltrexone Nalmefene

(Barash)

    c. **Endorphins = Mu**
    d. **Dynorphins = Kappa**
    e. **Enkephalins = Delta**
    f. **OLR-1 = Nociceptin**
    g. **Hydrophilic drugs =** early & late resp depression (morphine & dilaudid)
        Likes water



57. **Time for late onset resp depression d/t duramorph?**
    a. 6-12 hrs
    b. **Lipophilic drugs =** only early resp depression (fentanyl); Time for early resp depression = <6 hrs
    c. **Shivering receptor**: kappa; Tx = meperidine (stimulates kappa receptor)
    d. **Contraction of sphincter of Oddi**:↑biliary press; reversed by naloxone or glucagon, meperidine causes the lowest ↑in biliary press
    e. **Opioid receptor location**: substantia gelatinosa in dorsal horn of spinal cord (rexed's lamina II), medulla, midbrain, periaqueductal grey, medial thalamus, amygdala, limbic cortex
    f. **Pain is inhibited when**: spinal neurons release GABA & glycine (inhibitor NTs)
58. **Interscalene block**:
    a. anesthesia for procedures involving shoulder & proximal humerus; **NOT** for Surgery @ or distal to elbow; placed at level of trunks; roots C5-7 are most densely blocked; ulnar nerve (originating from C8 & T1) may be spared;
    b. landmarks = b/w middle scalene & anterior scalene muscles
59. **Identify ventral nerve on interscalene US**
    a.



60. **Drawing of Brachial Plexus: Which trunk is normally missed w/ interscalene block?**
    a. Trunk off of C8 **(ulnar)**
61. **Cervical levels blocked for CEA**:
    a. C2-C4 (Barash)
62. **"Arm patch" of lateral shoulder, can't abduct is d/t what nerve damage**:
    a. Axillary
63. **Sacral plexus nerves**:
    a. Sciatic nerve (branches into tibial & common peroneal nerves) & Posterior femoral cutaneous
64. **Which spinal level allows rotation but not flexion?**
    a. Thoracic
65. **TURP causes (2)**
    a. **Instilling all the saline during TURP causes what?**
        i. HypoNa, HTN, Bradycardia, Pulm edema/resp. distress
66. **Best antisialagogue in patient w/ AS**
    a. Scopolamine? (d/t least affect on HR)
67. **Tx central anticholinergic syndrome**
    a. Physostigmine

FLUID OVERLOAD + HYPOOSMO

TABLE 17-15 Signs and symptoms of transurethral resection of the prostate (TURP) syndrome

System	Signs and Symptoms	Cause
Cardiovascular	Hypertension, reflex bradycardia, pulmonary edema, cardiovascular collapse, ECG changes (wide QRS, elevated ST segment, ventricular arrhythmias)	Rapid fluid absorption, reflex bradycardia (secondary to hypertension or increased ICP), third-spacing secondary to hyponatremia and hypo-osmolality
Respiratory	Tachypnea, hypoxemia, Cheyne-Stokes breathing	Pulmonary edema
Neurologic	Nausea, restlessness, visual disturbances, confusion, somnolence, seizure, coma, death	Hyponatremia and hypo-osmolality causing cerebral edema and increased ICP, hyperglycinemia (glycine is an inhibitory neurotransmitter that potentiates NMDA receptor activity), hyperammonemia
Hematologic	Disseminated intravascular coagulation, hemolysis	Hyponatremia and hypo-osmolality
Renal	Renal failure	Hypotension, hyperoxaluria (oxalate is a metabolite of glycine)
Metabolic	Acidosis	Deamination of glycine to glyoxylic acid and ammonia

68. **During mediastinoscopy pt develops dilated vessels to chest/neck, edema to face/orbits - what syndrome is most likely cause?**
    a. **Superior Vena Cava syndrome**
    b. A constellation of signs that develops in pts w/ a mediastinal tumor that obstructs venous drainage in upper thorax. ↑venous pressure → 1. Dilation of collateral veins in the thorax & neck 2. Edema & cyanosis of face, neck, & upper chest 3. Edema of conjunctiva 4. Evidence of ↑ICP including HA, AMS, nausea, seizures. Dyspnea is common. CA accounts for nearly all cases of SVCS. Worsening may occur as a result of generous intraop fluid replacement. Diuretics may ↓tumor volume, but reduction in preload may result in significant hypotension (d/t already compromised venous return). Surgery bleeding is often ↑d/t an ↑CVP. Most often associated w/ lung CA. (Stoelting)
    c. Placement of IV lines in lower extremities is preferred. Insertion sites could delay drug effect d/t slow distribution. Administer fluids w/ caution because lg vol can worsen s/s. (Nagelhout)
69. **Parturient plasma volume is ↑d/t**
    a. Is likely the result of greater circulating levels of progesterone & estrogen resulting in enhanced renin-angiotensin-aldosterone activity.
    b. Stoelting mentioned something about Na++ retention
70. **Blood volume in a parturient is ↑by what %?**
    a. 35% (Type in question…round to nearest 5th %)
71. **Cardiac ouput in 3rd trimester ↑by what %?**
    a. 40%
72. **Aspiration prophylaxis med that interferes w/ amine LA metabolism (slows metabolism of lidocaine)**:
    a. Cimetidine
73. **What is it called when vapor pressure equals atmospheric pressure?**
    a. Boiling point VAPOR PRESSURE = ATMOSPHERIC PRESSURE 1 ATM = 760 mmHg
74. **Cardioselective beta blockers (good for pts w/ *asthma*)**
    a. BE A MAN: Bisoprolol, Esmolol, Acebutolol, Metoprolol, Atenolol, Nebivolol
75. **Nonselective beta blockers**:
    a. Carvedilol, labetalol, nadolol, pindolol, propranolol, & timolol
    b. Cause ↑airway resistance → AVOID W/ ASTHMATICS
76. **Hearing loss associated w/ N2O d/t**:
    a. Changes in middle ear pressure, rupture of tympanic membrane? Something about inability of middle ear to ↑volume
77. **Sevo causes fire w/ which absorbent**:
    a. Baralyme
78. **Inhibitory NTs (2-3 questions)**:
    a. GABA, Glycine, norepi, serotonin
79. **Excitatory NTs**:
    a. Glutamate, Ach, Aspartate **glutes Ass and a seat.. make me excited SNAG a deal = EXCITED**
80. **Body's response to pain**:
    a. release of NTs from nociceptive nerve endings (sub P, glutamate, bradykinin, histamine, serotonin, prostaglandins, cytokines, catecholamines, aspartate, ATP, Ach, enkephalins, B-endorphins, NE, adenosine, GABA, glycine, calcitonin gene-related peptide)

↳ INHIBITORY OR EXCITATORY

81. **MOA osmotic diuretics**
    a. ↑osmotic pressure in renal tubule which ↓ reabsorption of water
    b. ↑serum osmolarity & pull water across the BBB; if BBB is disrupted, mannitol enters brain & ↑cerebral edema
    c. Osmotically active diuretics are filtered at the glomerulus & undergo limited/no reabsorption in the PCT. Their presence in the PCT limits passive water reabsorption that normally follows active Na+ reabsorption. Although their major effect is to ↑water excretion, in large doses, osmotically active diuretics also ↑e-lyte (Na+ & K+) excretion. The same mechanism also impairs water & solute reabsorption in the LOH. Mannitol is the MOST COMMON used osmotic diuretic. It is a 6-carbon sugar that normally undergoes little/no reabsorption. In addition to its diuretic effect, mannitol appears to ↑RBF. The latter can wash out some of the medullary hypertonicity & interfere w/ renal concentrating ability. Mannitol appears to activate the intrarenal synthesis of vasodilating prostaglandins. It also appears to be a free radical scavenger. (M&M)
    d. Portions of the renal tubules that are highly permeable to water, namely, the PCT & more importantly, the LOH, represent the principal site of action of osmotic diuretics. By increasing tubular fluid osmolality, it ↓ water reabsorption & promotes water diuresis. Na+ is diluted in the retained water in the renal tubules → less reabsorption of Na+. However, hyperNa may result from the water diuresis. In addition to causing renal tubular effects, IV admin of mannitol also ↑plasma osmolarity, thus drawing fluid from intracellular to extracellular spaces. This ↑plasma osmolarity may result in an acute expansion of the intravascular fluid volume which could be poorly tolerated in pts w/ borderline cardiac function. ↑plasma osmolarity allows water to move along osmotic gradient from tissues, including brain, into the intravascular space, → ↓ICP. Mannitol is a scavenger of O2-free radicals, which may prevent cellular injury. (Stoelting's Pharm)

DESCENDING LOOP OF HEN

82. **Mannitol**
    a. **Dose to ↓ICP**: 0.25-1 gram/kg IV
    b. 74 kg patient mannitol @ 0.5 mg/kg; concentration is 20 g/dL, give how many mL?
        i. 37 mg
        ii. 1 dl = 100 ml; 1 g = 1000 mg
        iii. 20g/100 ml; 20,000 mg/ 1000 mg = 20 mg/ml
        iv. 1.85 ml

Pharmacokinetics of Opioid Agonists

	pK	Percent Nonionized (pH 7.4)	Protein Binding (%)	Clearance (mL/min)	Volume of Distribution (L)	Partition Coefficient	Elimination Half-Time (h)	Context Sensitive Half-Time: 4-Hour Infusion (min)	Effect-Site (Blood/Brain) Equilibration Time (min)
Morphine	7.9	23	35	1,050	224	1	1.7–3.3		
Meperidine	8.5	7	70	1,020	305	32	3–5		
Fentanyl	8.4	8.5	84	1,530	335	955	3.1–6.6	260	6.8
Sufentanil	8.0	20	93	900	123	1,727	2.2–4.6	30	6.2
Alfentanil	6.5	89	92	238	27	129	1.4–1.5	60	1.4
Remifentanil	7.3	58	66–93	4,000	30		0.17–0.33	4	1.1

83. **Opioid w/ fastest elimination equilibrium (*Remi not an option*)**
    a. (Remi, Alfentanil, Morphine, Sufentanil, Fentanyl) *RAMS F
84. **Syndrome w/ rigid neck**
    a. Tetanus, ruptured aneurysm, torticollis
85. **Which of these have unstable c-spine?**
    a. Klippel-Feil (Kids Tri Gold)
86. **Age HgbF goes away**
    a. Starts at 2-3 months; Complete at 6 months (Apex says month 4-6)
87. **Degree C = hypothermia**
    a. 35 (type in question); <36°C (Nagelhout)
88. **Thyroid**
    a. Joined by: isthmus
    b. **Hormones**:
        i. T4: thyroxine (prohormone synthesized from tyrosine); is converted to T3
        ii. T3: triiodothyronine (active thyroid hormone)
        iii. Cacitonin
        iv. When iodine not available, thyroid is unable to produce sufficient quantity of T3 & T4
        v. Think of thyroid hormone as an "on switch" that stimulates many metabolic functions
        vi. ↑thyroid → ↑BMR → ↑O2 consumption + ↑CO2 production
    c. **Cretinism**: neonatal hypothyroidism → physical & mental retardation

$T_4 \rightarrow T_3$

89. **Thyroidectomy**:
    a. **Complications**: hypoCa (can cause laryngospasm), RLN damage (MOST COMMON: unilat = hoarse; bilat = stridor), Neck hematoma, Thyroid storm, SLN injury, Infection, Pneumo, Tracheomalacia,
    b. **What is a risk after thyroidectomy? (more than 1 question)** hypoparathyroidism, RNL injury, airway edema
    c. **What is the likely cause of respiratory distress afterwards** hematoma? ← said something about bleeding/compression of thyroid
    d. **Post-op question about a complication** laryngospasm, muscle weakness, another about weakness, & another random choice
    e. **Thyroid storm**: MOST COMMON occurs 6-18 hrs after Surgery
        i. **Tx**:
            1. **Block *synthesis***: Methimazole, PTU, carbimazole, potassium iodide
            2. **Block *release***: radioactive iodine, potassium iodide
            3. **Block *T4 to T3 conversion***: PTU, propranolol
            4. ***Beta* blocker**: propranolol, esmolol
        ii. **Anesthetic Considerations**: don't proceed to elective Surgery until patient is euthyroid, goiter = awake intubation, Avoid sympathomimetics, anticholinergics, ketamine, & pancuronium
        iii. **Surgical Complications**: RLN injury, HypoCa
    f. **Hypothyroidism**
        i. **Anesthetic Considerations**: goiter = awake intubation; hemodynamic support best provided w/ sympathomimetics that improve myocardial performance (**NOT** Neo); Tx hypotension unresponsive to catecholamines w/ corticosteroids

$T_{1/2}$ LIFE T_3 – 1.5 DAYS
$T_{1/2}$ LIFE T_4 – 7 DAYS

THYROID RECEPTORS MORE $T_4 > T_3$
T_3 → MORE POTENT, LESS PROTEIN BOUND

TSH → IODIDE TRAPPING

90. Thyroid hormone

a. Principal hormonal secretions: thyroxine (T4) & triiodothyronine (T3)). T4, a prohormone synthesized from tyrosine, represents 80% of the body's thyroid hormone production. T3, 5 Xs more active than T4, is produced directly from tyrosine metabolism or from conversion of T4 in peripheral tissues. 2 distinct deiodases (located in liver, kidneys, & CNS) metabolize T4 & T3 to inactive compounds. The half-lives of endogenously/exogenously administered T3 & T4 are 1.5 & 7 days, respectively. T3 & T4 are both highly protein bound to albumin, thyroid-binding prealbumin, & thyroid-binding globulin w/ only 0.2% of T3 & 0.3% of T4 freely circulating unbound & pharmacologically active. Iodine present in thyroid hormones is not necessary for biologic activity. In addition, the thyroid gland secretes calcitonin, which is important for Ca^{2+} use. The thyroid hormones ↑O_2 consumption in nearly all tissues, except the brain. Failure of thyroid hormones to greatly alter O_2 consumption of the brain is consistent w/ minimal changes in MAC that accompany hyperthyroidism or hypothyroidism. CV changes are often the earliest clinical manifestations of abnormal thyroid hormone levels. Absence of thyroid gland hormones ↓ minute O_2 consumption to approx 40% < normal; excesses of thyroid hormones can expand O_2 consumption as much as 100% > normal. Thyroid hormones stimulate carb metabolism & facilitate the mobilization of free fatty acids. Despite the latter effect, plasma concentrations of cholesterol usually ↓, reflecting stimulation of LDL receptor synthesis by thyroid hormones. Thyroid hormones are stored in combination w/ thyroglobulin. Stimulation of proteases by TSH results in cleavage of hormones from thyroglobulin & their release into the systemic circulation. When thyroid hormones enter cells, T3 binds to nuclear receptors. T4 also binds to these receptors but not as avidly. Indeed, T4 serves principally as a prohormone for T3, so that the biologic effects of T4 are largely a result of its intracellular conversion to T3. Thyroid hormones exert most, if not all, of their effects through control of protein synthesis. Thyroid hormones activate the DNA transcription process in the cell nucleus to form new cell proteins & enzymes. Sympathomimetic effects that accompany thyroid hormone stimulation most likely reflect a greater # & sensitivity of β-adrenergic receptors to release of T4 & T3. It has been proposed that thyroid hormones modulate conversion of α-adrenergic to β-adrenergic receptors. Cardiac cholinergic receptor #s are ↓by thyroid hormones, which is consistent w/ an ↑in HR that is out of proportion to the ↑in CO. When thyroid hormones accelerate metabolism, tissues vasodilate & blood flow delivers necessary O_2 & carries away metabolites & heat. As a result, CO often ↑but systemic BP is unchanged because peripheral vasodilation offsets the impact of more blood flow. Excess protein catabolism associated w/ greater secretion of thyroid hormones is the mechanism behind skeletal muscle weakness characteristic of hyperthyroidism. The fine muscle tremor that accompanies hyperthyroidism stems from the sensitivity of neuronal synapses in the area of the spinal cord that controls skeletal muscle tone. Diarrhea reflects additional motility in the GI tract w/ excessive activity of the thyroid gland. (Barash)

b. The 1st stage of thyroid hormone formation is the transport of iodides from the extracellular fluid into the thyroid cells & follicles. About 1/5 of the circulating iodide is removed from the blood by the thyroid cells & used for the synthesis of thyroid hormones, a process called iodide trapping. The iodide pump normally concentrates the iodide to about 30 times its concentration in the blood. Iodide trapping is the rate-limiting step in thyroid hormone synthesis & is under the control of TSH from the anterior pituitary. Once inside the thyroid gland, iodide ions are oxidized back to iodine.

c. Thyroid Hormone Formation: are formed in the follicles of the thyroid gland under the control of TSH. Thyroglobulin contains the amino acid tyrosine, which combines w/ iodine to form various iodotyrosines, including the 2 major thyroid hormones. Thyroxine & triiodothyronine remain part of the thyroglobulin molecule, stored as colloid within the thyroid follicle until release. Enough hormone is synthesized & stored under basal conditions to supply the body w/ its normal hormone requirements for 2-3 months. As a result, if complete thyroid hormone synthesis ceases, the physiologic effects are not seen for several months. Although iodine is required for hormone synthesis, paradoxically, excess iodine can cause the gland to recede & inhibit production of thyroid hormone.

d. Release of Thyroxine & Triiodothyronine: TSH controls the release of hormones from thyroid gland. On release, T4 & T3 are cleaved from the thyroglobulin molecule & secreted into the circulating blood. Thyroglobulin remains within the colloid. Under normal conditions, about 93% of the hormone released from the thyroid gland is T4 & 7% is T3. The secretion of T4 from the thyroid gland is 80-100 mcg/day. When thyroid hormones reach their target tissues, most of the T4 is deiodinated to T3-T4 serving mostly as a hormone precursor. Triiodothyronine is more potent & less protein bound than T4 & is the primary metabolically active hormone that stimulates target tissues.

e. Transport of Thyroxine & Triiodothyronine to Tissues: Thyroid hormone exists in circulation in both free & bound forms. The amount of free hormone, which is the metabolically active fraction, is extremely small, < 0.03% of total circulating T4 & 0.3% of total circulating T3. The majority of circulating hormone (99.9%) is bound to plasma proteins. Most of the thyroid hormones bind to the circulating proteins thyroxine-binding globulin, transthyrotin, & albumin. Because of the very high affinity of the plasma-binding proteins for thyroid hormones, the hormones are released to the tissue cells very slowly. The half-life of T4 in the circulation is 6-7 days, & the half-life of T3 is about 24 hrs.

f. Functions of Thyroid Hormones: ↑Cellular Metabolic Activity- Thyroid hormones initiate protein formation in virtually all cells of the body. Consequently, the level of enzymes, structural proteins, transport proteins, & other substances ↑considerably under the direction of T4 & especially T3. The net result is a generalized ↑in metabolic activity, heat production, & O_2 consumption of all or almost all tissues in the body. The BMR can ↑by as much as 60-100% above normal when large quantities of thyroid hormones are secreted. The rate & depth of respiration ↑d/t the enhanced metabolic rate & ↑O_2 use & CO_2 formation by cells. The use of energy substrates is greatly accelerated. Protein synthesis is ↑; however, protein catabolism also is ↑. When the quantity of thyroid hormone is slightly ↑, the muscles react w/ vigor; however, when the quantity is excessive, muscles become weakened from excess protein catabolism.

g. Effect of Thyroid Hormone on Growth: Thyroid hormones are necessary for normal growth in infants & children. In a hypothyroid state, the rate of tissue growth is greatly reduced. Thyroid hormone is required for normal growth & development of the brain during fetal life & for the first few years of postnatal life.

h. Effect of Thyroid Hormone on Specific Systems: Thyroid hormones have direct & indirect effects on the excitability of the heart. HR & the force of contraction are augmented w/ increasing thyroid hormone production. Vasodilation occurs from ↑cellular O_2 consumption & results in ↑blood flow to most tissues. Thyroid hormones ↑the rate of hormone secretion from most endocrine glands, especially the pancreas. The heightened cellular requirement for glucose mandates higher insulin secretion. Thyroid hormones also enhance the secretion of digestive juices & the motility of the GI tract, in addition to increasing an individual's appetite & food intake. In adults, thyroid hormones enhance the rapidity of cerebration.

i. Regulation of Thyroid Hormone Secretion: Specific feedback mechanisms operate through the hypothalamus & anterior pituitary gland to precisely control the rate of thyroid secretion. Thyrotropin-releasing hormone (TRH) from the hypothalamus causes cells of the anterior pituitary lobe to produce & secrete TSH. Potent stimuli to TRH & TSH release include low levels of T3 & T4, stress, & exposure to cold. TSH ↑all known activities of thyroid gland cells, resulting in ↑gland size & vascularity & ↑hormone synthesis & release. High circulating levels of primarily T3, & to a lesser extent T4, inhibit the secretion of TRH & TSH through a negative feedback effect on the hypothalamus & anterior pituitary.

C° TO F° = 1.8(C) + 32 KELVIN = C + 273

F° TO C° = F − 32 / 1.8 C° = K − 273

91. **CEA**
    a. **MOST COMMON complication after CEA:**
        i. HTN
    b. **Postop CVA**:
        i. usually result of an embolic phenomena, not hypotension or hyperperfusion
92. **MOST COMMON CAUSE mortality after CEA**:
    a. #1 = MI → EMBOLIC EVENT??
    b. #2 = CVA
93. **Bone probs w/ Cushing's d/t** → TOO MUCH ALDOSTERONE
    a. Osteoporosis
94. **Anion gap calculation**
    a. *Anion ion gap = Major Cations - Major Anions or [Na++] - ([Cl--] + [HCO3-]) = 8 -12 mEq/L*
    b. *Answer was 11* ↳ ANION GAP
95. **ABG: identify compensated respiratory alkalosis**
    a. pH within normal range, ↓PCO2, ↓HCO3 (NA) − (CL + HCO3)
96. **Clears 2-3 hrs before Surgery does what (2)** → CLEAR LIQUIDS **
    a. ↓postop insulin resistance, ↑postop insulin sensitivity, Shifts cellular metabolism to anabolic
    b. Picked ↓ gastric volume (stimulates emptying) & ↑gastric pH

ERAS

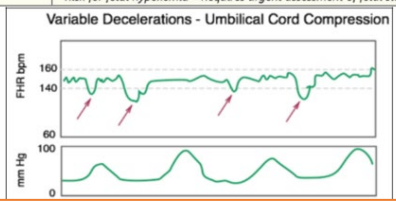
97. **CV changes w/ scoliosis**: MVP (most common), MR, PVR (d/t 1. arterial hypoxemia resulting in pulm vasoconstriction 2. Changes in pulm arteriols as a consequence of ↑pulmonic pressure may cause narrowing & irreversible PAH 3. Compressed chest wall may ↑vascular resistance 4. Development of scoliosis at an early age inhibits growth of pulm vascular bed) & ensuing PAH → development of RVH
98. **Ginseng, Kava kava, St. John's wort (see chart on top right of page**)**
    a. **Ginseng**: interacts w/ *warfarin*; causes coagulopathy
    b. **Kava kava**: interacts w/ benzos, ETOH, & anesthetics
    c. **St. John's wort**: interacts w/ dig, OCPs, serotonin antagonists, & anesthetics

    C's → ANTICOAGULATION
99. **MOST COMMON tracheoesophageal fistula (picture)**
    a. Key diagnostic indicator: Maternal polyhydramnios
    b. S/S: unable to pass OG tube, choking/cyanosis w/ feeding, recurrent pneumonia
    c. Precordial stethoscope on left chest will immediately detect a right mainstem intubation
100. **2nd stage labor pain originates from what nerves**:
    a. S2-S4: somatic afferent fibers carried via *pudendal* nerves; from end of 1st stage until baby delivered
    b. 1st stage = T10-L1: visceral afferent unmyelinated C fibers
101. **Flow volume loop (Patient sitting → normal loop, but tracheal stenosis when supine)**
    a. Mediastinal mass
102. **Fetal heart decels & causes (8 Q's)**
    a. ***Know VEAL CHOP**

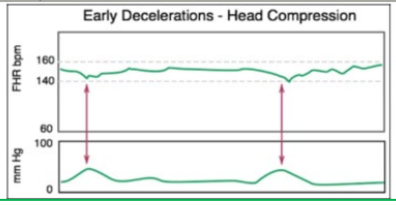
Here's a mnemonic to tie it together: VEAL CHOP

• **V**ariable decels:	**C**ord compression
• **E**arly decels:	**H**ead compression
• **A**ccelerations:	**O**K or give O2
• **L**ate decels:	**P**lacental insufficiency

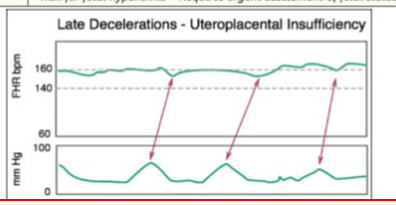
Variable	No consistent pattern between FHR and uterine contraction Umbilical compression causes baroreceptor mediated reduction in FHR Decelerations are usually self-limiting Maintains variability during deceleration Fetal compromise prolongs FHR recovery time *Risk for fetal hypoxemia – Requires urgent assessment of fetal status*



Early	Uterine contractions compress the fetal head Head compression increases vagal tone (HR < 20 bpm from baseline) Onset and offset parallels uterine contraction Loses variability with each deceleration *No risk of fetal hypoxemia*



Late	Decreased uteroplacental perfusion leads to fetal compromise FHR falls after peak of contraction then returns to baseline after contraction Occurs with each contraction Gradual (not abrupt) reduction in FHR Cause by maternal hypotension, hypovolemia, acidosis, preeclampsia *Risk for fetal hypoxemia – Requires urgent assessment of fetal status*



b. **Signs of fetal distress or a healthy baby:** healthy = beat to beat variability VARIABILITY = NORMAL SINUSOIDAL = BAD
c. Sinusoidal rhythm: ominous; caused by maternal hemorrhage (abruption)

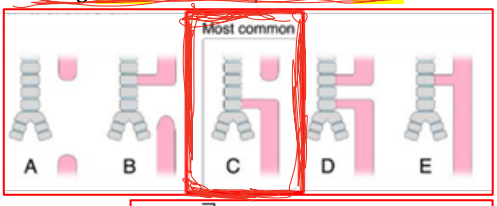
103. **PO versed dose for 44kg 6 yo**:
    a. 0.25-1 mg/kg = 11-44 mg
104. **MOST COMMON hematologic problem w/ abruption**:
    a. DIC (↓platelets, factor 5 & 8; ↑fibrin split products)
105. **Mg load for Tx of PreE** (type in question):
    a. 4 grams → 4 - 6 GRAMS
106. **MgSO4**
    a. Relaxes uterus & ↑risk of PP hemorrhage → *
107. **Gas MOST COMMON w/ emergence delirium**:
    a. Sevo (most common) & Des (Nagelhout)
108. **Acute onsent mental status change after Surgery in geriatric patient**:
    a. Delirium

POST-OP DELIRIUM

POST-OP COGNITIVE DYSFUNCTION → WEEKS TO MONTHS POST-OP

Suggested Uses, Potential Toxicities, and Drug Interactions of Dietary Supplements and Herbal Remedies

	Suggested Uses	Potential Toxicity	Drug Interactions
Black cohosh	Menopausal symptoms	Gastrointestinal discomfort	
Chaste tree berries	Premenstrual symptoms	Pruritus	Dopamine-receptor antagonist
Cranberry	Urinary tract infections	Nephrolithiasis	
Dong quai	Menopausal symptoms	Rash	Warfarin
Echinacea	Upper respiratory infections	Hypersensitivity reactions Hepatic inflammation	
Evening primrose	Eczema Irritable bowel syndrome Premenstrual symptoms Rheumatoid arthritis	Nausea Vomiting Diarrhea Flatulence	Antiepileptic drugs
Feverfew	Prevent migraine Arthritis Allergies	Hypersensitivity reactions Inhibits platelet activity	Warfarin
Garlic	Hypertension Hypertriglyceridemia Hypercholesterolemia	Gastrointestinal discomfort Hemorrhage	Warfarin
Ginger	Motion sickness Vertigo		Warfarin
Ginkgo biloba	Dementia Claudication Tinnitus	Gastrointestinal discomfort Headache Dizziness Bleeding Seizures	Warfarin
Ginseng	Fatigue Diabetes	Tachycardia Hypertension	Warfarin
Goldenseal	Laxative	Hypertension Edema	
Kava-kava	Anxiety	Rash Sedation Liver toxicity	Benzodiazepines Alcohol Anesthetic drugs
Kola nut	Fatigue	Irritability Insomnia	Stimulants
Licorice	Gastric ulcers	Hypertension	
Saw palmetto	Prostatic hyperplasia	Gastrointestinal discomfort	
St. John's wort	Depression Anxiety	Headache Insomnia Dizziness Gastrointestinal discomfort	Digoxin Oral contraceptives Serotonin antagonists Anesthetic drugs
Valerian	Insomnia	Headaches	Benzodiazepines Anesthetic drugs Antiepileptic drugs



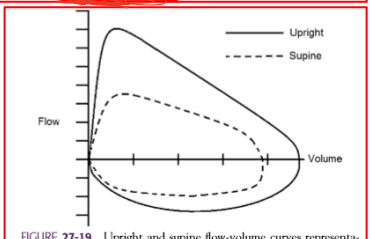


FIGURE 27-19 Upright and supine flow-volume curves representative of anterior mediastinal mass. Note abnormality appears in the supine position, when gravity causes the mass to impinge on the lower airway. The greatest decrement affects the expiratory limb of the loop, which is generally indicative of variable *intra*thoracic obstructions. Thoracic excursion during spontaneous inspiration relieves the pressure. This effect is lost when ventilation is changed to exclusively positive pressure input from above.

TABLE 48-3 Commonly Prescribed Pediatric Premedicants	
Drug Name	**Pediatric Dose**
Opioids	
Morphine sulfate	0.1-0.3 mg/kg IM 0.05-0.1 mg/kg IV
Fentanyl	0.5-1.0 mcg/kg IV 10 mcg/kg oral transmucosal
Benzodiazepines	
Midazolam	0.07-0.1 mg/kg IM 0.2 mg/kg nasally 0.025-0.05 mg/kg IV 0.25-1.0 mg/kg PO 0.5-1 mg/kg rectally
Ketamine	5-10 mg/kg PO 2-10 mg/kg IM 1-2 mg/kg IV
Clonidine	2.5-5.0 mcg/kg PO 1-2 mcg/kg IV
Dexmedetomidine	0.5-1.0 mcg/kg intranasally 2-5 mcg/kg PO 0.4-0.6 mcg/kg IV infusion in 100 mL saline
Anticholinergics	
Atropine	0.02 mg/kg PO, IV, IM
Glycopyrrolate	0.01 mg IV

MGSO4

Serum Level (mg/dL)	Clinical Effect
1.8 – 3	Normal level
4 – 8	Tocolysis
7 – 9.5	Anticonvulsant
10 – 12	Patellar reflex abolished
12 – 18	Respiratory depression
> 18	Apnea
> 25	Cardiac arrest

109. Anest mgmt of patient on lithium

a. Consider toxicity during preop eval (review most recent concentration); to prevent significant renal reabsorption, admin Na+-containing IV solutions during periop period. Stimulation of UO w/ thiazide diuretics must be avoided. Monitor ECG for lithium-induced conduction defects/dysrhythmias. Association of sedation w/ lithium therapy suggests anesthetic requirements may be ↓. Monitoring effect of NMBs is indicated because DOA of both depolarizing & NDNMBs may be prolonged. (Stoelting)

b. T-wave smoothing, ventricular arrhythmias, myocarditis; sinus dysfunction can → extreme atropine-resistant sinus brady; dehydration will → ↑in lithium blood levels; Continue pre & periop; check thyroid function if indicated; monitor serum Na+ (Nagelhout)

110. Nonlap Surgery MOST COMMON w/ ↓FRC (Thoracic, Upper Abd, & 2 other choices):

a. thoracic

111. OB pneumoperitoneum max pressure for nonOB Surgery:

a. Maintain low pressures or use a gasless technique (Nagelhout): 15

112. Plug in faulty equipment =

a. OR becomes grounded ??

113. Full O2 tank @ 2 L/min will last how many min?

a. 330 ↳ HOLDS 660 mL

114. PISS 2,5 =

a. O2 → $N_2O = 3,5$; AIR = 1,5

115. Des in a variable vaporizer will do what?

a. 1. Delivery would be unpredictable (s/t temp compensation being almost impossible) & 2. The amount of vapor produced would be uncontrollable (would need much high FGF to achieve sufficient dilution of the large volume of Des vapor) (Barash); over deliver

116. Everything about Tech 6 vs variable bypass vaporizers:

a. Has an injector to inject precise amt into FGF, No vaporizing chamber, Only for Des, Heats to 39° C, Pressurizes to 2 atm, Doesn't compensate for changes in elevation

117. Piston ventilator

a. Ventilator substitutes an electrically driven piston for the bellows; requires either minimal or no pneumatic (O2) power.

b. The major advantage: ability to deliver accurate Vt to pts w/ very poor lung compliance & to very small pts.

c. During VC ventilation the piston moves at a constant velocity whereas during PC ventilation the piston moves w/ decreasing velocity. As w/ the bellows, piston fills w/ gas from the breathing circuit. To prevent generation of significant negative pressure during the downstroke of the piston the circle system configuration has to be modified. The ventilator must also incorporate a negative-pressure relief valve or be capable of terminating the piston's downstroke if negative pressure is detected. Introduction of a negative-pressure relief valve to the breathing circuit may introduce the risk of air entrainment & the potential for dilution of O2 & VA concentrations if patient breathes during mechanical ventilation & ↓FGFs. (M&M)

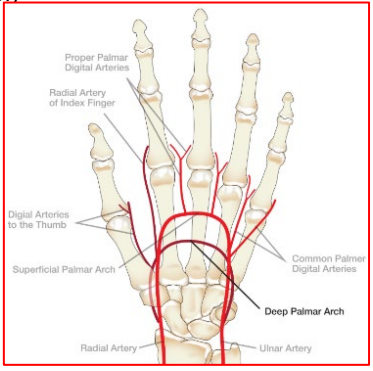
d. More recently, w/ introduction of circle breathing systems that integrate fresh gas decoupling, resurgence has been seen in the utilization of mechanically driven anesthesia ventilators. These "piston"-type ventilators utilize a computer-controlled stepper motor instead of compressed drive gas to actuate gas movement in the breathing system. In these systems, rather than having dual circuits, a single patient gas circuit is present. Thus, they are classified as piston-driven, single-circuit ventilators. The piston operates much like the plunger of a syringe to deliver the desired Vt or airway pressure to the patient breathing circuit. Sophisticated computerized controls are able to provide advanced types of ventilatory support such as SIMV, PCV, & PS-assisted ventilation, in addition to VC & PC. Since the patient's mechanical breath is delivered w/o use of compressed gas to actuate a bellows, these systems consume dramatically less compressed gas during ventilator operation than traditional pneumatic ventilators. This improvement in efficiency may have clinical significance when the anesthesia workstation is used in a setting where no pipeline gas supply is available (e.g., remote locations or office-based anesthesia practices). (Barash)

118. Pumping effect is d/t

a. Intermittent back pressure that results from either PPV or use of the O2 flush valve may result in ↑vaporizer output. This phenomenon, known as the pumping effect, is more pronounced at low FGF rates, low concentration dial settings, & low levels of liquid anesthetic in the vaporizing chamber. In addition, the pumping effect is ↑by rapid RR, high PIP, & rapid ↓ in pressure during exhalation. Newer variable bypass vaporizers such as the GE-Datex-Ohmeda Tec 4, Tec 5, & Tec 7, & Dräger Vapor 19.n & 20.n (Vapor 2000 series) are relatively immune from the pumping effect. 1 proposed mechanism for the pumping effect is dependent on retrograde pressure transmission from the patient circuit to the vaporizer during the inspiratory phase of PPV. Gas molecules are compressed in both the bypass & vaporizing chambers. When the back pressure is suddenly released during the expiratory phase of PPV, vapor exits the vaporizing chamber via both the vaporizing chamber outlet & retrograde through the vaporizing chamber inlet. This occurs because the output resistance of the bypass chamber is lower than that of the vaporizing chamber, particularly at low concentration dial settings. The ↑output concentration results from the increments of vapor that flow in a retrograde direction to the bypass chamber. To ↓the pumping effect, the vaporizing chambers of contemporary variable bypass systems are smaller than those of older model vaporizers (e.g., Fluotec Mark II: vaporizing chamber volume 750 mL). Consequently, no substantial volumes of vapor can be discharged from the vaporizing chamber into the bypass chamber during the expiratory phase of ventilation. The Dräger Vapor 19.1 & 20.n have a long spiral tube that serves as the inlet to the vaporizing chamber. When the pressure in the vaporizing chamber is released, some of the vapor enters this tube but does not enter the bypass chamber because of the tube's length. The Tec 4 has an extensive baffle system in the vaporizing chamber, & a one-way check valve has been inserted at the common gas outlet (outlet check valve) to minimize the pumping effect. This check valve attenuates, but does not eliminate, the pressure ↑because gas still flows from the flowmeters to the vaporizer during the inspiratory phase of positive-pressure ventilation. (Barash)

119. Which finger shows desat the fastest w/ pulse ox

a. Index finger



120. EtCO2 not returning to baseline is d/t:

a. Incompetent valve; Exhausted soda lime & patient is rebreathing exhaled CO2

121. O2 analyzer

a. **Location**: resides in the inspiratory limb

b. **Types**: 1. Galvanic fuel cell, 2. Paramagnetic, 3. Polarographic (Clark electrode)

c. **Monitors**: O2 concentration

122. O2 supply failure alarm:

a. Shuts off N2O if O2 pressure falls below 25 psig, no mixing of N2O & O2 occurs here

123. O2 check valve does what

a. If 1 tank of gas in back of machine runs out, other can't go back & fill it

b. Ensures unidirectional forward flow, so that a machine running on cylinder supplies, w/ the hoses D/Cd at the wall outlet, doesn't leak.

124. Common gas outlet location

a. low pressure system; Receives the gases & vapors from the machine; Connected by high pressure hose to the soda lime canister

125. CO2 laser - Why is it used?

a. Far infrared; ↓↓↓tissue penetration & can be used where great precision is needed. Also absorbed by H2O, so that minimal heat is dispersed to surrounding tissues. Used primarily for procedures in the oropharynx & in & around the vocal cords; Use suction to get rid of smoke (prevent fire & spread of microbes); used extensively in Gen Surgery, ortho, gyne, urology, & ENT & is associated with ↓EBL

126. Avagadro's #:

a. 1 mole of gas = 6.023 X 10^{23} molecules; 1 mole of gas at standard temp (**273 K+**) & pressure occupies 22.4 L

127. 1 mole of any gas will expand to how many mL at 0° C:

a. 22,400 ml

b. Conversion Chart

Name	Symbol	Factor
tera-	T	10^{12}
giga-	G	10^{9}
mega-	M	10^{6}
kilo-	k	10^{3}
hecto-	h	10^{2}
deca-	da	10^{1}
deci-	d	10^{-1}
centi-	c	10^{-2}
milli-	m	10^{-3}
micro-	μ	10^{-6}
nano-	n	10^{-9}
pico-	p	10^{-12}

128. Test w/ least ionization risk (MRI, CT, fluoro, xray):

a. MRI

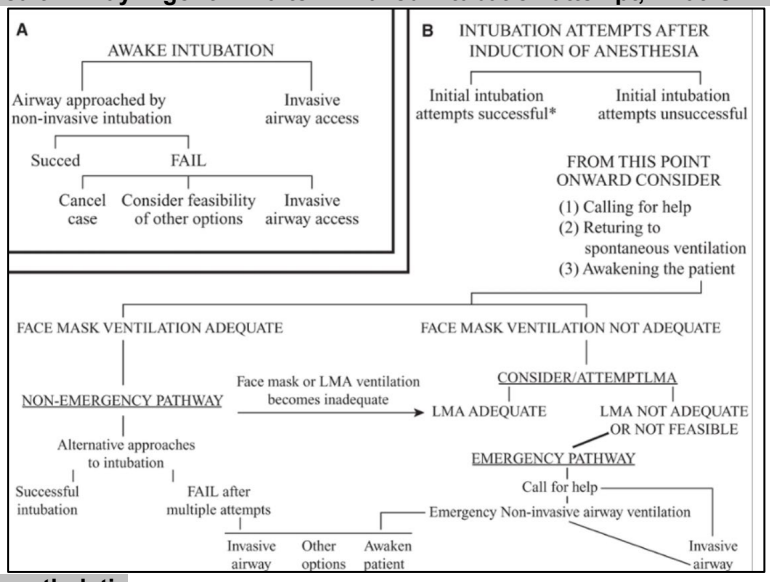
129. Endocarditis Abx prophylaxis in PCN Ax patient

a. Clinda or Vanc? Clinda per AHA guidelines

130. Robotic Prostatectomy positioning:

a. Lithotomy & steep T-burg, both arms tucked (remind circulator to lower legs in lithotomy before going T-burg)

131. Difficult Airway Algorithm: after 1st failed intubation attempt, what is BEST (select 2)



132. Sympatholytic

a. cause dissolution of SNS at adrenergic receptor sites; Ex: alpha blockers (phenoxybenzamine, phentolamine, tolazoline, prazosin), beta blockers (-olol), CCBs (verapamil, nifedipine, nicardipine, nimodipine, diltiazem)

133. Drug in OB when need complete uterine relaxation:

a. NTG?

134. Conscious sedation definition:

a. drug-induced sedation (moderate sedation/analgesia); patient responds purposefully to verbal commands either alone or w/ light tactile stimulation, patient maintains a patent airway & spontaneous ventilation, CV function maintained

135. Amaurosis fugax:

a. Unilateral loss of sight; Sign of *impending CVA*; emboli travel from internal carotid to ophthalmic artery & impairs perfusion of optic nerve & causes retinal dysfunction; Occurs in 25% of pts w/ high grade carotid stenosis

136. Murmurs & locations

a. Heart Murmur association to valvular disease: 2 ICS midsystolic R sternal border; AS **(ASSS)**

b. AR: diastolic, R sternal border **(ARDS)**

c. MS: diastolic, apex of L axilla **(MSDA)**

d. MR: systolic, apex of L axilla **(MRSA)**

137. Questions about MDAs supervising CRNAs:

a. 1 MDA & 5 CRNAs is billed: supervising

138. Joint Commissions purpose:

a. To check on Medicare regulations

139. Interventions when cerebral oximeter <25 % of baseline:

a. remove clamp; a > 25% change from baseline suggests a reduction in cerebral oxygenation.

b. Package insert

Observe	Consider
rSO, directly correlated with change in BP	Dysautoregulation
rSO, and ↑ BP inversely correlated	Vasoconstrictor hypoperfusion
rSO, and ↓ BP uncorrelated	• Airway inadequacy • Ventilation abnormality (i.e., hypocapnia) • Anesthetic delivery inadequacy • Cardiopulmonary/ CPB dysfunction • Blood loss/hemodilution • Nonpulsatile perfusion • Brain temperature increase • Intracranial hypertension
rSO, ↑ BP uncorrelated	• Cerebral hyperemia • Brain temperature decrease • Pulsatile perfusion reestablished • Low O, reserve with rSO, increase >5%
rSO, asymmetry appearance	• Patient malposition • Heart malposition • Cannula, catheter, clamp, or vent malposition • Low CO, reactivity with rSO, increase <5%
rSO, trend rapid oscillation	• Seizure-like activity • Cerebral vasospasm

140. Where to aim fire extinguisher:

a. Base of fire

141. Know anatomy of nephron, what is absorbed, where drugs work, & what part is in the medulla vs cortex

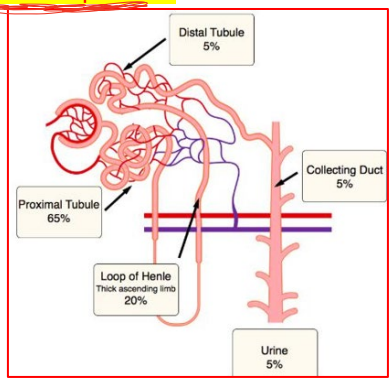
a. **Anatomy**
   i. **Renal cortex**: outer section; contains glomerulus, Bowman's capsule, proximal tubules, & distal tubules (juxtaglomerular apparatus is located in distal tubule); receives 90% of RBF
   ii. **Renal medulla**: inner section; contains LOH & collecting ducts; receives 10% of RBF; more sensitive to ischemia
   iii. Receives 20-25% of cardiac output
   iv. RBF: autoregulated 50-150; ↓ 10% per decade after age 50

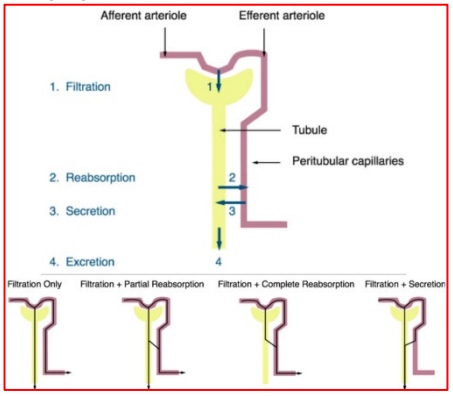
b. **Renal Vasodilation Pathways**: prostaglandins, ANP, Dopamine receptors
   i. **DA1**: function: vasodilation, ↑RBF, ↑GFR, diuresis, Na+ excretion
      1. **Fenoldopam**: selective DA1 receptor agonist that ↑RBF w/o affecting arterial BP; 0.1-0.2 mcg/kg/min

c. **Most important determinant of GFR**: glomerular hydrostatic pressure

d. **Most useful indicator of GFR**: creatinine clearance

e. **Na++ handling**:





   i. **PCT**: in cortex; bulk reabsorption of solutes & water; K+, Cl-, & bicarb follow Na+ in direct proportions by Na+ co-transport mechanism; 65% of these ions are reabsorbed in PCT; proteins, glucose, uric acid, urea
   ii. **LOH (Descending)**: in medulla; countercurrent mechanism + high permeability to H2O; 20% water is reabsorbed; vasa recta are the peritubular capillaries that run parallel to LOH & are essential for the countercurrent mechanism
   iii. **LOH (Ascending)**: in medulla; countercurrent mechanism + ***NO PERMEABILITY TO H2O***; Na+(20%), K+, & Cl- are actively pumped into the interstitium via Na+-K+-2Cl co-transporter; since water can't follow Na+ the ultrafiltrate becomes hypotonic & interstitial space becomes hypertonic; H is excreted via Na+ countercurrent mechanism
   iv. **DCT**: in cortex; fine tunes solute concentration; Na+ (5%) is reabsorbed & K+,Cl-, & bicarc follow via Na+ co-transport mechanism; the *late distal tubule is impermeable to H2O except in presence of aldosterone or ADH*; adjusts urea concentration
   v. **Collecting duct**: in medulla; *regulates final concentration of urine*; reabsorbs Na+ (5%); aldosterone works here & DCT

142. Diuretics (COLT PA)
   i. Carbonic anhydrase inhibitors: acetazolamide & dorzolamide; associated w/ hyperchloremic metabolic acidosis; works in PCT (cortex); bicarb isn't reabsorbed (net loss of bicarb & Na+ w/ net gain of H & Cl-)
   ii. Osmotic diuretics: mannitol, glycerin, & isosorbide; works in PCT (cortex) as well as LOH (medulla)
   iii. Loop diuretics: lasix, bumetanide, & ethacrynic acid; works in ascending LOH (medulla); poison the Na+-K+-2Cl transporter, amount of Na+ that remains in tubule overwhelms the distal tubule's reabsorption capability thus a large volume of dilute urine is excreted; K+, Ca2+, Mg, & Cl- are also lost to the urine
   iv. Thiazide diuretics: HCTZ, chlorhalidone, metolazone, & indapamide; works in DCT (cortex); inhibits Na+-Cl- exchanger which ↑Ca2+ reabsorption & serum Ca2+; cause hyperglycemia
   v. Potassium sparing diuretics: amiloride & triamterene; works in collecting ducts (medulla); inhibit K+ secretion & Na+ reabsorption
   vi. Aldosterone antagonists: spironolactone; works in collecting ducts (medulla); subclass of K+-sparing diuretics; blocks aldosterone at mineralocorticoid receptors thus inhibiting K+ secretion & Na+ reabsorption

b. **Periop AKI**:
   i. **Risk factors**: pre-existing kidney disease, prolonged renal hypoperfusion, CHF, advanced age, sepsis, jaundice, high risk Surgery (use of aortic cross clamp & liver transplant)
   ii. **Classification based on insult location**:
      1. **Prerenal – hypoperfusion**: hypovolemia, ↓CO, systemic vasodilation, renal vasoconstriction, ↑intraabdominal pressure
      2. **Intrinsic – Parenchymal**: ischemia (renal medulla is more susceptible to ischemic insult), nephrotoxic drugs (IV contrast, nephrotoxic Abx, NSAIDs)
      3. **Postrenal – Obstruction**: obstruction anywhere b/w collecting system & urethra

c. **Use of hydroxyethyl starches are associated w/**: ↑risk of renal morbidity

d. **Vasopressin**: preferentially constricts the EFFERENT arteriole; maintains GFR & UOP better than NE or Neo

e. **CKD**:
   i. **Most common cause of CKD?** DM
   ii. **2nd Most common cause of CKD?** HTN
   iii. **Why do chronic renal failure pts get anemic?** reduced EPO production (normally stimulates stem cells in bone marrow to produce erythrocytes) → normochromic normocytic anemia
   iv. 1st stage: ↓ GFR with slight ↑in serum Cr & urea
   v. 2nd stage: when GFR approaches 10% of normal, ↑K+, Na+, & aldosterone
   vi. 3rd stage: Na+ homeostasis & ECF volume regulation
   vii. Complications: uremic syndrome; S/S: anemia, N/V, pruritis, anorexia, fatigue, & coagulopathy;
   viii. Tx = dietary protein restriction to ↓protein catabolism & urea production

f. **Methoxyflurane**: metabolism liberates a significant amount of free fluoride ions; this is directly r/t a high output RF

g. **Benzlisoquinolines**: cisat is better than atracurium (because atra produces more laudanosine)

h. **Aminosteroids**: Roc is primarily eliminated by liver; associated w/ unpredictably ↑DOA

i. **Opioids**: fentanyl, Sufentanil, alfentanil, & Remi don't produce active metabolites & are better choices in pts w/ RF

j. **Precedex**: biotransformed by liver & may be used safely in pts w/ RF

k. **Nephrotoxic Agents**
   i. **Sevo**: compound A (produced in breathing circuit), FGF 1L/min safe up to 2 MAC hrs; free fluoride ions (produced in liver), hepatic metabolism liberates inorganic fluoride ions are nephrotoxic
   ii. **Aminoglycosides (Gent, Tobra, & Amikacin)**: bind in PCT then transported to cytosol where they induce free radical damage
   iii. **NSAIDs**: nephrotoxic
   iv. **Calcineurin Inhibitiors (Cyclosporins & Tacrolimus)**: cause HTN & renal vasoconstriction

143. Which factor of VAs is onset?
   a. Blood:gas coefficient?

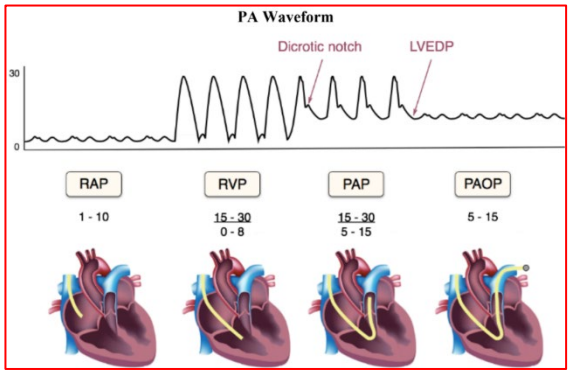
144. Trauma plus super bloody airway & unstable c spine; which airway method is least appropriate?
   a. Video laryngoscopy?

145. ECT
   a. What is highest risk?
      a. MOST COMMON CAUSE death = MI & cardiac dysrhythmias
   b. Which of these pts can't have ECT?
      a. (MI within 2 wks, pregnant, AICD, & ?)
   c. What are the side effects?
      a. 1st → **PARASYMPATHETIC** response: ↓BP, brady to asystole *can preTx w/ glycopyrrolate
      b. 2nd → ↑↑↑**SYMPATHETIC** response: HTN, tachy, dysrhythmia, MI)
   d. Absolute CONTRAINDICATED
      a. pheo, MI <6 wks, recent CVA <3 months, recent crani (<3 months), intracranial mass/lesion, unstable C-spine)
   e. Relative CONTRAINDICATED:
      a. angina, CHF, PPM, AICD, severe pulm disease, major bone Fx, glaucoma, retinal detachment, thrombophlebitis, pregnancy
   f. Initial: ↑PNS = ↓HR & BP (15 sec); **Secondary**: ↑SNS = ↑HR & BP (several min)
   g. Good drugs: Methohexital, Precedex, Clonidine, Esmolol

146. VWB disease:
   a. give desmopressin

147. Match which is released from which part of body
   a. Somatotropin (GH): ant pituitary
   b. Vasopressin (ADH): post pituitary
   c. Somatostatin: hypothalmus/delta cells in pancreas
   d. Cortisol: adrenal cortex → zona fasciculata

148. Know distances from RIJ to different areas in heart & waveforms when floating a Swan
   a. Choose waveform in area 25-35 cm from RIJ (RV)
   b. What depth would you expect PAOP? 50 cm



Insertion Site → Vena Cava & Right Atrial Junction		Vena Cava & Right Atrial Junction → Catheter tip	
Subclavian	10 cm	Right atrium	0 – 10 cm
Right IJ	15 cm	Right ventricle	10 – 15 cm
Left IJ	20 cm	Pulmonary artery	15 – 30 cm
Femoral	40 cm	PAOP position	25 – 35 cm
Right median basilic	40 cm		
Left median basilic	50 cm		

149. Which adds electrons so drug can be eliminated?
   a. Phase 1 reduction

150. Lots of High/Intermediate/Low system questions
   a. **High**: begins at cylinder & ends at cylinder regulators; Components: Hanger yoke, Yoke block & check valves, Cylinder pressure gauge, Cylinder pressure regulators
   b. **Intermediate**: exposed to pipeline pressure (50 psi); begins at pipeline & ends at flowmeter valves; Components: Pipeline inlets, Pressure gauges, O2 pressure failure device, O2 2nd stage regulator, O2 flush valve, Ventilator power inlet, Flowmeter valves
      i. O2 flush valve: 35-75 L/min at 45-55 psi; risk barotrauma & inhaled anesthetic dilution; avoid use during inspiration
   c. **Low**: distal to flowmeter needle valve; begins at flowmeter tubes & ends at common gas outlet; Components: flowmeter tubes (Thorpe), Vaporizers, Check valve (if present), Common gas outlet

151. Trace O2 path from supply to patient (pick 6):
   a. O2 tank, Check valve, Thorpe tube, Variable bypass, Common gas outlet, Circuit

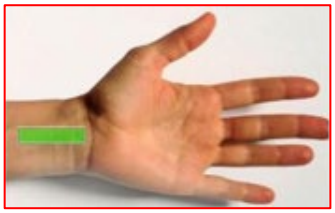
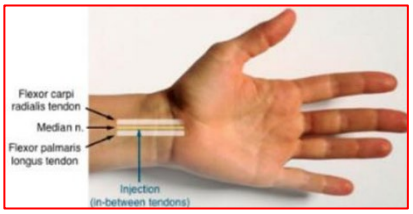
152. CPP
   a. **Calculation:** MAP-ICP (or CVP, whichever is higher)
   b. **Ketamine**: ↑CPP

153. MAP calculation:
   a. (SBP + 2DBP)/3 or [(CO X SVR/80)] +CVP

154. Pediatric ETT size cuffed & uncuffed:
   a. Age/4 + 4 = diameter (subtract 0.5mm for cuffed); then triple it = length

155. Median nerve block:

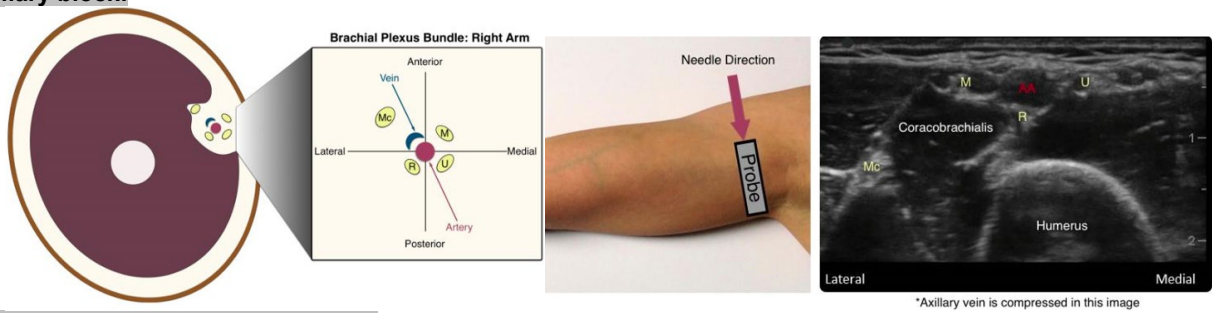
a. Provides sensory innervation to ventral portion of thumb, 1st & 2nd fingers, lateral half of 3rd finger, & palm of hand.
b. **Wrist anatomy:** median nerve is medial to radius longus tendon, lateral to adductor pollicus
c. **CONTRAINDICATED**: pts w/ carpal tunnel
d. **Hotspots**:

156. BP block

a. **Complications:** interscalene- verterbral artery injection, pneumo, horner's syndrome (stellate ganglion block; located @ C7), block phrenic nerve, injection into epidural/SA space; supraclavicular- highest risk = pneumo (R>L), SCA puncture; infraclavicular- SCA/Axillary artery puncture, technically difficult; Axillary- axillary artery puncture, miss musculocutaneous
b. **Infraclavicular**: targets cords; Landmarks: Pectoralis major & pectoralis minor

157. Axillary block:



*Axillary vein is compressed in this image

158. O2 tank calculations for E cylinders:

a. Tank capacity (L)/full tank pressure (psi) = contents remaining (L)/gauge pressure (psi)
b. O2 tank = 1900 psi/660 L
c. N2O tank = 745 psi/1590 L (wt. full = 20.7 lbs; wt empty = 14.1 lbs)
d. Air tank = 1900 psi/625 L

159. Most common O2 sensor piezo, infrared etc?:

a. Infrared

160. Pulse ox wavelengths:

a. Deoxy is 660 (red light)
b. Oxy is 990 (near-infrared)
c. Based on Beer Lambert Law; relates intensity of light transmitted thru a solution & the concentration of the solute within the solution

161. O2 failsafe: how does it work?

a. Pressure

162. Temperature effect on EEG:

a. Hypothermia → ↓frequency, burst suppression, slowing, eventually isoelectric

163. Carbon monoxide production:

a. ↑w/ dessicated soda lime (Des > Iso >>> Sevo)

164. Bain circuit vs Jackson Rees vs Mapleson D - what is the difference, what does it do w/ the work of breathing?

a. Bain is modified D; low resistance so ↓WOB

When to Use a Mapleson Circuit?

By knowing the advantages and disadvantages of the Mapleson designs, you can easily determine when to use them.

Advantages	Disadvantages
Less airway resistance (good for peds)	Increased apparatus dead space
Convenient	Requires high FGF to prevent rebreathing
Easily scavenged	Loss of heat and humidity
Bain circuit prevents heat loss	Inefficient use of inhaled anesthetics
	Risk of environmental pollution
	Unrecognized kinking of fresh gas hose in Bain circuit
	• Use Pethick test

165. Benefits of the mapleson system used in pediatrics:

a. APL is right next to bag?

166. Mapleson system- which is most common in peds:

a. F (is a modified E aka Jackson Reese) *↓WOB but heat loss

167. Best mapleson circuit for spontaneous breathing:

a. A

168. Coronary artery ostium: where are they located?

a. Above/proximal/cephalad to the cusps *answer option said posterior to cusps, so not sure

169. Geriatric lung volume changes:

a. **Increased:** Minute ventilation, FRC, Closing capacity, RV, lung compliance
b. **Decreased**: lung elasticity, Chest wall compliance, Vital capacity, ERV, IRV, FEV, FEV1, protective airway reflexes, upper airway tone, response to hypercarbia & hypoxia
c. **No change**: TLC

170. Geriatric CV changes:

a. **Increased**: myocardial hypertrophy, BP, pulse pressure
b. **Decreased**: arterial compliance, venous compliance, myocardial compliance, cardiac conduction, diastolic function, HR, CO
c. **No change**: systolic function

171. Geriatric Renal changes:

a. **Increased:** BUN, ADH response to hypertonic saline, ability to develop hyper/↓K+
b. **Decreased**: kidney mass, RBF, renal plasma flow, GFR, renal function, muscle mass, creatinine production, fluid handling, Na+ handling, concentrating ability, response to ADH, response to aldosterone
c. **No change**: serum creatinine

172. Bariatric lung volume changes: restrictive ventilatory defect
- a. **Increased**: O2 consumption, CO2 production, Minute ventilation, pulm BF
- b. **Decreased**: FRC, ERV, Lung compliance, Vital capacity, Total lung capacity, PaO2
- c. **No change**: closing capacity, RV, PaCO2

173. Bariatric CV changes:
- a. **Increased**: CO ↑100 ml/min for every extra 1 kg of fat, blood volume, SV, O2 consumption, LV workload, pulm BF, RV workload, BP
- b. **Decreased**: ventricular compliance
- c. **No change**: HR
- d. **HTN**: is result of hyperinsulinemia, SNS, & RAAS activation, as well as elevated cytokine concentration in plasma

174. Change in NIBP in forearm vs upper arm:
- a. Extra-large thigh cuffs can be used on the upper arm/lower leg (over the posterior tibial artery; A regular-size or large BP cuff can be used on the forearm over the radial artery; Cuffs w/ bladders that encircle a minimum of 75% of upper arm circumference (preferably entire arm) should be used. Forearm measurements w/ a standard cuff overestimate both SBP & DBP in obese pts.

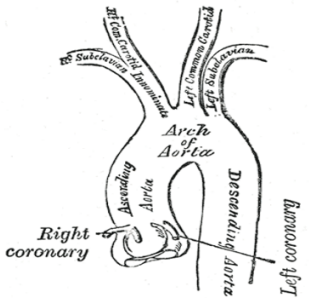
175. Complications of bariatric Surgery & which are most common:
- a. Anastomotic leak, Ventilation abnormalities postop (pneumonia/atelectasis), Rhabdo, Thromboembolism, Wound infection/dehisence, Tachycardia (unexpected HR >120 is alarming), Fever, Abd pain
- b. Barash says MOST COMMON = respiratory, vascular, & wound; M&M says MOST COMMON = respiratory failure

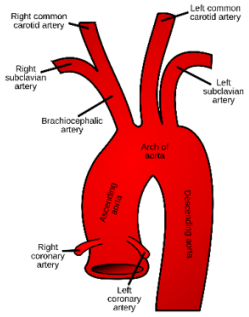
176. Vent settings for bariatric Surgery
- a. PEEP is the only vent setting shown to improve resp function (<15 cm H2O); recruitment maneuvers followed by PEEP
- b. Vt: 10-12 ml/kg
- c. RR: 12-14 bpm

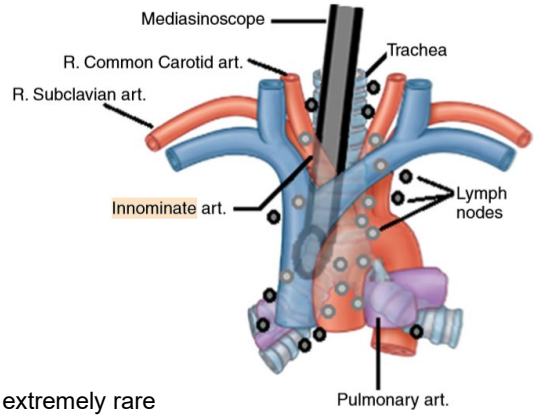
177. Sniffing Position Modifications for Obesity:
- a. Head Elevated Laryngoscopy Position = sternum & external auditory meatus are in same horizontal plane; put bed in reverse T-burg (unloads diaphragm & may prolong time b/w apnea & desat)

178. Hyperkalemia:
- a. **Interventions (tall, peaked T waves)**: Ca2++2, gluc , CaCl, Hyperventilate, Bicarb, IV insulin/D50, Diuretics, Kayexalate, βagonist, HD
- b. **ECG changes**: Tall peaked T waves, Wide QRS, prolonged PRI

179. Hot spot mediastinoscopy compression of which vessel:
- a. innominate/brachiocephalic







180. Postop liver failure could be related to which anesthetic:
- a. MOST COMMON w/ halothane d/t ↑trifluoroacetic acid (TFA); Possible w/ others but extremely rare

181. VA & Hepatic Biotransformation: Sevo > Iso > Des > N2O

182. Alveolar ventilation: (Vt-Dead space) X RR
- a. **Begins where**:
  - i. respiratory bronchioles (resp zone)
- b. Determines rate of removal of CO2 from body (MV does not)

Lung Region	Alveolar Ventilation	Perfusion	V/Q	$P_{A}O_2$	$P_{A}CO_2$	$P_{A}N_2$
Non-Dependent	Low	Low	High	High	Low	~ Same
Dependent	High	High	Low	Low	High	~ Same

183. Things that ↑dead space:
- a. extending neck of intubated patient
- b. ↑**dead space**: ↓CO, Atropine/Anticholinergics (bronchodilation), Neck extension, FM, HME, PPV, ↓pulm BF, COPD, PE, Sitting
- c. ↓ **dead space**: ETT, LMA, Trach, Neck flexion, Supine, Head down

184. During tidal breathing, intrapleural pressure is:
- a. always negative (keeps lungs inflated)
- b. *Only time it becomes positive is during forced expiration (Valsalva)
- c. *Intrapulmonary pressure = negative during inspiration, + during expiration

185. Cricothyrotomy CONTRAINDICATED (other than kids):
- a. goiter/mass; Age <6 yrs, Laryngeal Fx, Laryngeal neoplasm, Infection/abscess, Tumor, Distortion of anatomy, Bleeding/coagulopathy

186. Afferent branch of the oculocardiac reflex: V (trigeminal)

187. Know how to convert 20 cm of distance in to a change in millimeters of mercury for blood pressure.
- **a. For example BP is 120/80 at phlebostatic axis. If you're measuring BP 20cm below that point, what is BP:**

188. Henry's, Boyle's, Charles's laws:
- a. **Henry's**: at constant temp, amt gas dissolved in solution is directly proportional to partial pressure of that gas
  - i. Temp affects solubility (↑temp = ↓solubility; ↓temp = ↑solubility)
  - ii. DO2 = CO X [(1.34 X Hgb X SpO2) + (PaO2 + 0.003)} X 10
- b. **Boyle's**: at constant temp, pressure is inversely proportional to volume
  - i. V1P1 = V2P2
  - ii. Ex: squeeze ambu bag, diaphragm contraction ↑Vt
- c. **Charles's**: at constant pressure, volume is directly proportional to temp
  - i. V/1T1 = V2/T2
  - ii. Ex: LMA cuff expands in autoclave

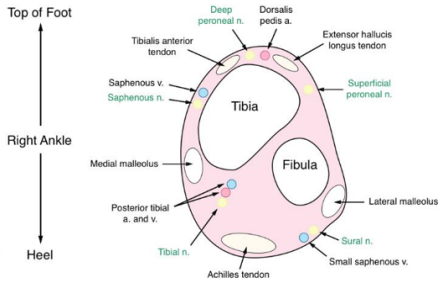
189. Law of Laplace:
- a. relationship b/w wall tension, internal pressure, & radius
- b. T = P X r

190. Poisuielle's Law:
- a. adaptation of Ohm's law that incorporates vessel diameter, viscosity, & tube length
- b. $Q = (\pi r^4 \Delta P)/8\eta l$
- c. Q = blood flow; r = radius; ΔP = pressure gradient; η = viscosity; l = length
- d. Blood viscosity: determined by Hct & body temp; inversely proportional to temp
- e. Doubling r ↑flow by 16-fold & tripling r ↑flow by 81-fold
- f. Laminar flow: travel in parallel path thru tube
- g. Turbulent flow: travel in nonlinear path & will create eddies; energy is lost via heat & vibration

191. Ohm's Law:
- a. states that current passing thru conductor = directly proportional to voltage & inversely proportional to resistance; law for BP & *SVR

192. Fick's Law of Diffusion:
- a. describes transfer rate of gas thru tissue; explains 2nd gas law
- b. Directly proportional to: partial pressure, diffusion coefficient, membrane surface area
- c. Indirectly proportional to: membrane thickness & molecular weight

193. Ankle block:
- a. **Know the five nerves, their location in the leg & where the needle goes to block them;** know nerves & ligaments/tendons
  - i. Deep peroneal, Superficial peroneal, Sural, Tibial, Saphenous (*3 S's are purely sensory)
    1. Deep peroneal: sensory innervation to lateral side of hallux & medial side of 2nd digit
    2. Superficial peroneal: innervation to dorsum of foot
    3. Sural: sensory innervation to posterior portion of heel & lateral sole of foot as well as part of Achilles tendon above ankle
    4. Posterior Tibial: direct continuation of the tibial nerve & branches into calcaneal, lateral plantar, & medial plantar nerves. It provides sensory innervation to the heel, medial sole, & part of the lateral sole of the foot, as well as the tips of toes.
    5. Saphenous: terminal branch of femoral nerve & courses SQ along the medial aspect of the knee. Pressure on medial aspect of knee = saphenous nerve injury; follows saphenous vein distally along anterior aspect of medial malleolus; sensory to medial aspect of lower extremity below knee
- b. **Which is best for bunionectomy?**: Sural, Saphenous, Obturator, or Deep Peroneal???
- c. **Originate from**: femoral or sciatic
- d. **TIPPED**: **T**ibial **I**nversion, **P**lantar flexion, **P**eroneal **E**version, **D**orsiflexion
- e. **Sensation of foot**:
  - i. Plantar aspect of foot = posterior tibial
  - ii. Dorsum = superficial peroneal
  - iii. Posterior sole/heel = sural
  - iv. Lateral hallux/median/2nd toe = deep peroneal
  - v. Medial malleolus = saphenous





194. Popliteal block:
- a. landmarks = popliteal crease, medial border of femoris biceps muscle laterally, & tendon of the semitendinous muscle medially

195. Sciatic Nerve:
- a. originates from lumbosacral trunk & is composed of nerve roots L4-5 & S1-3; Largest nerve trunk of the body; supplies muscles of back of thigh, skin of leg, & muscles of lower leg & foot

196. 3-in-1 Block:
- a. anesthetizes femoral, lateral femoral cutaneous, & obturator nerves w/ single injection; AKA inguinal perivascular technique

197. Know the nerves of the wrist & where to block them:
- a. **Radial block @ elbow**:
- b. terminal branch of posterior cord; biceps tendon is identified in the AC fossa; insert needle lateral to tendon & direct toward lateral epicondyle until wrist or finger extension is elicited; US- radial n. identified proximal to AC fossa b/w biceps & brachioradialis muscles
- c. **Median Nerve Block @ elbow:**
  - i. derived from lateral & medial cords; Forearm & hand w/ combo of this & ulnar; don't use in pts w/ carpal tunnel; Brachial artery identified in the AC crease just medial to biceps insertion; Needle is inserted just medial to the artery & directed toward the medial epicondyle until wrist flexion or thumb opposition is elcited. US- median nerve identified in cross-section just medial to the brachial artery, inject local around it
- d. **Median Nerve Block @ wrist**:
  - i. palmaris longus tendon identified by having patient flex the wrist against resistance; needle inserted just medial & deep to palmaris longus tendon b/w long palmar muscle & radial flexor muscle; 2 tendons: flexor carpi radialis & flexor palmaris longus (inject b/w)

198. Why do we use a high volume low pressure cuff on our ETT:
- a. ↓ pressure on trachea; easy to pass things around cuff (esophageal stethoscope, OGT, temp probe); protection against aspiration isn't as good as w/ low volume high pressure cuff

199. Low volume High pressure ETT cuff causes:
- a. ↑risk of ischemia

200. Where to put the strap on a patient & lateral decubitus position:
- a. between the iliac crest & the head of the femur rather

201. Which nerve goes along the back of the elbow & can be damaged:
- a. Ulnar

202. When can you give a premature baby O2 & not damage the retina:
- a. 44 weeks post conception

203. OLV/DLT (at least 15 questions)

- **a. Absolute indications**:
  - **i.** *Isolation of 1 lung to avoid contamination:* (infection/hemorrhage),
  - **ii.** *Control of distribution of ventilation:* (bronchopleural fistula/cyst/bulla/life-threatening hypoxia d/t lung disease),
  - **iii.** *Unilateral bronchopulmonary lavage:* (alveolar proteinosis)
- **b. Relative indications**:
  - **i.** *Surgical exposure*
    1. **High priority**: TAA, pneumonectomy, thoracoscopy, upper lobectomy, mediastinal exposure
    2. **Lower priority**: middle/lower lobectomy, esophageal resection, thoracic spine surgery
  - ***ii.*** *Pulm edema s/p CABG*
  - ***iii.*** *Severe hypotension r/t lung disease*
- **c. Contraindications:**
  - **i.** 5-10 cm PEEP to dependent; CPAP nondependent - single most effective means to ↑PaO2 w/ OLV
- **d.** What do you do after you go through the cords w/ a left DLT?
  - **i.** rotate 90°
- **e.** What happens when you insert L DLT too distal?
  - **i.** Obstruct upper lobe, lower lobe, mainstem, trachea?
- **f. OLV measures to ↑ventilation**

BOX 27-9

Management of One-Lung Anesthesia

- Ventilate:
  - Tidal volume: 6-8 mL/kg
  - Rate: 12-15 (permissive hypercapnia okay)
  - Fio_2: 0.4-0.8; maintain Spo_2 > 90%
  - PEEP: 5-10 cm H_2O (2.5-5 if COPD)
  - I:E ratio: 1:2 (1:3 if COPD or PEEPi)
- Consider alveolar recruitment maneuver prior to OLV
- Assess ABG 15 minutes after OLV initiated
- Volatile anesthetics < 1-1.5 MAC or IV agents
- Respond to hypoxemia:
  - Confirm tube position with fiberscope
  - Remedy detrimental effects of anemia or vasodilators
- Perform alveolar recruitment maneuver to DL
- Titrate PEEP in NDL
- Perform gentle recruitment, then CPAP 5-10 cm H_2O to NDL
- Increase Fio_2
- Intermittent or continuous two-lung ventilation
- Low-or no-pressure oxygen insufflation to NDL or selected lobe of NDL
- Reposition to lateral decubitus position if supine
- Alter perfusion with almitrine to NDL; nitric oxide to DL

TABLE 37-4. CLINICAL APPROACH TO ONE-LUNG VENTILATION (OLV) MANAGEMENT

1. Use Fio_2 of 1
2. Ventilate with a V_T of 6–8 mL/kg with PEEP 5 cm H_2O
3. Respiratory rate to maintain $Paco_2$ between 35 and 40 mm Hg
4. Check the DLT/endobronchial blocker position subsequent to the lateral decubitus positioning
5. If peak airway pressure exceeds 40 cm H_2O during OLV, DLT/endobronchial blocker malposition should be excluded
6. For hypoxemia, apply CPAP 10 cm H_2O to the nondependent lung (not during VAT)
7. If additional correction of hypoxemia is necessary add PEEP 5–10 cm H_2O to the ventilated lung
8. Frequent recruiting maneuvers
9. Avoid fluid overload
10. TIVA may be preferable to inhalation anesthetics
11. If necessary, intermittently inflate and deflate the operated lung

- **g.** Also ligation of pulm artery; Neo or almitrine might augment HPV (Barash)
- **h.** PEEP to dependent, CPAP to nondependent (Nagelhout)
- **i. If using L DLT & clamp bronchial & still have sounds on L, what is wrong?**
  - **i.** Still above the carina
- **j. HPV: which will negatively impact HPV? (select 2)**: MAC 1.5 or > & don't remember
  - **i.** ↓**HPV**: VAs >1.5 MAC, vasodilators, PDE inhibitors, dobutamine, CCBs, excess PEEP/Vt; *IV anesthetics don't affect HPV
- **k. OLV vent settings**:
  - **i. Vt on L**: 6 ml/kg
  - **ii. Vt on R**: 8 ml/kg
  - **iii. Plateau Inspiratory Pressure**: <25 cm H2O
  - **iv. permissive hypercapnia**: (supports HPV & also directly reduces cytokine response); Maintain PaO2 below 60-70 to reduce incidence of dysrhythmias, hypotension, & pulm HTN
  - **v. FiO2**: lowest level to maintain adequate SpO2
- **l. DLT complications**: Malpositioning of tube (MOST COMMON), Hypoxemia d/t poor position, Traumatic laryngitis, Tracheobronchial rupture, Inadvertent suturing of tube to bronchus, Rupture of thoracic AAA w/ L DLT, Damage to vocal cords, Barotrauma if inserted too deep, Hoarseness
- **m. DLT sizing by age:**
  - **i.** Formula:18+age
  - **ii.** 13 yo female: 32

204. Which nerve causes laryngospasm?

- **a.** External SLN stimulation causes closure of cricothyroid

205. Posterior cricoarytenoid =

a. ABduct ("Please Come Apart")

206. Lateral cricoarytenoid =

a. ADDuct ("Let's Close")

207. Cricothyroid =

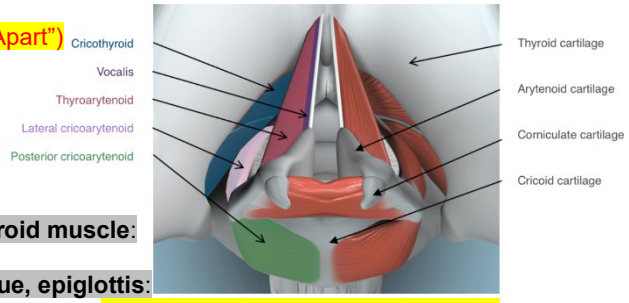
a. cords tense/elongate

208. Thyroarytenoid =

a. they relax/shorten

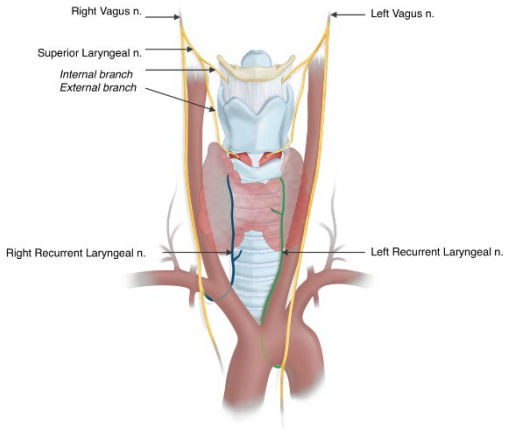
209. Motor function to cricothyroid muscle:

a. external SLN



210. Nerve for arytenoids, tongue, epiglottis:

a. **SLN** internal: sensory; innervates underside of epiglottis up to the cords
b. **SLN** external: motor; innervates cricothyroid muscle
c. **RLN**: *sensory* below cords to trachea; *motor* to **ALL** muscles **EXCEPT** cricothyroid
d. Anterior tongue: trigeminal (mandibular branch – V3)
e. Posterior tongue, soft palate, oropharynx, vallecula, anterior epiglottis: glossopharyngeal
f. Posterior epiglottis: SLN internal
g. Vocal cords: SLN internal & RLN
h. Trachea: RLN



211. Ways to ↓likelihood of laryngospasm:

a. iGel, LMA, King/Combi, ETI

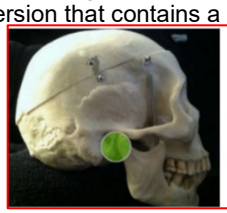
212. Supraglottic Devices: Combitube vs King

a. **Combi**: double lumen tube; blindly placed in hypopharynx; useful prehospital & in difficult airway management; Size 37 (4'-6' tall); Size 41 (>6' tall); no options if shorter than 4'; Oropharyngeal cuff (proximal) inflated first to occlude hypopharynx; Distal cuff occludes esophagus; 2 ventilation ports: Since tip usually enters esophagus, attempt ventilation thru blue (proximal) lumen; If tip enters trachea, ventilated thru clear (distal) lumen; cuff pressures <60 cm H2O (overinflation can rupture esophagus)
   i. **Benefits**: aspiration protection, can decompress stomach, useful for obese, blind insertion technique, doesn't require neck extension, allows high ventilatory pressure (up to 50 cm H2O), doesn't need to be taped to patient
   ii. **CONTRAINDICATED**: Intact gag reflex, prolonged use (>2-3 hrs) d/t risk of ischemia from oropharyngeal balloon, esophageal disease (Zenker's diverticulum- pouches form in pharyngeal mucosa), ingestion of caustic substances

b. **King LT** → (also known as the laryngeal tube airway); is a minimally invasive airway device similar to the Combi.
   i. This device is a single lumen that has a ventilation outlet b/w the low-pressure oropharyngeal & esophageal cuffs.
   ii. The device is inserted in a manner similar to the Combi but has only 1 ventilation port.
   iii. The King LT is a reusable supraglottic airway device created as an alternative to tracheal intubation or mask ventilation.
   iv. This airway is designed for PPV and for spontaneously breathing pt, thereby allowing max versatility as an airway management tool.
   v. The King LT has the capacity to achieve a ventilatory seal of 30+ cm H2O. It is easy to insert, & a randomized comparison of the laryngeal tube has shown it to be as effective as the classic LMA during anesthesia w/ controlled ventilation.
   vi. There is evidence that King LT may be useful in situations where ventilation & intubation are difficult or have failed altogether.
   vii. However, the research is limited & further inquiries into its usefulness have been suggested.
   viii. The King LT is 100% latex free & can be autoclaved up to 50 cycles, or is available in a disposable model.
   ix. The LT model has a single opening for ventilation.

c. King LTS-D → disposable version that contains a 2nd port that accommodates 18F NG suction tube for insertion into the stomach.
   i. 1 ventilation port

213. Larson's maneuver (hotspot):



Borders
Posterior: Mastoid Process
Anterior: Ramus of mandible
Superior: Skull base

214. Best way to obtain secured airway w/ laryngeal infection:

a. not trach/cric

215. Oxyhemoglobin dissociation curve - something about hyperthermia causes what other changes: right shift

a. **Left shift**: ↓PCO2, ↓H, ↓temp, ↑pH, ↑CO poisoning, HgbF, metHgb, smoking
b. **Right shift**: ↑PCO2, ↑H, ↑temp, ↑2,3 DPG, ↓pH, sickle cell

216. Haldane effect:

a. how body carries CO2; when blood is mostly deoxygenated, curve shifts to LEFT, ↑O2 carrying capacity

217. Bohr effect:

a. how body carries O2

218. What muscles assist w/ inspiration?

a. **Primary** → diaphragm; Intercostals contribute
b. **During forceful inspiration** → SCM & scalene muscles contract

219. CONTRAINDICATED & indications for FOB & light wand:

a. **FOB:**
   i. ***Indications***: difficult airway (awake spont. vent. is gold standard), C-spine instability/limitations, Limited mouth opening
   ii. ***Contraindications***: No absolutes, Hypoxia (lack of time), Excessive secretions, Hemorrhage, Uncooperative patient, LA Ax

b. **Light Wand:**
   i. ***Indications*:** Small mouth opening, Min. neck manipulation, Can ventilate but not intubtate
   ii. ***Contraindications***: Can't intubate can't ventilate, Upper airway anomaly, Short thick neck, Redundant soft tissue

220. What does a beta-2 antagonist do to insulin production/glucose levels?

a. ↑blood glucose levels & insulin levels (which drives K+ into cells causing transient ↓K+)

221. Beta2 agonist:

a. albuterol (short acting), metaproterenol, salmeterol; ↑cAMP → ↓Ca2+

222. Tx of intraop Bronchospasm

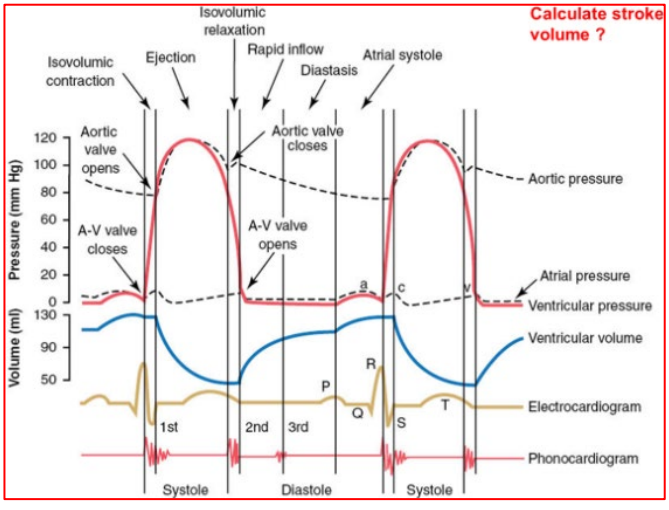
a. short acting B2 agonist (albuterol), inhaled ipratropium, epi 1 mcg/kg IV, hydrocortisone 2-4 mg/kg IV (doesn't Tx acute s/s, but prevents potential problems later), aminophylline (theophylline doesn't work well for acute bronchospasm), heliox

223. Arterial waveform - steep upstroke, widened pulse pressure, low diastolic

a. Aortic regurg

224. The pic w/ all the waveforms on it - identify a specific area

a. Systole = contraction
b. Diastole = relaxation



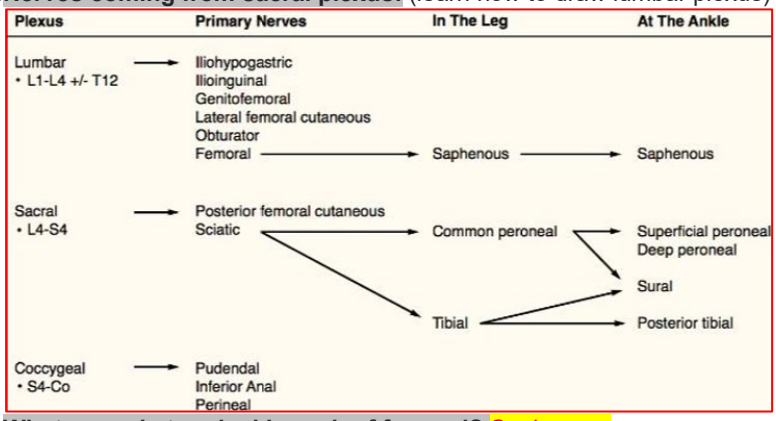
225. **What to do w/ hemodynamics during renal transplant Surgery?**
    a. PaO2 > 100, Normocapnia, Hct 30+%, SBP 100+, Maintenance of UO >100 ml/hr, CVP > 10, MAP > 60, BG < 200
226. **PA cath lung zone:**
    a. Zone III
227. **Cardiac lead questions (matching & various questions over them) - what leads monitor what part of the heart**:
    a. **Inferior, Posterior**: II, III, aVF = ***RCA***
    b. **Lateral**: I, aVL, V5, V6 = ***Left Circumflex***
    c. **Anterior, Septal**: V1-V2 (septum) V3-V4 (anterior) = ***LAD***
228. **Pediatric surgeries for heart anomalies – match:**
    a. **Ross procedure** → replace your aortic w/ pulmonic
    b. **Ebstein's** → Glenn's shunt, Fontan
    c. **Tricuspid Atresia** → Blalock-Taussig shunt, Glenn's shunt, Fontan
    d. **Transposition** → Mustard's operation, Senning's operation, Rash Kind's procedure
    e. **HLHS** → Norwood
229. **Cerebral steal question**: AKA **luxury perfusion**
    a. Vasodilators/hypoventilation allow vessels in **non-ischemic** area to dilate & ↑blood flow to areas → ↓ **blood flow to ischemic area**
230. **Hyperventilation to ↓ICP:** only works short term (6-20hr); causes constriction of cerebral vessels which ↑CVR, which ↓ CBF, which ↓ ICP
    a. PaCO2 30-35; lowering PaCO2 <30 ↑risk of cerebral ischemia d/t vasoconstriction & L-shift → which ↓s O2 offloading
231. **CSF: where made, pathway, & reabsorbed**:
    a. **Produced**: by ependymal cells in the choroid plexus
    b. **Pathway**: Lateral, Monroe, 3rd, Sylvius, 4th, Luschka, Magendie, Subarachnoid Space, Arachnoid Villi, Superior Sagittal Sinus
    c. **Reabsorbed**: by arachnoid vili in superior sagittal sinus
232. **Factors governing passage across BBB (multiple questions)**: small size, nonpolar, lipophilic, nonionized (freely enter = CO2, O2, H2O, lipid soluble *IV anesthetics) (poorly penetrates = ions, proteins, large molecules *Mannitol)
    a. BBB has tight junctions; separates CSF from plasma; Is immature in neonates
233. **Vasospasm**: peak is 4-9 days after **SAH**, resolves in 2-3 weeks; **Triple H Therapy**: Hemodilution, Hypervolemia, HTN; Nimodipine (CCB)
234. **Goal for post-op care of a subdural hemorrhage**: prevent vasospasm
235. **Match the blood flow from aorta to circle of Willis**: Aorta → Carotid a. → Internal carotid a. → Circle of Willis
236. **SSEPs**:
    a. **What meds interfere w/ SSEP the least?** Ketamine, Etomidate, Opiates, Barbs, Propofol (no Δ latency or amplitude)
    b. **Things that could interfere w/ SSEP?** **HYPO**thermia (↑***latency***), **HYPER**thermia (↓ ***amplitude***)
    c. **When we use/type of Surgery?** posterior cord, posterior spinal arteries,
        i. **SOMATOSENSORY**,
        ii. **SAFE to use NMBs**.
        iii. Monitors integrity of dorsal column (DCML)
237. **MEPs**
    a. **When we use/type of Surgery**: anterior cord, anterior spinal artery,
        i. **MOTOR**
        ii. Do **NOT** use NMBs!
        iii. Monitors integrity of corticospinal tract
238. **A delta fibers:** Rexed's lamina I & V, NT: glutamate
239. **C fibers**: rexed lamina II & III, NT: substance P
240. **What is first line tx for neuropathy?** Antidepressant (Gabapentin, Amitriptyline)
241. **What do you tx intra-op w/ for a chronic pain patient:** Ketamine, Methadone
242. **Hyperalgesia - what do you use intra-op:** Ketamine, MgSO4
243. **Ionized vs non-ionized questions - maternal/fetal passage:** nonionized goes to fetus
244. **When pKa & pH are the same:** 50% of drug ionized & 50% unionized
245. **Weak acid is paired w/ a positive ion such as Na+, Ca2+, or Mg**: Ex = Na+ Thiopental
246. **Weak base is paired w/ a negative ion such as Cl- or SO4**: Ex = Lidocaine HCl
247. **Protein binding questions - lots of them dealing w/ LAs & maternity**
    a. **Higher protein binding** = longer DOA
        i. Levobupivicaine = 98%
        ii. Bupivicaine = 96%
        iii. Ropivicaine = 94%
        iv. *Chloroprocaine = 0%
    b. **Potency**: Chloroprocaine > Mepiv = Lido > Tetra > Bupivacaine > Ropi > Etidocaine
248. **Bier block question on tourniquet time:** 20 min
249. **Something about a special laser for urology**: →

TABLE 47.1

Common Surgical Lasers

Laser	Wavelength	Applications
Far-Infrared		
CO_2	10,600 nm	Multiple uses: general surgery, orthopedics, gynecology, urology, otolaryngology, plastic surgery
Mid-Infrared		
Nd:YAG	1064 nm	Multiple uses: gastroenterology, pulmonology, urology, ophthalmology, dermatology
Ho:YAG	2070 nm	Orthopedics, urology
Er:YAG	2940 nm	Dermatology
Near-Infrared		
Diode	800-900 nm	Multiples uses: ophthalmology, otolaryngology, periodontics, cosmetic surgery, pain management. Multiple nonmedical applications
Visible		
Argon	488 and 514 nm	Multiple uses: ophthalmology, plastic surgery, dermatology, gynecology, otolaryngology
Krypton	476, 521, 568 nm	Dermatology
Copper bromide	511 and 577 nm	Dermatology, photosynthesizer
KTP	532 nm	Dermatology
Pulsed dye	577–585 nm	Dermatology
Gold	578–628 nm	Oncology
Ruby	694 nm	Dermatology
Alexandrite	755 nm	Dermatology
Ultraviolet (Excimer)		
Argon-fluoride	193 nm	Multiple uses: ophthalmology, dermatology
Krypton-fluoride	249 nm	Dermatology
Xenon-chloride	308 nm	Multiple uses: dermatology, ophthalmology, dermatology
Xenon-fluoride	351 nm	Multiple uses: angioplasty, ophthalmology, dermatology

250. Nerves coming from sacral plexus: (learn how to draw lumbar plexus)



251. What nerve is terminal branch of femoral? Saphenous

252. What nerve is combo of lumbar plexus & sacral plexus? Sural

253. Lumbar plexus block (Psoas compartment): *lateral* to vertebral column; risk of spinal/epidural injection

254. Lumbar plexus block complications: Epidural or spinal injection, Sympathectomy, hematoma, Renal capsular injection, IV injection

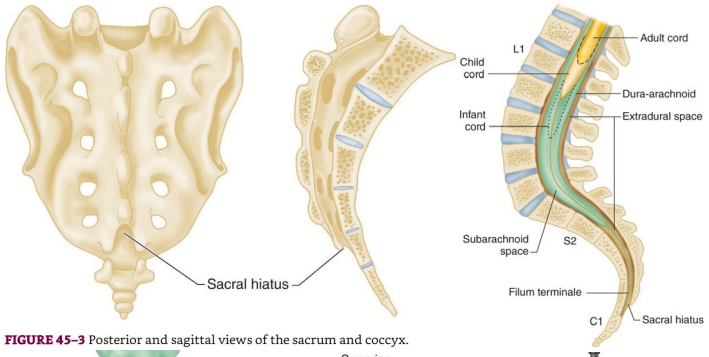
255. Nerve injuries - matching for leg positioning & masking:

a. **Crossing legs**: top leg = sural
b. **Bottom** leg = superficial peroneal
c. **Mask** = facial
d. **T-burg w/ shoulder brace & wrist suspension** = brachial plexus upper C5 & C6
e. **Abduction of arm w/ hyperextension, external rotation** (magnified w/ contralateral head rotation) = lower C8 & T1

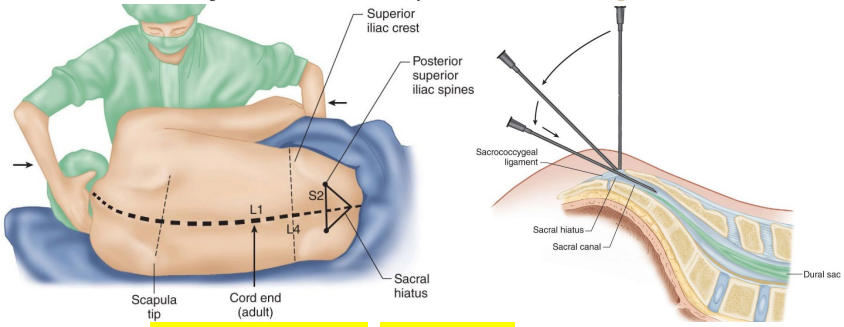
256. Nerve injury d/t fracture table: Pudendal

257. Caudal Block:

a. **Where do you inject LA? (hot spot)**:



i. FIGURE 45–3 Posterior and sagittal views of the sacrum and coccyx.



Block Height	Pediatric Dose	Adult Dose
Sacral	0.5 mL/kg	12 – 15 mL
Sacral to low thoracic (~ T10)	1 mL/kg	20 – 30 mL
Sacral to mid thoracic	1.25 mL/kg *Miller says to avoid this dose range	n.a.

Infrequently used in adolescence & adulthood:

- Sacral anatomy is more difficult to ID.
- A lumbar approach to epidural space is easier to perform and equally effective

Contraindications

Absolute

- Spina bifida
- Meningomyelocele
- Meningitis

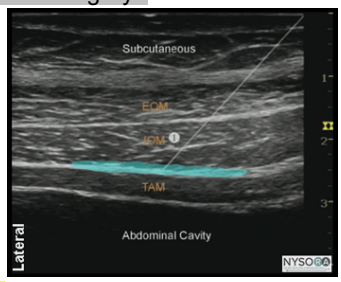
Relative

- Pilonidal cyst
- Abnormal superficial landmarks
- Hydrocephalus
- Intracranial tumor
- Progressive degenerative neuropathy

b. **Landmarks?** superior iliac spines / sacral hiatus
c. **Used to block what?** sacral, lumbar, & lower thoracic dermatomes (**can get up to T10 sensory block**)
d. How far to insert caudal epidural catheter in 5 yo patient for upper abd Surgery?

258. TAP block:

a. **Level of block: T9-L1**
b. **Target of injection is between which two planes? IO & TA**
c. **Landmarks**= EO, latissimus dorsi, & iliac crest
d. Identify the transverse abdominus muscle





259. Progesterone in CSF of parturient: may ↑spread of LA & ↑sensitivity to LAs

260. Patient is used cocaine the night before his ENT Surgery. What do you do? (pick 2):

a. Delay Elective Surgery: Avoid beta blockers (accentuate coronary artery vasospasm)
b. Probs w/ cocaine use: Tachycardia, Dysrhythmias, Myocardial ischemia, Acute use = ↑MAC, Chronic use = ↓MAC & thrombocytopenia, HTN, Beta blockers can cause HF if SVR is ↑, Hypotension doesn't respond to ephedrine, Pulm edema
c. Ester LA; inhibits NE reuptake in presynaptic SNS flooding synaptic cleft w/ NE which ↑SNS tone

261. **Vapor pressures of VAs? (matching)**:
    a. **Sevo**: **157**
    b. **Iso**: **238**
    c. **Des**: **669**
    d. **N2O**: **38,770**
262. **Bld:gas coefficient of VAs (matching)**:
    a. **Des** = **0.42**
    b. **N2O** = **0.46**
    c. **Sevo** = **0.65**
    d. **Iso** = **1.46**
263. **Concentration effect**: The lower the blood:gas solubility, the faster the rise in the lung & brain concentrations.
    a. However, N2O exhibits a slightly faster rate of rise compared w/ Des, despite a higher blood:gas coefficient.
    b. This is a result of the concentration effect. N2O is given at much higher concentrations (50-70%) than Des (3-9%).
264. **What VA has the highest VP at the lowest temp**: N2O > Des > Iso > Sevo?
265. **What % ↓in MAC for parturient**: 30-40% (returns to normal 3-4 days after delivery)
266. **Highest MAC is what age?** 6-12 months (peak @ **3 months**)
267. **Things that ↓or ↑MAC**:
    a. **Decrease**: N20, Opioids, IV anesthetics, Alpha2 agonist, Acute ETOH, Lithium, Lidocaine, Verapamil, Hydroxyzine, THC, HypoNa, Hypotension, PaO2 < 38, Anemia, CPB, Metabolic acidosis, Hypoosmolality, Pregnancy, Age (6% ↓per decade after 40)
    b. **Increase**: Chronic ETOH, Acute amphetamine/cocaine, MAOIs, Ephedrine, Levodopa, Cyclosporine, HyperNa, Hyperthermia, Red hair
268. **Acceptable level of volatile agent in OR w/o N2O: 2ppm**
    a. **VA alone:** ≤2 ppm
    b. **N2O alone:** ≤25 ppm
    c. **VA + N2O:** ≤0.5 ppm + ≤25 ppm
269. **When is the best time to check soda lime**: at the end of every case?
270. **Phase I & II biotransformation**:
    a. **Phase 1**: can be excreted w/o undergoing phase 2.
        i. **Oxidation:** adds O2
        ii. **Reduction:** adds electron
        iii. **Hydrolysis:** adds H2O to a compound to split it apart (usually an ester)
    b. **Phase 2**: involves conjugation reactions in which a drug/metabolite is conjugated w/ an endogenous substrate such as glucuronic, sulfonic, or acetic acid (adds on an endogenous, highly polar, water soluble substrate to the molecule).
    c. Can occur w/o phase 1. *Conjugation of variety of endogenous compounds to polar chemical groups of the drug
271. **elimination/circulation calculation**: ??
272. **Doses (matching) for propofol, ketamine, remifentanil, versed**:

	Induction	Maintenance	Other
Propofol	2-2.5 mg/kg	100-200 mcg/kg/min	25-75 mcg/kg/min (Conscious sedation)
Ketamine	1-2 mg/kg	15-45 mcg/kg/min (or 1-3 mg/min)	0.1-0.5 mg/kg (Analgesia)
Versed	0.1-0.2 mg/kg	--	0.01-0.1 mg/kg (Sedation)
Remi	0.5-1 mcg/kg	0.05-2 mcg/kg/min	
Fentanyl	5-10 mcg/kg	0.01-0.05 mcg/kg/min	

273. **What narcotics/benzos are best for outpt Surgery**:
    a. Ideal anesthetic agents should have appropriate pharmacokinetic traits (rapid onset & offset, short duration, inert metabolites, & insignificant side effects); Roc, Des, Sevo, Gabapentin, Tylenol, Propofol, NSAIDS, Precedex, Versed, Fentanyl, Alfentanil
274. **DOA benzos**:
    a. *Short* = versed; *Intermediate* = lorazepam; *Long* = diazepam
275. **Agonist-antagonist questions**:
    a. Butorphanol for labor pain: works on kappa receptors, weak @ mu, min @ delta; SE: sedation, nausea, diaphoresis, & resp. depression; used intranasally for migraines & postop pain; effective in Tx postop shivering but MOA unknown; ceiling effect (incrementally higher doses do not result in increasing resp depression); Dose for labor is 1-2 mg IV or IM. It is 5 times as potent as morphine & has a half-life of 3 hrs. Unlike morphine, it ↑PA pressure & myocardial work. Has no active metabolites.
    b. Nalbuphine: can reverse resp depression d/t opioid use & still maintain analgesia; analgesic response is = to morphine; has ceiling effect; used to antagonize pruritis induced by epidural & intrathecal morphine; antagonizes fentanyl-induced resp depression; doesn't produce adverse circulatory changes; Dose in labor is 5-10 mg IV, IM, or SQ & is equivalent to morphine 10 mg. It results in a greater reduction in FHR variability compared w/ meperidine.
    c. Butorphanol & nalbuphine are opioid agonist-antagonists may result in less N/V than pure opioids. Agonist-antagonists have typically been used in 1st-stage latent-phase labor to provide sedation & a period of rest. Because these compounds possess significant antagonist properties, their use in opioid-dependent pts should be avoided.
        i. Receptors: Mu: antagonist; Kappa: partial agonist; Delta: agonist
    d. Buprenorphine: binds mainly to Mu; DOA = 8 hrs; clinically significant resp depression can occur w/ therapeutic doses; exhibits a ceiling effect; minimal effect on GI motility & smooth muscle sphincter tone; can give transdermal for CA pts (provides analgesia w/ low incidence adverse events)
276. **Remi**: ester structure is susceptible to hydrolysis by nonspecific esterase in the blood → rapid elimination
277. **Why is succ a bad idea after a thermal burn**:
    a. ↑# of cholinergic nicotinic receptors; succ can cause K+ release from the muscle membrane → ↑K+ & possible cardiac arrest
278. **Antisialagogue effects**:
    a. Scopolamine: +++
    b. Glyco: ++
    c. Atropine: +

279. Class of antiarrhythmics – matching:
- a. **Class 1**: Na+ channel blockers; interfere w/ phase 0 depolarization of cardiac action potential **(-ine, -ide, -oin)**
  - i. 1a: quinid**ine**, procainam**ide**, disopyram**ide**
  - ii. 1b: lidoca**ine**, phenyt**oin**
  - iii. 1c: flecain**ide**, propafenone
- b. **Class 2**: Beta blockers **(-olol)**
- c. **Class 3**: K+ channel blockers (amiodarone, bretylium)
- d. **Class 4**: Ca2+ channel blockers (verapamil, diltiazem)

280. Dig tox in the absence of renal failure: non-potassium sparing diuretics

281. Transfusion reaction w/ the most common ↑of mortality:
- a. TRALI > hemolytic transfusion reaction > transfusion associated sepsis > **TACO** > babesiosis > anaphylaxis > GvH disease

282. Allowable blood loss calculation:
- **a. EBV** →
- b. **MABL** = EBV X (start Hct - target Hct)/start Hct *can interchange Hct w/ Hgb

	EBV mg/kg
Premature Neonate	95
Term Neonate	85
Infant	80
> 1 Year	75
Adult Male	75
Adult Female	65
Obese	50

283. Drugs dosing for obese pts:
- a. **LBW**: thiopental, propofol (induction), roc, vecuronium, atracurium, versed, cisatracurium, fentanyl (maintenance), Sufentanil (maintenance), remifentanil.
- b. **TBW**: succ, precedex, fentanyl (loading), Sufentanil (loading), propofol (maintenance), versed
- c. **IBW**: roc, vecuronium, cisatracurium, remifentanil

Drug	Dose	Recommendation	
Propofol	Induction Maintenance	LBW TBW	• The loading dose of propofol is based on LBW • Because the offset is caused by redistribution & **NOT** clearance (which would depend on Vd). • Pharmacokinetic modeling has determined that the maintenance dose is best based on TBW.
Succinylcholine	Intubation	TBW	• Although succs is **water-soluble**, the dose for intubation is calculated with TBW. • **This is a clear exception to the rule for water soluble drugs**** • ↑ BV/Vd + ↑ pseudocholinesterase activity/clearance → necessitates TBW dose
Rocuronium Vecuronium	Intubation Maintenance	LBW LBW	• Rocuronium and vecuronium are distributed throughout body water and are dosed on LBW
Cisatracurium Atracurium	Intubation Maintenance	TBW TBW vs LBW	• Loading doses of cisatracurium and atracurium should probably be dosed on TBW.
Fentanyl Sufentanil	Loading Maintenance	TBW LBW	• Because of their fat solubility & ↑ Vd → use TBW. • Maintenance dosing is based on LBW. • ↑ **Vd** correlates with a ↑ **T1/2**.
Remifentanil	Loading Maintenance	LBW LBW	• Rapidly cleared by plasma esterases • Does **NOT** behave like a high Vd drug
Midazolam	Loading Maintenance	TBW TBW	• Midazolam is administered by TBW because of an ↑ VD. • Dosing in this way will ↑ T1/2& DOA.
Epidural LA		75% of normal dose	• Engorgement of the epidural veins + ↑ epidural fat → ↑ spread of LA in the epidural space.
Oral Drugs		No Δ	• There is **no difference in absorption** of orally medications in the obese population.

284. Fluid deficit for a pediatric patient:
- a. Maintenance = 1st 10 kg = 4 ml/kg, 2nd 10 kg = 2 ml/kg, then 1 ml/kg for any over 20 kg;
- b. Deficit = NPO hrs X Maintenance
- c. **In peds, it is best calculated based off of**: BMI, BSA, Dry body weight, or Gross body weight???

285. Hespan is not used as much now because of?
- a. Produces a dilutional dysfunction of coagulation; directly inhibits clot formation by moving into fibrin clots

286. Most common cause of hypotension in PACU?: hypovolemia s/t inadequate replacement intraop

287. What causes the most heat loss in OR?: Radiation (60%) > Convection (15-30%) > Evaporation (20%) > Conduction (5%)

288. What hormones are produced in the neurohypophysis (posterior pituitary): ADH (vaso), oxytocin

289. What hormones are produced in the adenohypophysis (anterior pituitary): FSH, LH, ACTH, TSH, Prolactin, GH **(FLAT PiG)**

290. Causes of bone fragility in obese?
1. Causes OA from excess stress on weight-bearing joints
2. ↓ mobility → Bone reabsorption → lead to reduced bone density & contribute to stress fracture

291. Parathyroidectomy complications postop: Postop complications include RLN injury, bleeding, & transient/complete hypoparathyroidism.
- a. Unilateral RLN s/s: hoarseness (usually requires no intervention).
- b. Bilateral RLN injury: rare, producing aphonia & requiring immediate tracheal intubation.
- c. After successful parathyroidectomy, a ↓in the serum Ca2+ level should be observed within 24 hrs.

292. Cushing's: Glucocorticoid excess; S/S: muscle atrophy, thin atrophic skin, central obesity w/ thin extremities, glucose intolerance, plethoric facies, purplish striae, muscle weakness, easy bruising, OP, fluid retention, ↓K+ alkalosis, HTN, women (hirsutism, thin hair, acne, amenorrhea), men (gynecomastia, impotence); Causes: pituitary adenoma, ectopic secretion of ACTH by nonpituitary tumor, adrenal tumor/carcinoma, or MOST COMMON CAUSE = iatrogenic (pharmacologic admin of glucocorticoids for various autoimmune & inflammatory conditions; Diagnosis: measure plasma cortisol in AM after dose of dexamethasone (won't suppress CRH, ACTH, & plasma cortisol), elevated salivary cortisol, & urinary free cortisol; Tx: depends on cause, Tx of choice if microadenoma = transsphenoidal microadenomectomy; Anesthesia Management: normalize everything intraop, careful positioning, enforce aseptic technique, NMBs may have more exaggerated effect in pts w/ ↓K+, give stress dose steroid if adrenal resection

293. What is the only *irreversible* alpha blocker?: Phenoxybenzamine (long acting, nonselective, noncompetitive antagonist of alpha1 & alpha2 receptor; causes vasodilation by decreasing SVR; impairs NE regulating effect of presynaptic alpha2 receptor → reflex tachycardia; primary role = Tx pheo; 0.5-1 mg/kg PO; SE: orthostatic hypotension & nasal congestion)

294. What is a *reversible* alpha blocker?: Phentolamine

295. What is the precursor to norepi?:

a. **NE Synthesis**: Tyrosine (tyrosine hydroxylase – rate limiting step) → DOPA (DOPA decarboxylase) → Dopamine (dopamine b-hydroxylase) → NE (phenylethanolamine, N-methyltransferase) → Epi
b. **NE Termination**: Reuptake:80%; NE in synaptic cleft metabolized by MAO; NE in bloodstream is metabolized by COMT

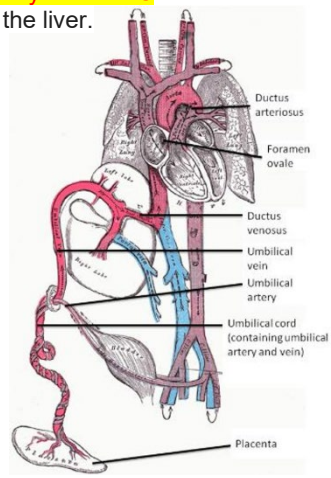
296. **SNS & PNS preganglionic both:** release Ach to a nicotinic receptor
297. **Macro- something that has to do w/ the head - post-op tx after transsphenoidal removal of tumor:**
298. **What gastric cells produce hydrochloric acid**: Gastric distension → G cells release Gastrin → parietal cells secrete hydrochloric acid
299. **Carcinoid Syndrome**:
    a. **Drugs to give**: somatostatin (octreotide or lanreotide; inhibits release of vasoactive substance from carcinoid tumors), antihistamines, 5HT3 antagonists, steroids, Neo, Vaso, propofol is best induction agent (suppresses SNS response to intubation best), vec or roc, fentanyl, N2O is safe, sevo or des
    b. **Drugs to avoid**: morphine, meperidine, atracurium, thiopental, succ (fasciculations can cause hormone release from tumor), sympathomimetics (ephedrine, ketamine), exogenous catecholamines (can potentiate hormone release)
    c. **Caused by**: secretion of vasoactive substance from enterochromaffin cells (histamine, kinins, kallikrein, serotonin)
    d. **S/S**: MOST COMMON = flushing & diarrhea; Least = palpitations, diaphoresis, HA
        i. **Histamine**: flushing (head & neck)
        ii. **Kinins & Kallikrein**: flushing (head & neck)
        iii. **Serotonin**: HTN, SVTs
    e. **Carcinoid Crisis**: life threatening; S/S: tachycardia, HTN, hypotension, intense flushing, abd pain, diarrhea
    f. Anesthetic Management

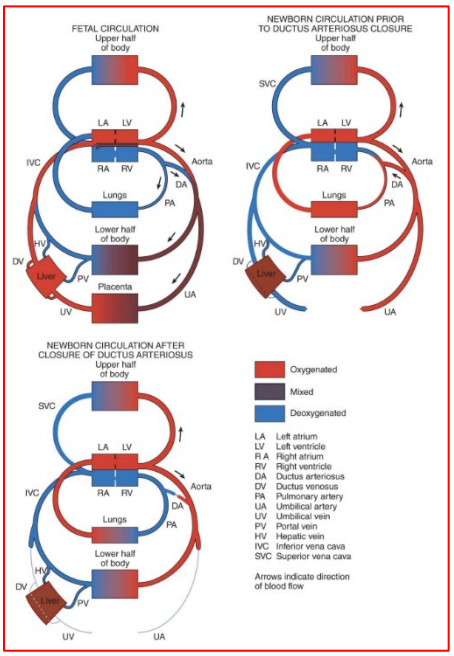
Anesthetic Considerations in Carcinoid Syndrome	
• The most common clinical signs are cutaneous flushing of the head, neck, and upper thorax, wheezing, blood pressure and heart rate changes, and diarrhea. • Preoperative assessment should include complete blood count, measurement of electrolytes, liver function tests, measurement of blood glucose, electrocardiogram, and determination of urine 5-HIAA levels. Echocardiography should be performed to determine the extent of carcinoid heart disease. • Confirmation: > 30 mg of 5-HIAA per 24-hour urine sample (normal = 3 - 15 mg/24 hr) • Optimize fluid and electrolyte status and pretreat with octreotide as noted. Continue octreotide throughout the postoperative period. Aprotinin (kallikrein inhibitor) may be used to treat hypotension refractory to octreotide. Interferon-a has shown success in controlling some symptoms. • Both histamine-1 and histamine-2 receptor blockers must be used to fully counteract histamine effects. • Perioperative blockade of serotonin receptors	• Avoid histamine-releasing agents such as morphine and atracurium. Avoid sympathomimetic agents such as ketamine and ephedrine. • Treat hypotension with an a-receptor agonist such as phenylephrine. • General anesthesia is preferred over regional anesthesia. Patients with high serotonin levels may exhibit prolonged recovery; therefore, desflurane and sevoflurane, which have rapid recovery profiles, may be beneficial. • Aggressively maintain normothermia to avoid catecholamineinduced vasoactive mediator release. • Monitor intraoperative plasma glucose, because these patients are prone to hyperglycemia. Treat with insulin as is customary. • Pay attention to procedures, treatments, and drugs that may stimulate release of vasoactive substances from tumor cells. These include tumor-debulking surgery or hepatic artery embolization to reduce tumor size and biotherapy (interferon for tumor shrinkage) or chemotherapy for systemic spread.

300. **Rheumatoid Arthritis - what do you have to be mindful of when starting a case?** I think it was asymptomatic axial issues
    a. Airway impaction
        i. TMJ: limited mouth opening
        ii. Cricoarytenoid joint: ↓glottic opening
        iii. C-spine: MOST COMMON complication is Atlantoaxial subluxation & separation of atlanto-odontoid articulation (d/t weakening of transverse axial ligament which allows odontoid to directly compress the SC @ level of foramen magnum, patient @ risk quad/para-plegia), limited extension
301. **Eaton-Lambert syndrome**: IgG mediated destruction of presynaptic voltage gated Ca2+ channel at presynaptic nerve terminal
    a. Proximal muscle weakness worse in AM & gets better throughout the day Myasthenic syndrome
    b. ANS dysfunction causes orthostatic hypotension, slowed gastric motility, & urinary retention
    c. Sensitive to succ & NDNMBs
    d. Reversal may be inadequate
    e. 60% have small cell (oat cell) carcinoma of lung

Fetal circulation/hemodynamics - SVR, PVR, shunt:

f. 1st breath → lung expansion → ↑PaO2 & ↓PaCO2 → ↓PVR
g. Placenta separates from uterine wall (or cord clamp) → ↑SVR → ductus venosus closure
h. ↓PVR + ↑SVR → LAP > RAP → flap valve of foramen ovale closes
i. ↓PVR → reversal of blood flow thru ductus arteriosus → exposes DA to ↑PaO2 → DA closure
j. ↓circulating PGE1 (released from the placenta) → DA closure
k. **Which shunt closes d/t ↓portal resistance?**: ductus venosus
l. **Anatomic closure of DV**: umbilical cord clamping (on picture)
   i. DV → shunts a portion of the left umbilical vein blood flow directly to the IVC.
   ii. Thus, it allows oxygenated blood from the placenta to bypass the liver.





302. Maternal resp changes:

a. **Increases**: PaO2 (104-108), P50 (right shift), Minute ventilation (50%), Vt (40%), RR (10%), Mallampati, O2 consumption (term- 20%, 1st stage labor- 40%, 2nd stage labor- 75%)
b. **Decreases**: PaCO2 (28-32), HCO3 (20), FRC (20%), ERV (20-25%), RV (15-20%)
c. **No change**: arterial pH, TLC, VC, CC
d. **Maternal hyperventilation**: left shift oxyHb curve = ↓ O2 delivery to fetus

303. Maternal CV Changes:

a. **Increases**: CO (40% by 3rd trimester), O2 consumption (20%), HR (15%), SV (30%)
b. **Decreases:** DBP (15%), SVR (15%), PVR (30%)
c. **No change**: CVP, PAOP, MAP, SBP
d. Left axis deviation d/t gravid uterus

304. Maternal Hematologic Changes:

a. **Increases**: intravascular fluid (35%), plasma volume (45%), erythrocyte (20%), fibrin breakdown, factors 1 (fibrinogen), 7, 8, 9, 10, 12
b. **Decreases:** anticoagulants (protein C & S), antifibrinolytic system (factor 11 & 13)
c. **No change**: platelets, factors 2 & 5, bleeding time

305. Matching placenta accrete, placenta abrupto, & previa w/ symptoms:

a. **Accrete**: abnormal implantation to/in/beyond myometrium
b. **Previa**: attaches to lower uterine segment either partially or completely covers cervical os; S/S: painless vaginal bleeding, potential for hemorrhage, PP bleeding ↑b/c lower uterus won't contract as well; Risk factors: prior C/S, H/o multiple births
c. **Abruption**: S/S: Painful vaginal bleeding, uterine tenderness, ↑uterine activity, atypical presentation; Risk factors: PIH, PreE, chronic HTN, cocaine use, smoking, ETOH

306. Placental transfer of meds:

a. **Do cross**: VAs, Opioids, Benzos, LAs, IV anesthetics, Atropine, Beta blockers, Mg
b. **Don't cross**: succ, insulin, glyco, NDNMBs, heparin ("SIGN Here")
c. **LAs that cross (most to least)**: Mepivacaine > Lidocaine > Ropivacaine > Bupivacaine (***Chloroprocaine doesn't cross**)

307. What is the most protein bound LA:

Amide-Type	pKa	Ionization % at 7.4	Protein Binding %
Bupivacaine	8.1	83	96
Levo-bupivacaine	8.1	83	98
Ropivacaine	8.1	83	94
Lidocaine	7.9	76	65
Prilocaine	7.9	76	55
Mepivacaine	7.6	61	78
Ester-Type	**pKa**	**Ionization % at 7.4**	**Protein Binding %**
Procaine	8.9	97	6
Chloroprocaine	8.7	95	0
Tetracaine	8.5	93	76

a. Low pKa = fast onset; High pKa = slow onset

308. PreE: SBP >140 or DBP > 90 after 20 wks gestation accompanied w/ proteinuria (>300 mg/dl);

a. Tx = bedrest, antihypertensive drugs, MgSO4 (4 grams over 10 min, then 1-3 g/hr), therapeutic Mg level = 4-6 meq/L
b. *Definitive tx for severe preE*: delivery
c. **Severe:** BP > 160/110, proteinuria excess of 5 g/dl, oliguria (<500 ml/day)

309. Uterine rupture w/ an epidural s/s: fetal distress is most definitive; Others: frank hemorrhage, loss of uterine tone or hypotension w/ occult bleeding into abdomen; Tx: volume resuscitation & immediate laparotomy

310. s/s given that describes HELLP or DIC & you have to know that the platelets drop:

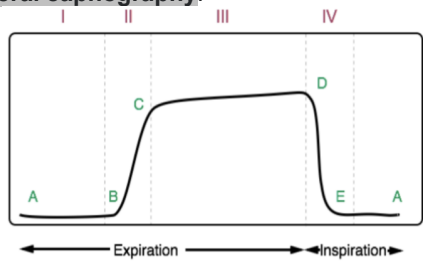
a. HELLP = **H**emolysis, **E**levated **L**iver **E**nzymes, **L**ow **P**latelets
b. S/S: epigastric pain, upper abd tenderness, proteinuria, HTN, jaundice, N/V; patient at risk for DIC; definitive Tx = delivery

311. **OB sympathomimetics**:
    a. Neo is equally efficacious to ephedrine for Tx maternal hypotension & results in less fetal acidosis.
    b. Ephedrine crosses the placenta to a greater extent than Neo & undergoes less early metabolism.
    c. Stimulation of fetal *β*-adrenergic receptors by ephedrine ↑ fetal metabolic rate; however, clinical significance of this effect is not known.
    d. Neo may be administered as a bolus to Tx hypotension (100-150 *μ*g) or continuous prophylactic infusion (starting rate, 25-50 *μ*g/min).
    e. Ephedrine can ↑both FHR & variability if given to parturient in large enough doses. Maternal HR should guide choice.
    f. Administration of Neo by infusion during C/S to maintain maternal SBP at baseline is associated w/ ↓ incidence of fetal acidosis
312. **Dopamine vs dobutamine – difference**:
    a. **Dopamine**: *naturally* occurring catecholamine; stimulates dopamine/beta/alpha receptors in dose dependent manner; indirect effect by eliciting release of NE via B1 stimulation; inhibits aldosterone; ↑CO, contractility, renal blood flow, GFR, Na+ excretion, & UO
    b. **Dobutamine**: *synthetic* sympathomimetic amine; primarily stimulates B1 w/ some B2 effects; strong inotropic response, minimal chronotropy; unlike dopamine, it doesn't ↑filling pressures & may be associated w/ less tachycardia; CO often ↑w/o significant BP change; ↓ SVR; good for cardiogenic shock
        i. Compared to dopamine, dobutamine: causes tachycardia (no), is mostly alpha (no), ?, ?

	DOPAMINE	**DOBUTAMINE**
Source	Natural catecholamine (Precursor of NE & also release NE from cardiac nerve terminals)	Synthetic catecholamine (Does not release NE)
Mechanism	Adrenergic + Dopaminergic Da 2, α, β	Pure adrenergic (β1 > β2 > α)
Effect on BP	↑, sustained	Increase/neutral/decrease
Effect on PO/VP	No favorable effect	Reduced/optimized/favorable
Effect on renal blood flow and diuresis	Improves at low dose by direct action	No direct action / May improve due to improved cardiac function
Arrhythmogenicity	More (Less than dobutamine)	More, (Generally safe)
Inotropic action	Less	More, Consistent,
Clinical utility*		
Shock	• Yes	• No
Cardiac failure	• No	• Yes
Ischemic LVF	• No	• Yes
Renal failure	• Yes	• No

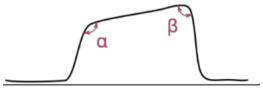
313. **Geriatric regional response**:
    a. Spinals: ↑sensory block; Longer DOA
    b. Epidurals: ↓segmental dose; Epidural volume tends to result in more extensive cephalad spread of anesthetics, but accompanied by a shorter duration of analgesia & motor block
314. **Geriatric volume of distribution changes & effects**
    a. **↑total body fat** = ↑Vd for lipophilic drugs (prolongs elimination); ↓Vd for hydrophilic drugs d/t ↓lean body mass (higher plasma concentration w/ dose)
315. **Geriatric arrythmias**: loss of SA node tissue; ↑likelihood of Afib, 1st degree AV block, 2nd degree block, & SSS
316. **Geriatric metabolism changes**: ↓phase I, no change in phase 2
317. **Geriatric response to catecholamines**: ↓response
318. **2 populations highest risk for ulnar injury**:
319. Male, Age >50, preexisting neuropathy, prolonged bedrest/hospitalization; extremes of body habitus (very thin or obese)
320. **What should you do post-op for a pedi patient having tonsillectomy for severe OSA?**
    a. Surgical edema may cause obstruction; Positioning & adequate oral suctioning can help prevent obstruction & possible laryngospasm
321. **Why should you not do Surgery on a preemie outpt center?**
    a. Because of potential abnormalities such as: anemia, underdeveloped gag, immature temp control → hypothermia → **postop apnea**
    b. Immature brainstem → **pathologic resp conditions**
322. **Nitroprusside - arterial or venous dilation**: **ARTERIAL=VENOUS**; reduces afterload & preload
    a. **NTG**: **VENOUS** > arterial
    b. **Hydralazine**: **ARTERIAL** > venous
323. **What do vasopressors do to CBF**: decreases?
324. **CBF**:
    a. **Increases**: resp. acidosis, PaO2 <50-60, (CMRO2 is ↑by: hyperthermia, seizures, ketamine & N2O); for q1mmHg ↑PaCO2 CBF will ↑1-2 ml/100g brain tissue, NTG, nitroprusside
    b. **Decreases**: resp. alkalosis, Precedex, (CMRO2 is ↓by: hypothermia, halogenated anesthetics, propofol, etomidate, barbs, & hyperthermia beyond 42°), hyperventilation, for q1mmHg ↓PaCO2 CBF will ↓1-2 ml/100g brain tissue, jugular compression s/t improper head positioning (head needs to be >30° to facilitate venous drainage from brain), ↑intrathoracic pressure s/t coughing/PEEP, vena cava thrombosis, vena cava syndrome
325. **Best scenario for awake intubation**: not an urgent situation; awake intubation takes time & a cooperative patient
326. **Which device can you ventilate w/ - fiberoptic, bougie, light wand, bronchial blocker**: FOB
327. **How many atmospheres for something**: 1 atm = 760 mm Hg = 760 torr = 1 bar = 100 kPa = 1033 cm H2O = 14.7 lb/in^2
328. **Convert mm Hg to cm H2O**: 1 mm Hg = 1.36 cm H2O; 1 cm H20 = 0.74 mm Hg
329. **Therapeutic dose calculation?**: Therapeutic index = Toxic dose/Effective dose
330. **End product of CO2 absorber**: end products include: heat, H2O, & Ca2+ carbonate
    a. **Soda lime reactions**:
        i. $CO_2 + H_2O \rightarrow H_2CO_3$
        ii. $H_2CO_3 + 2NaOH \rightarrow Na_2CO_3 + 2H_2O$ + Heat (fast reaction) **final product
        iii. $Na_2CO_3 + Ca^{2+}(OH)_2 \rightarrow CaCO_3 + 2NaOH$ (slow reaction)
        iv. **the activator (NaOH or KOH) are regenerated in step 3 (written out)**
331. **Soda lime**: consists primarily of Ca2+ hydroxide, Na+ hydroxide, water, & K+ hydroxide [Ca2+(OH)2 + NaOH + KOH]
    a. Small granule = high surface area with high resistance
    b. Large granule = low surface area with low resistance
    c. 4-8 mesh granules are used. Each granule is between 1/8-1/4" in diameter and will pass through a mesh screen with 4-8 holes per square inch

332. Several capnography:

a. 

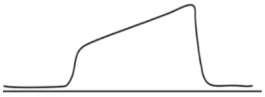
Phase I (A-B)	Exhalation of anatomic dead space
Phase II (B-C)	Exhalation of anatomic dead space + alveolar gas
Phase III (C-D)	Exhalation of alveolar gas
Phase IV (D-E)	Inspiration of fresh gas that does not contain CO2

b. EtCO2 is measured at point D

c. 

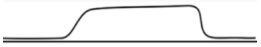
d. Alpha angle is measured at point C; normal is 100-110°
   i. ↑angle: expiratory airflow obstruction (COPD, bronchospasm, kinked ETT)

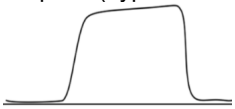
e. Beta angle is measured at point D; normal is 90°
   i. ↑angle: specific to rebreathing caused by a faulty unidirectional valve (won't be ↑in case of CO2 absorbent exhaustion)

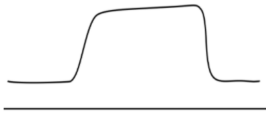
f. 
   i. Airflow Obstruction: prolonged upstroke w/ ↑alpha angle; Ex: COPD, bronchospasm, kinked ETT

g. 
   i. Cardiac Oscillations: more common in kids d/t close proximity of heart to lungs

h. 
   i. Curare Cl-eft: spont breathing during mechanical ventilation; if present during spontaneous ventilation, suggests inadequate NMB reversal (lack of synchronization b/w intercostal muscles & diaphragm)

i. 
   i. Low EtCO2: occurs w/ hyperventilation (light anesthesia, metabolic acidosis), ↓CO2 production (hypothermia), or ↑alveolar dead space (hypotension, PE)

j. 
   i. Elevated EtCO2 w/ Normal Plateau: occurs w/ ↑CO2 production (MH, sepsis, fever, hyperthyroidism) or ↓alveolar ventilation (hypoventilation, narcotics)

k. 
   i. Inspired CO2: rebreathing; occurs w/ exhausted CO2 absorbent, incompetent expiratory valve, hole in inner tube of a Bain system, inadequate FGF w/ Mapleson circuit

l. 
   i. Incompetent Expiratory Unidirectional Valve: ↓lope w/ widened beta angle; CO2 in limb w/ faulty valve; may or may not reach 0 depending on FGF

m. 
   i. Sample Line Leak during PPV: beginning of plateau low because dilution of alveolar gas as atmospheric air is aspirated into sample line; Inspiratory PPV pushes CO2 rich gas thru sample line, resulting in peak at end of plateau; not seen w/ spontaneous ventilation because there is no PPV; may also occur in obese & pregnant pts.

n. 
   i. Patient w/ Single Lung Transplant: alveolar gas from transplanted lung & diseased lung have different time constants; 1st peak is alveolar gas from transplanted lung (has normal time constant); 2nd peak is alveolar gas from diseased lung (longer time constant because air is trapped in sick lung)

333. Sympathomimetics:

Drug	Dosages		Site of Activity						Hemodynamics (↑ Increase; ↓ Decrease; – = No Change)			
Listed from α to β	IV Push adults	IV Infusion[a]	α_{1A}	α_{1V}	β_1	β_2	DA	CO	INOTROP	HR	VR	TPR	RBF
Phenylephrine	50–100 μg	a. 10 mg/250 mL b. 40 μg/mL c. 0.15–0.75 μg/kg/min d. 0.15 μg/kg/min	++++	+++++	0	0	0			Reflex Reflex			
Norepinephrine	N/R	a. 4 mg/250 mL b. 16 μg/mL c. 0.01–0.1 μg/kg/min d. 0.1 μg/kg/min	+++	+++	++++	?+	0	–↓	–	↓ Reflex	↑↑↑	↑↑	––↓
Epinephrine	0.3–0.5 mL 1:1000 (0.3–0.5 mg) SC—Asthma IV—Anaphylaxis 5 mL 1:10,000 (0.5 mg) cardiac arrest every 5 min	a. 1 mg/250 mL b. 4 μg/mL 0.01–0.03 μg/kg/min c. 0.03–0.15 μg/kg/min 0.15–0.30 μg/kg/min d. 0.015 μg/kg/min	+ +++ +++++ +	+ +++ +++++ +	++++ ++++ ++++	++++	0		↑–↓	↑ Reflex	↑↑↑	↑↑↑	↓↓↓
Ephedrine	5–10 mg	N/R + +	+++	+++	++++	+++	↑↑	↑↑	↑	↑↑	↑	↑	↑
Dopamine[c]	N/R	a. 200 mg/250 mL b. 800 μg/mL 0.05–5 μg/kg/min c. 2–10 μg/kg/min 10 μg/kg/min[b] d. 2 μg/kg/min				++	0 0	↑– ↑–↓ ↑↑ ↑	↑↑ ↑↑ ↑↑	↑↑ ↑ ↑ ↑	↑ ↑ ↑ ↑ ↑↑	↑↑ ↑↑↑ ↑ ↑ ↑	↑ ↓– ↓ ↑ ↑–↓
Dobutamine[c]	N/R	a. 250 mg/250 mL b. 1,000 μg/mL c. 2–30μg/kg/min d. 5 μg/kg/min	+ +++++	++++ ++++ ++++ ++++	+++ +++++								
Isoproterenol	0.004 mg (0.2 mL of 0.2 mg/mL solution) Third-degree heart block	a. 1 mg/250 mL b. 4 μg/mL c. 0.15 μg/kg/min to desired effect d. 0.015 μg/kg/min	0 – +	?	++++	+++++	+++++ +++++	↑ ↑↑ ↑–↓ ↑	↑ ↑↑ –	– –↑ ↑↑ –	– ↑ ↑ ↑	–↓ –↑ ↑↑	↑ ↑ –↓

334. Unidirectional Valve: minimizes the likelihood of transfilling or of leakage to the atmosphere.
- a. If 2 cylinders are open, transfilling occurs when gas flows from cylinder w/ higher pressure into cylinder w/ lower pressure

335. What analyzes gases:
- a. **Infrared Absorption Spectrophotometry**: each gas absorbs different wavelength; has replaced other methods
- b. **Mass Spectrometry**: creates ion fragments; system is large
- c. **Raman Scatter Spectrometry**: argon laser produces photons that collide w/ gas molecules
- d. **Piezoelectric Crystals**: can't identify multiple gases at once; not practical for clinical

336. Elements of malpractice: duty, breach (of duty), causation (connection b/w breach & damages), damages

337. Battery vs assault - patient demanded GA, but provider did regional instead:
- a. **Assault**: placing someone in fear of unwanted touching (verbal)
- b. **Battery**: unwanted touching of a patient (physical)

338. Agencies
- a. **DOT**: sets standards for compressed gas cylinders
- b. **ASTM**: sets standards for components of anesthesia machine
- c. **FDA**: sets gas standards
- d. **OSHA**: sets standards for acceptable exposure to VAs

339. Lower extremity nerve stimulation responses:
- a. **Sciatic**: plantar flexion (preferred) or dorsiflexion
- b. **Popliteal**: plantar flexion

340. Cardiac Output to Organs:
- a. Lungs: 100%
- b. Liver: 30%
- c. Kidney: 25%
- d. CNS: 15%
- e. Heart: 5%

341. Transtracheal Block: blocks RLN thru cricothyroid membrane; aspirate air = good

342. Ester LAs
- a. Most potent: Tetracaine d/t slow hydrolyzed by plasma cholinesterase & highly protein bound (most toxic)
- b. Least potent: Procaine
- c. **Prilocaine is metabolized to**: ortho-toluidine which converts Hgb→metHgb

343. Long acting LAs: Bupivacaine, Etidocaine, Tetracaine, Ropivacaine *BETR*

344. Effects of Ketamine: targets NMDA receptors in brain; can enhance opioid-induced analgesia & prevent hyperalgesia; metabolized in liver (P450); CV effects: ↑MAP, CI, PAP, CVP, & HR; causes bronchodilation; avoid w/ head injury d/t ↑ICP; avoid in pts w/ seizures; undergoes smallest amount of plasma protein binding (12%)

345. S/S Intraop PE:
- a. Sudden CV collapse, Hypoxemia, ↓EtCO2 (often 1st s/s), Hypotension, Tachycardia, Bronchospasm, PVR > 300, S/S RV strain on ECG (negative T wave, R axis deviation, Pulm P wave, R > S or Q, RBBB)

346. Peds airway anatomy: Anterior & Cephalad (↑) larynx located at C3-C4 (adult is C4-C5)
Infants = Obligate nose breathers & Na+ excreters
Cricoid cartilage is narrowest point in kids < 5 yo
Tongue large relative to oral volume

347. Sulfur hexafluoride (SF6): (intraocular gas bubble) →no N2O at least 10 days
Perfluoropropane (C3F8) →30 days
Air bubble →5 days
Silicone oil → 0 days

348. False high pulse ox readings: carboxyhemoglobin, methemoglobin

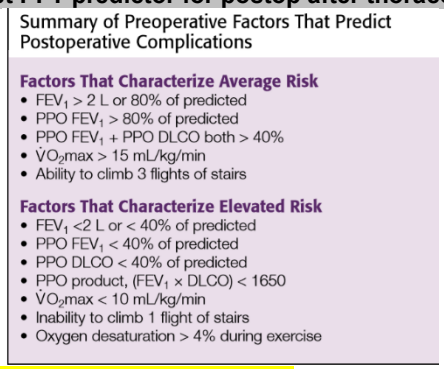
349. Postop complication of Parathyroidectomy: RLN damage, hematoma, acute hypoCa (Chvostek's & Trousseau's sign), laryngospasm, seizure, AMS, hypotension, prolonged QT, paresthesia

350. Adrenal Gland: **Go Find Rex, Make Good Sex or GFR salt, sugar, sex

a. **Zona glomerulosa**: mineralocorticoids (aldosterone); Na+ reabsorption in exchange for K+ & H ions; aldosterone regulates intravascular volume, **NOT** Na+ concentration or osmolarity (osmolarity is controlled by ADH)
b. **Zona fasciculata**: glucocorticoids (cortisol); stress can ↑cortisol production upwards of 100 mg/day; cortisol improves myocardial performance by increasing # & sensitivity of beta receptors on myocardium; cortisol is also required for the vasculature to respond to vasoconstrictive effects of catecholamines
c. **Zona reticularis**: androgens (dehydroepiandrosterone)
d. **Medulla**: catecholamines (epi & NE)

351. **CO2 abosorbent size**: irregular granules are 4-8 mesh
352. **R-L shunt (cyanotic)**: slows inhalation induction d/t blood bypassing lungs
    a. Tetralogy of Fallot, Transposition, Tricuspid (Ebstein), Truncus arteriorsus, TAPVC, HLHS
    b. PVR is high & SVR is low
353. **SLN block**: Thyrohyoid membrane just below cornu of hyoid; air = too deep
    a. Airway block landmarks: SLN = cornu of greater hyoid bone; Transtracheal = cricothyroid membrane; Glossopharyngeal = glossopalatine

354. **Best PFT predictor for postop after thoracotomy**:

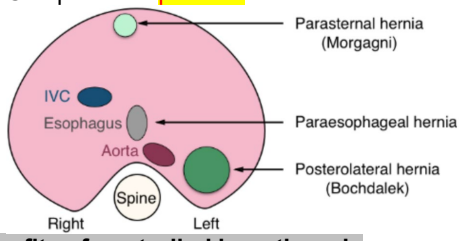

Summary of Preoperative Factors That Predict Postoperative Complications

Factors That Characterize Average Risk
- FEV_1 > 2 L or 80% of predicted
- PPO FEV_1 > 80% of predicted
- PPO FEV_1 + PPO DLCO both > 40%
- $\dot{V}O_2$max > 15 mL/kg/min
- Ability to climb 3 flights of stairs

Factors That Characterize Elevated Risk
- FEV_1 <2 L or < 40% of predicted
- PPO FEV_1 < 40% of predicted
- PPO DLCO < 40% of predicted
- PPO product, (FEV_1 x DLCO) < 1650
- $\dot{V}O_2$max < 10 mL/kg/min
- Inability to climb 1 flight of stairs
- Oxygen desaturation > 4% during exercise

    a.
    b. PPO FEV1 <40% predicted
355. **O2 Delivery Systems**
    a. **Binasal Cannula**: 24-44% FiO2 (3-4%/1 L ↑)
    b. **Partial rebreather**: 65-75% FiO2
    c. **Non-rebreather**: 85-95% FiO2
    d. **Venturi**: 24-50% FiO2
    e. **PP** → ↑100%
356. **Myasthenia Gravis – which trimester has greater risk for symptom exacerbation (select 2)**: 1st trimester or 1st 10 days PP
357. **H2 blockers**: may cause bronchospasm in asthmatics (cimetidine, ranitidine, famotidine)
358. **Drug most likely to cause allergic reaction**: NMB > latex > Abx
359. **When to stop warfarin preop**: Hold 5 days
360. **Lower extremity nerves & muscle innervation**:
    a. **Lateral femoral cutaneous** (L2-L3): purely sensory to skin of anterior & lateral regions of thigh to knee, & also lateral portion of thigh, from greater trochanter in hip, to mid-thigh, just above the knee
    b. **Femoral** (L2-L4): sartorius & quadriceps femoris; flex the hip joint (pectineus, iliacus, sartorius) & extend the knee (quadriceps femoris: rectus femoris, vastus lateralis, vastus medialis, vastus intermedius)
    c. **Obturator** (L2-L4): hip adductors; obturator externus, adductor longus, adductor brevis, adductor magnus & gracilis
    d. **Saphenous**: purely sensory to medial aspect of the leg & the foot.
    e. **Common peroneal**: biceps femoris
361. **Res Ispa Loquitur**: "the thing that speaks for itself"; allows plaintiff to use circumstantial evidence to prove negligence; shifts burden of proof from plaintiff to defendant
362. **O2 consumption**: Adult = 3.5 ml/kg/min (about 250 ml/min for 70 kg adult); Newborn = 7 ml/kg/min
    a. Neonate: O2 Consumption & CO2 production are twice those of the adult
    b. O2 content (CaO2) = (1.34 X Hgb X SaO2) + (PaO2 + 0.003); normal = 20 ml O2/dL
363. **Determinate of preload & things that affect it**: volume (venous return), venous tone, compliance (ventricular)
364. **BIS monitor**
    a. **Drug effects:**
    deep anesthesia ↓ frequency & ↑amplitude
    N2O (when used solo) doesn't ↓BIS
    All IV anesthetics & VAs will ↓BIS (except Ketamine: ↑frequency)
    b. When to not use BIS: less accurate in kids; hypothermia; encephalopathy; EMG interference
        i. Answer options: H/o seizures, prone position, ?, ?
    c. Absence of recall if value <80; GA value = 40-60; 100 = fully awake
365. **Electrostatic attachment to AchE**: edrophonium; no true chemical binding = shorter onset & DOA; competitive inhibition
366. **Best time to use Eschmann introducer (bougie)**: G3V or G2bV; G4V = worst time; advance tip into trachea (about 25 cm); "hold up" at carina 35-40 cm; feel clicks along tracheal rings

367. **Congenital diaphragmatic hernia**: allows extrusion of abd contents into thoracic cavity; MOST COMMON site = foramen of Bochdalek (on left side)
    a. s/s: dyspnea, tachycardia, cyanosis, absence of breath sounds on affected side, severe retractions, scaphoid abd, barrel chest, associated w/ other heart lesions
    b. Tx: emergent repair required; give anticholinergic to prevent bradycardia; ventilate gently (pressure <30 cm H2O); HFOV, inhaled NO, avoid N2O, keep them spontaneously breathing
    c. Complication: pneumo
    d.

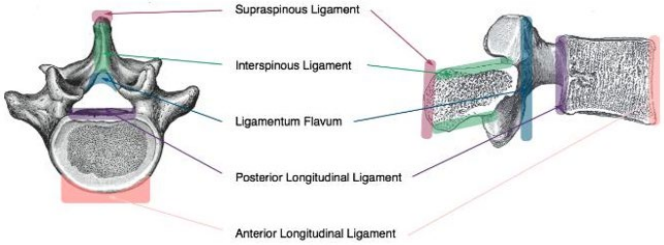


368. **Benefits of controlled hypothermia**:
    1. O2 consumption ↓ 5-7% for q1°C ↓in body temp 2. ↓free radicals & other mediators of ischemia injury
369. **Hypothermia stimulates release of (select 2)**: not NE or Ach; possibly serotonin? Answers options: Epi, NE, serotonin, Ach
370. **Hepatic blood flow**:
    a. CO = 25%
    b. Portal vein = 75% BF; receives venous blood that has passed thru the splanchnic circulation
    c. Hepatic artery = 25% BF
    d. O2 delivery: 50/50
371. **Neonate dosing**: have ↑Vd, so require ↑doses of water soluble drugs
372. **Lowest LA in fetus**: Chloroprocaine d/t rapid metabolism by plasma cholinesterase (keeps plasma levels low & ↓ fetal transfer)
373. **Precedex**:
    a. **Effects:** ↓MAC requirements, sedation, analgesia (enhances N2O), anxiolysis
    b. **Loading dose:** 1 mcg/kg over 10 min
    c. **Infusion:** 0.2-0.7 mcg/kg/hr
374. **Parasympathetic CN**: Craniosacral: 3, 7, 9, 10; S2-S4
375. **Complications of TIPS procedure**: portal vein rupture, liver perforation, portal vein thrombosis, encephalopathy, sepsis, bleeding, RBBB/complete heart block, CHF
376. **DKA**: more common in DM1 than DM2; Labs: BG >250, pH <7.3, bicarb <18, Anion gap >10, Hyperosmolarity >300, Ketones >5
377. **Hyperglycemic Hyperosmolar State**: more common in DM2; BG >600
378. **↓ventricular compliance**: ↑PAOP = overestimation of LVDEV & underestimation of LVEDP
379. **Wood's metal**: used to seal leaks in cylinder; melts when exposed to ↑temp to prevent explosions
380. **VAE**
    a. **Doppler placement to best hear VAE**: precordial placement induction/emergence; right side of heart; right sternal border b/w 3rd & 6th intercostus
    b. **S/S**: air on TEE, mill wheel murmur, ↓EtCO2, ↑EtN2, ↑PAP, Hypotension, dysrhythmias, Pulm edema, hypoxia, cyanosis
381. **Most significant side effect of neuraxial sufenta**: sedation & resp depression s/t sedation
382. **Flowmeter principles**:
    a. Viscosity (Poiseuille's law) = Laminar →Vice President
    b. Density (Graham's law) = Turbulent →Dense Grams
383. **CONTRAINDICATED for PA catheter**: LBBB (can cause RBBB = complete heart block)→
384. **Intrathoracic mass**: inhale = flow is normal; exhale = collapses airway (↓flow)
385. **Hct of PRBCs**: 70%

Pulmonary artery rupture
Right bundle branch block
Complete heart block (if pre-existing LBBB)
Dysrhythmias

386. **Things that cause bronchodilation**: NO, ketamine, anticholinergics, Epi, B2 agonists (albuterol), Sevo
387. **Etomidate**: causes dose dependent adrenocortical suppression of HPA axis (hypothalamus links CNS to the endocrine system) by inhibiting 11-beta-hydroxylase
388. **TBW/ICF/ECF:**
    - Adult/Child = 60%/40%/20%
    - Neonate = 75%/35%/40%
    - Premie = 85%/25%/60%
389. **Potentiate NMBs (residual in PACU)**: VAs (D>S>I>N2O/fentanyl), aminoglycoside Abx, hyperMg, hypoCa, hypoK, hypothermia, LAs, quinidine (antiarrhythmic), ketamine, CCBs (including dantrolene), lithium; Eaton Lambert??
390. **NMB in pts w/ liver failure**: Succ → prolonged; NDNMBs: titrate carefully
    Roc & Vec →Hepatic elimination= prolonged DOA
    Best option: Cisat (Hofmann elimination)
391. **Kupffer cells**: macrophage lining hepatic sinuses; cleans blood of over 99% of bacteria from gut; drains clean blood
392. **Hyperbaric O2 therapy**:
    a. Tx for: decompression sickness, gas embolism, gas gangrene, CO poisoning
    b. SE: hypoventilation, absorption atelectasis, O2 toxicity, ROP, fires
393. **Things/Drugs that ↓CSF production**: Loop diuretics (acetazolamide, lasix), Dehydration, Carbonic anhydrase inhibitors, Corticosteroids, Spironolactone, Iso, vasoconstrictors
394. **CONTRAINDICATED for Intra-aortic balloon pump**: Aortic insufficiency, Descending aortic disease, severe PVD, Sepsis
395. **Tachyphylaxis**: multiple does of ephedrine can give progressively smaller response to given dose; implies acute intolerance of drug following multiple rapid injections→common w/ Neo & ephedrine

396. **Epidural Placement**
   a. **Paramedian approach**: skin, SQ, ligamentum flavum, epidural space
   b. **Midline approach**: skin, SQ, supraspinous ligament, interspinous ligament, ligamentum flavum, epidural space
   c. Needle should NEVER pass thru anterior or posterior longitudinal ligament
   d.



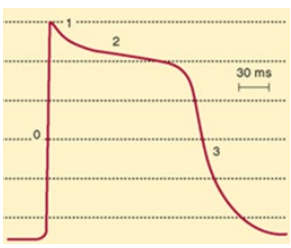
397. **Type 1 hypersensitivity**: rapidly developing reaction that results from antigen-antibody interaction in an individual who has be previously exposed & sensitized to the antigen. Responsible antigen (allergan) reacts w/ specific IgE antibodies on tissue mast cells & circulating basophils to trigger mediator release & allergic response
   a. **Anaphylaxis**: occurs when inflammatory agents are released from basophils & mast cells as a result of an antigen interacting w/ IgE
398. **RAAS**: sympathetic stimulation & ↓delivery of both Na+ & Cl- to the macula densa →cause the juxtaglomerular cells release renin. Renin clears angiotensinogen from the liver to form angiotensin 1. In the lung, angiotensin 1 is changed to angiotensin 2 under the influence of a converting enzyme. In addition to having a generalized vasoconstricting effect, angiotensin 2 causes constriction of the efferent arteriole. This causes the pressure in the glomerulus to ↑& the GFR to return to normal
399. **Cranial Nerves**

Oh-Olfactory-Some
Oh-Optic-Say
Oh-Oculomotor-Marry
To-Trochlear-Money
And-Abducens-But
Touch-Trigeminal-My
Feel-Facial-Brother
Virgin-Vestibulocochlear-Says
Girl-Glossopharyngeal-Big
Vagina-Vagus-Boobs
So-Spinal Accessory-Matter
Hot-Hypoglossal-More

(LR6,SO4)

400. **Obstructive Resp Disease**: Acute URI, Asthma, COPD, Bronchiectasis, CF, Primary ciliary dyskinesia, Bronchiolitis obliterans, Tracheal stenosis
401. **Adiabetic process**: when an air canister explodes/catches fire; rapid expansion/compression of a gas w/o equilibration of energy w/ surrounding environment & entails no ↑/↓in a system's energy. Compressing a gas quickly will intensify the kinetic injury such that thermal measurement of the gas will be higher. The high temp could cause a burn or fire.
402. **P50**: PaO2 at which 50% of the Hgb is saturated. Under normal conditions, adult human blood has a P50 of 26-27 mm Hg.
403. The basic breathing rhythm originates in the medulla.
   - 2 medullary groups:
     1. DRG (primarily active during inspiration→ pacemaker of normal breathing)
     2. VRG (primarily active during expiration)
        The lower pons contains the apneustic center
        Upper pons contains the pneumotaxic center (thought to be the inspiratory cut off switch which functions to smooth the resp pattern.
404. **Methadone**: prolongs QTi (lethal VT & Torsades de pointes)
   metabolized into inactive metabolites & does **NOT** contain an active metabolite
405. **Autonomic Hyperreflexia**:
   Sudden sympathetic discharge resulting from stimulation BELOW level of spinal cord injury
   Seen in 85% of pts w/ SCI ABOVE T6
   - S/S: Paroxysmal HTN
   - Bradycardia
   - Cardiac dysrhythmias in response to stimuli below level of SCI (Ex: bladder cath)

406. **Arterial supply to SA node**: RCA 60% or LAD 40%
407. **Plasma Colloid Osmotic Pressure**: Force is determined primarily by plasma protein concentration & serves to maintain the circulating fluid volume within the intravascular space. Hydrostatic pressure favors fluid movement out of the capillaries.
408. **Oncotic pressure**: balances the hydrostatic pressure tendency to push water out of capillaries. Normal = 28 mm Hg
409. **Methemoglobinemia**: methylene blue 1-2 mg/kg IV over 5 min w/ MAX of 7 mg/kg
410. **Protein C & S deficiency**: causes hypercoagulability & thrombosis
411. **Cryoprecipitate**: does not require ABO compatibility because there is limited antibody content.
412. **FFP**: must be ABO compatible.
413. **Platelets**: does not absolutely require ABO compatibility, but it is ideal.
414. **Hemophilia A Tx**: DDAVP (desmopressin)
   a. Factor 8 concentrate
   b. FFP
   c. Cryo (better than FFP)
415. **Ventricular Action Potential**

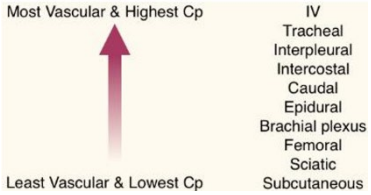


a. **Relative refractory period:** extends from middle of phase 3 to beginning of phase 4 membrane potential ranges from -60 to -90 mV.

416. **Omphalocele**: external herniation of abd viscera thru the base of the umbilical cord
    a. midline sac
    b. defect > 4cm
    c. associated w/ congenital disease
417. **Half Time**:
    - 0: 0 drug eliminated; 100% drug remaining
    - 1: 50% eliminated; 50% remaining
    - 2: 75% eliminated; 25% remaining
    - 3: 87.5% eliminated; 12.5% remaining
    - 4: 93.75% eliminated; 6.25% remaining
    - 5: 96.875% eliminated; 3.125% remaining
418. **Orlistat**: lipase inhibitor→ hinders the absorption & digestion of consumed fats
    - can inhibit absorption of vitamins A, D, E, & K+
    - Insufficient quantities of vitamin K+ will impair synthesis of clotting factors 2, 7, 9, & 10→ may cause coagulopathy.
419. **Encephalopathy could indicate**: uremia
420. **Abx**:
    - **Imipenem**: beta lactum antibiotic; hypersensitivity MOST COMMON adverse reaction
    - **Cefazolin**: broad spectrum cephalosporin
    - **Sulfonamide**: hepatotoxic & can cause AKI
    - **Vancomycin**: causes histamine release & hypotension w/ rapid infusion; administer at rate of 10-15 mg/kg over 1 hr
421. **Patient w/ TBI avoid**: steroids (cause hyperglycemia which is associated w/ worse outcomes in setting of cerebral ischemia) & hypotonic solutions (0.45% saline, 0.25% saline, Distilled Water)→(will worsen cerebral edema)
422. **Can induces seizure activity**: ketamine, LAs (reduce the seizure threshold)
423. **Scoliosis wake up test**: testing for paralysis
    - monitors anterior spinal cord motor pathways (ventral nerve root)
424. **Carotid sinus**: send afferent impulses to the nucleus tractus solitatrius in the medulla
425. **Baroreceptor reflex**:
    a. **Control center:** Vasomotor center in medulla & the pons
    b. **VAs:** ↓the effectiveness of reflex in dose dependent fashion; because Iso has mild beta1 agonist properties, it causes the least impairment
    c. **IV induction agents:**
        - thiopental preserves baroreceptor function
        - Propofol usually depresses baroreceptor function (can cause bradycardia & asystole)
        - Ketamine activates SNS & will cause ↑HR w/ minimal change in SVR (in catecholamine depleted patient, myocardial depressant effects may be unmasked)
        - Etomidate usually causes unchanged HR w/ small ↓in SVR (may cause hypotension in hypovolemic pts)
426. **Mechanism of fade during nerve stimulation**: caused by antagonism of presynaptic Nn receptors
    a. 2 supplies of Ach vesicles:
        i. Ach that's available for immediate release
        ii. Ach that must be mobilized before it can be made available for immediate release (stockpile that must be transported to frontline before it can be used)
        iii. AP arrives at nerve terminal, voltage-gated Ca2+ channels open & Ca2+ enters nerve terminal which destabilizes the proteins that hold the Ach vesicle in place. Next, vesicles exit the nerve via exocytosis & release Ach into synaptic cleft; Ach diffuses toward postsynaptic Nm receptors on motor endplate, where it depolarizes the motor endplate & initiates skeletal muscle contraction.
            - Not all Ach released diffuses towards the motor endplate, a fraction binds to prejunctional Nn receptors on the nerve terminal – causes mobilization of Ach vesicles inside presynaptic nerve. Moves stockpiled Ach to frontline for immediate release the next time the nerve is stimulated.
        iv. NDNMBs competitively antagonize presynaptic Nn receptors which impairs mobilization process, so now only vesicles that are available for immediate release are able to be used. Since this is a limited quantity, nerve stimulation can quickly exhaust this supply. W/ each successive stimulation, less Ach is released. Manifests as fade w/ TOF, double burst, & tetanus
        v. Succ stimulates prejunctional receptors – has same effect as Ach. When it binds to prejunctional Nn receptors, it facilitates mobilization process, so there's always Ach available for immediate release. Explains why no fade w/ depolarizers.
    b. Best mode to monitor for fade = double burst

Half-Time	Amount of Drug Eliminated %	Amount of Drug Remaining %
0	0	100
1	50	50
2	75	25
3	87.5	12.5
4	93.75	6.25
5	96.875	3.125

427. **Phase 1 & Phase 2 Block**:
    a. **Phase 1 (succs):** succ produces phase 1 block (agonist of presynaptic Nn receptor); mobilization of Ach functions as it normally does – there is ample supply of Ach. Stimulation results in diminished but = response; no post-tetanic potentiation
    b. **Phase 2 (NDMB):** NDNMBs competitively antagonize the presynaptic Nn receptor & cause a phase 2 block; Ach mobilization is impaired, so nerve terminal can only release a fraction of Ach that is immediately available. Because this supply exhausts quickly, fade is observed; post-tetanic potentiation is present
        i. High doses of succ (>7-10 mg/kg or 30-60 min continuous infusion) can cause phase 2 block; likely inhibits presynaptic Nn receptor, impairs Ach mobilization & release from presynaptic nerve terminal, &/or creates a conformational change in postsynaptic receptor
428. **Sugammadex**: Roc > Vec > Panc; has no effect on succ or benzylisoquinolines (Cisatracurium, Atrcurium, Mivacurium "CAM")
429. **Zero Order Kinetics**: constant amount removed per unit time; Ex: ASA, phenytoin, ETOH
430. **1st Order Kinetics**: constant fraction removed per unit time
    a. If give 80 mg & have 40 mg left after 4 hrs, how many mg will be left after 8 more hrs?



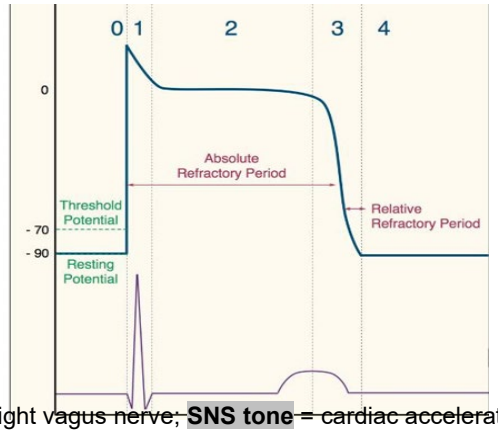
431. **Mnemonic: I Think Illogical Imposters Can't Educate But Fabulous Schools Should**
432. **Mixing propofol w/ lidocaine ↑risk of**: instability & risk of fat emboli
433. **Thiopental & EEG**: can cause burst suppression &/or isoelectric EEG
434. **ADH**: controls serum osmolarity by ↑H2O reabsorption but not Na+; produced in supraoptic & paraventricular nuclei of hypothalamus
    a. 2 mechanisms that control ADH release:
        1. ↑osmolarity of the ECF
        2. ↓blood volume
435. **Aldosterone**: water follows Na+ in direct proportion when it's reabsorbed into the peritubular capillaries
436. **Largest internal organ**: liver
437. **Liver functions**:
    a. **Synthetic Function**:
        i. **Proteins NOT produced by hepatocytes**:
            - immunoglobulin (humoral immune system)
            - Coag factors: VWB factor & Factor 3 aka tissue factor (vascular endothelial cells)
            - Factor 8 (liver sinusoidal cells & endothelial cells)
        ii. **Tests**: PATIENT & Albumin
    b. **Metabolic Function**: carbs, proteins, lipids, bilirubin

Cirrhosis causes:
ETOH (Fatty infiltration)
A1A deficiency (genetic disease causes emphysema)
Biliary obstruction (inflammation & tissue destruction)
Chronic hepatitis (inflammation & tissue destruction
Hemochromatosis (iron overload)
Right sided heart failure (↑hepatic vascular resistance)
Wilson's disease (genetic disease where copper accumulates in tissues)

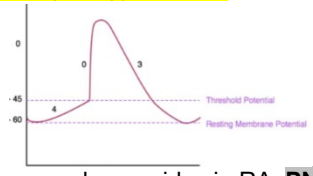
438. **Liver Dysfucntion**
    a. **CV Changes**:
        - **Hyperdynamic circulation**: ↑CO; ↓response to vasopressors
        - **Portal HTN**
        - **Ascites**: ↑Vd
439. **Pituitary Gland**: resides in sella turcica; connected to hypothalamus by pituitary stalk
440. **SIADH**: too much ADH
441. **DI**: too little ADH
442. **Mineralocorticoid Excess (Conn's Syndrome = Hyperaldosteronism)**
    a. **S/S**: HTN (Na+ & H2O retention), HypoK, Metabolic **alkalosis**
    b. **Tx**: Aldosterone antagonists (spironolactone or eplerenone)
443. **Adrenal Insufficiency**: ↓production of mineralocorticoids, glucocorticoids, & androgens
444. **Pancreas**
    a. **Insulin receptor**: 2 alpha (glucose) & 2 beta (insulin) receptors
    b. **Liver & brain don't require insulin for**: glucose uptake
    c. **Drugs that ↑serum glucose**: beta agonists
    d. **Drugs that ↓serum glucose**: beta antagonists & VAs
    e. **Somatastatin**: aka growth hormone-inhibiting hormone; regulates hormone output from islet cells
        - released by delta cells
        - inhibits insulin & glucagon
        - inhibits splanchnic blood flow, gastric motility, & gall bladder contraction
445. **Metformin**: biguanide; may cause vit B12 deficiency; D/C > 48 hrs before Surgery
446. **Sulfonylureas**: glyburide, glipizide, glimepiride, gliclazide, tolbutamide, chlorpropamide, acetohexamide; D/C 24-48 hrs before Surgery
447. **Insulin**
    a. Normal Secretion: total output is 40 units/day
    b. Beta-2 & PNS stimulation ↑insulin secretion
    c. Alpha-1 stimulation ⃠ secretion
    d. Drug Interactions:
        i. **Counter hypoglycemic effect of insulin**: epi, glucagon, estrogen, ACTH
        ii. **Extend &/or enhance hypoglycemic effect of insulin**: MAOIs, salicylates, tetracycline

448. Cardiovascular

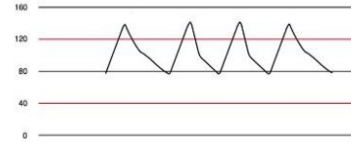
a. **Myocyte**: contains actin & myosin; T-tubule system & SR work to maintain Ca2+ homeostasis; Tight junctions (gap junctions); Contain more mitochondria than skeletal muscle cells; Generates a resting membrane potential (established by Na+-K+-ATPase); Can initiate & propagate an action potential
b. **Threshold Potential**: depolarization is an all or none phenomenon; when RMP closer to TP = easier to depolarize; when RMP further from TP = harder to depolarize
c. **Hyperpolarization**: ↑polarity across a membrane; Larger charge difference b/w inside & outside of cell
d. **Na+-K+-ATPase**:
   i. **K+**: explains why K+ is the primary determinant of RMP
      ↓ K+ = RMP more negative & more resistant to depolarization
      ↑K+= RMP more positive & easier to depolarize
   ii. **Na+**: when RMP approaches TP, voltage-gated Na+ channels open & Na+ conductance ↑which depolarizes cell
   iii. **Pump**: removes Na+ that enters during depolarization; returns K+ that left during repolarization; 3 Na+ out & 2 K+ in; requires ATP
e. **Cardiac Action Potential: Ventricle** →
   i. **Bare Minimum:**
      1. **Depolarization =** Na+ in
      2. **Initial repolarization =** Cl- in & K+ out
      3. **Plateau =** Ca2+ in
      ii. **Phase 0 upstroke:** Na+ in
      iii. **Phase 1 initial repolarization:** Cl- in & K+ out
      iv. **Phase 2 plateau:** Ca2+ in & K+ out
      v. **Phase 3 final repolarization:** Ca2+ in (briefly) & K+ out
      vi. **Phase 4 resting phase:** Na+ out



f. **Cardiac Action Potential: SA Node**

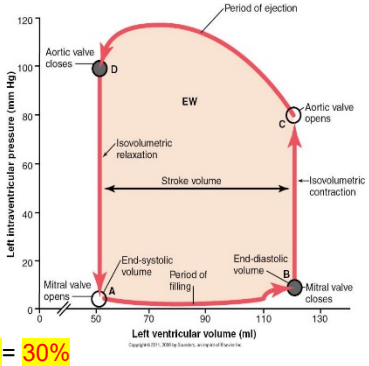


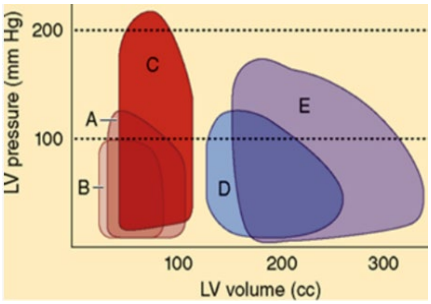
   i. **SA node (Keith-Flack node)**: usually the pacemaker; resides in RA; **PNS tone** = right vagus nerve; **SNS tone** = cardiac accelerator fibers (T1-T4); perfused by **SA nodal artery** (RCA in 50-60%; CxA in 40-50%)
   ii. **Phase 4 spontaneous depolarization**: Na+ in (I-f; funny current); Ca2+ in (T-type)
   iii. **Phase 0 depolarization**: Ca2+ in (L type)
   iv. **Phase 3 repolarization**: K+ out
   v. **HR Change**: can manipulate 3 variables:
      1. Rate of spontaneous phase 4 depolarization
      2. TP (Ca)
      3. RMP (K)
g. **O2 Delivery**: DO2 = CO X [(Hgb X SaO2 X 1.34) + (PaO2 + 0.003)] X 10 *Normal = 1000 ml/min
   i. **1.34 =** constant that tells us max occupancy of Hgb
   ii. **Whole body O2 extraction ratio**: 25%
   iii. **How fast is O2 being delivered to tissues**: this is cardiac output = 5-6 L/min in adult
h. **EDV (Preload) =** filling pressures + compliance
   i. **Factors that influence preload**: blood volume, atrial kick, venous tone, intrapericardial pressure, intrathoracic pressure, body position, skeletal muscle pumping action
   ii. **Ventricular Function Curve**: relationship b/w ventricular volume & ventricular output; Frank-Starling mechanism
      - Contractility is ability of myocardial sarcomeres to perform work; **independent of preload & afterload**; **affected by chemicals**
   iii. **Conditions that affect Ventricular Loading**:
      - Atrial kick = 20-30% of final LVEDV & by extension cardiac output; priming the pump; lost in patient w/ Afib; CO typically declines as a result
   iv. **Conditions associated w/ reduced myocardial compliance**: myocardial hypertrophy, fibrosis, & aging
i. **Coronary Arteries**: provide majority of coronary vascular resistance (not epicardial vessels); myocardial O2 extraction = 70%
   i. **Coronary BF**: 4-7% of CO
      - **Autoregulation**: b/w MAP of 60-120; effected by local metabolism, myogenic response, & ANS
   ii. **Lactic Acid**: presumed to be responsible for perception of chest pain; produced when aerobic metabolism shifts to anaerobic
j. **Periop MI**: occur 24-48 hrs after Surgery & carry 20% mortality
k. **Heart Sounds**: S1 = closure of mitral & tricuspid valves; S2 = closure of aortic & pulmonic valves; S3 = heart failure
l. **Concentric Hypertrophy =** stenosis → pressure overload; chamber wall becomes thicker
m. **Eccentric Hypertrophy =** regurgitation → volume overload; chamber dilates
n. **Aortic Stenosis**: severe = valve orifice < 0.8 cm^2; Triad of S/S: Syncope, Angina, Dyspnea (SAD)



   i. **Arterial Waveform**:
      - slower systolic upstroke (pulsus tardus)
      - delayed peak
      - narrow pulse pressure w/ waveform of small amplitude (pulsus parvus)
      - dampened waveform
   ii. **Anesthetic Management**: low & tight
o. **Aortic Regurgitation**
   i. **Arterial Waveform**
      - Sharp upstroke
      - low DBP
      - wide pulse pressure
      - biphasic systolic peaks (bisferiens pulse) because of an additional reflective wave from the periphery

p. **LV Pressure Volume Loops**

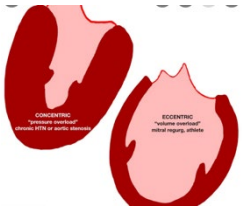




i. *Isovolumic contraction = systole; Ejection = systole
   1. A: normal
   2. B: Mitral stenosis
   3. C: Aortic stenosis (use Neo) *want ↓HR & ↑SVR
   4. D: Mitral regurgitation
   5. E: Aortic regurgitation

q. **Periop CV Risk**
   i. Risk of periop MI in patient w/ previous MI < 3 months = 30%
   ii. Highest risk of reinfarction is greatest within 30 days of an acute MI
   iii. Modified NY Association Functional Classification of HF: Class 4 = symptomatic @ rest

r. **Myocardial Ischemia**:
   i. **ECG**
      1. Best leads to monitor ST changes: V3 > V4 > V5 > III > aVF
      2. 5 lead: V3, aVF, & MCLS or III
      3. 3 lead: aVF & MCL5
      4. Lead II is best for monitoring dysrhythmias w/ narrow QRS where P wave analysis is critical for Diagnosis (junctional, Aflutter, Afib)
   ii. **Labs**
      - Cardiac troponins are more sensitive than CK-MB for Diagnosis of MI
   iii. **Tx:** make heart slower, smaller, & better perfused



s. **Heart Failure**
   i. **Systolic** (ventricle doesn't empty well); ↓EF w/ ↑EDV; eccentric hypertrophy
   ii. **Diastolic** (ventricle doesn't fill properly); normal EF; concentric hypertrophy
      - **Remodeling**: can be reversed w/ ACEIs (-pril) & Aldosterone inhibitors (spironolactone)
   iii. **MOST COMMON CAUSE of Right HF =** left HF

t. **HTN**
   i. **Diagnosis**: BP > 140/90
   ii. **Etiology**: primary HTN has no identifiable cause (most common)
   iii. **Effects on Cerebral Autoregulation**: chronic HTN shifts autoreg curve to right; malignant HTN ↑risk of hemorrhagic CVA & cerebral edema; hypotension ↑risk of cerebral hypoperfusion
   iv. **Antihypertensive drugs that target kidney**
      1. **ACEIs (-pril)**: inhibits ATII mediated vasoconstriction & aldosterone release
      2. **Thiazide diuretics** (HCTZ, metolazone, indamide, chlorthalidone): inhibits Na+-Cl- transporter in DCT; diuresis → ↓venous return (preload)
   v. **CCBs**
      1. **Reduce HR**: verapamil & diltiazem
      2. **Control contractility**: Verapamil > Nifedipine > Diltiazem > Nicardipine
         a. In patient w/ ↓contractility: Diltiazem is better than verapamil
      3. **Best for HTN**: Nifedipine & Nicardipine (vasodilators)
      4. **Best for coronary antispasmodic**: Nicardipine
      5. **Only CCB to reduce M&M from cerebral vasospasm**: Nimodipine
      6. **Nicardipine**: greatest vasodilating effects of all CCBs & has minimal myocardial depression

u. **Cardiac Tamponade**
   i. **S/S**: Beck's triad (not syndrome) =(hypotension, JVD, muffled heart tones), pulsus paradoxus (SBP ↓> 10 mmHg during inspiration)
   ii. **Anesthetic Management**: safe = Ketamine; best to maintain spontaneous ventilation until tamponade is relieved

v. **Obstructive Hypertrophic Cardiomyopathy**
   i. **Systolic Anterior Motion (SAM)**: can also occur after mitral valve repair (not replacement)
   ii. **Tx**: Neo, Beta blockers, CCBs (verapamil or diltiazem)

w. **PCI & Duration to wait until Elective Surgery**
   - Angioplasty (no stent): 2-4 weeks
   - Bare metal stent: 6 weeks minimum (12 weeks preferred)
   - Drug eluding stent DES: 1 year ; Current generation: 6 mo
   - CABG: 6 weeks minimum (12 weeks preferred)

x. **Protamine to reverse Heparin**: 1 mg protamine will reverse every 100 units of heparin that was given

y. **Intra-Aortic Balloon Pump**: inflation = diastole; deflation = systole

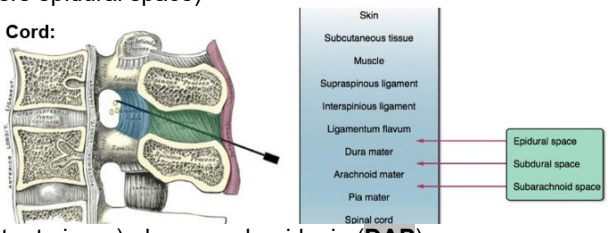
z. **AAA**: mortality signif. ↑once AAA reaches 5.5 cm, so Surgery is recommended when aneurysm >5.5 cm or if it grows > 0.5-0.8 cm/yr

aa. **Spinal Cord Circulation**
   i. **Posterior Spinal Arteries (2)**: perfuses the posterior ⅓ (sensory) of spinal cord
   ii. **Anterior Spinal Artery (1)**: perfuses the anterior ⅔ (motor) of spinal cord
   iii. **Artery of Adamkiewicz**: most important radicular artery; in 75% of population, it originates b/w T8-T12, & in another 10% it arises at L1-2

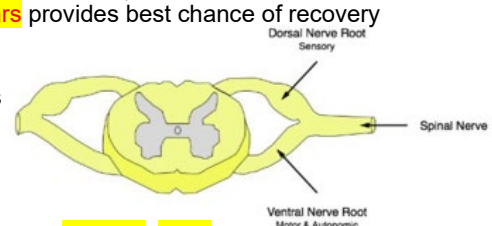
bb. **Anterior Spinal Artery Syndrome S/S**:
   - flaccid paralysis of lower extremities
   - bowel & bladder dysfunction
   - loss of temp & pain sensation
   - *preserved touch & proprioception*

cc. **Subclavian Steal Syndrome**: usually occurs on left side; arterial blood is stolen from posterior cerebral circulation when it's diverted to the ipsilateral arm (results in reversal of blood flow, where vertebral blood goes to ipsilateral arm instead of brain)

449. **Spinal Cord Anatomy**: 33 vertebrae (7 cervical, 12 thoracic, 5 lumbar, 5 sacral, 4 coccygeal)
   a. **Ligaments**: supraspinous & interspinous ligaments join w/ the spinous processes; ligamentum flavum (piercing contributes to LOR when needle enters epidural space)
   b. **From Skin to Spinal Cord:**



   c. Layers of SC (outer to inner): dura, arachnoid, pia (**DAP**)
      i. Dura begins at foramen magnum & ends at dural sac
      ii. Pia: should never be punctured during spinal anesthesia
   d. **Epidural Space**: contains nerve roots; epidural veins (Batson's plexus)
      i. **Epidural Hematoma**: Surgery decompression within 8 hrs provides best chance of recovery
   e. **Sacral Anatomy**
      i. **Sacral hiatus**: covered by sacrococcygeal ligament
      ii. **Sacral cornu**: boney nodules that flank the sacral hiatus
   f. **Dermatomes**
      i.



Spinal Nerve Root	Cutaneous Innervation
C6	1st digit (thumb)
C7	2nd & 3rd digits
C8	4th & 5th digits
T4	Nipple line
T6	Xiphoid process
T10	Umbilicus
T12	Pubic symphysis
L4	Anterior knee

   g. **Site of Action**
      i. **SA Block**: Nerve roots
         - **Factors that affect LA spread**: baricity of LA, patient position, dose, site of injection, volume of CSF, density of CSF
      ii. **Epidural**: must diffuse thru dural cuff before blocking nerve roots
         - **Factors that affect LA spread**: Volume is the primary determinant
   h. **Differential Blockade**
      Spinal:
      i. level of sensory block is 2 dermatomes higher than motor
      ii. Autonomic block is 2-6 dermatomes higher than sensory
   i. **Systemic Effects of Neuraxial Anesthesia**
      i. **CV**: predominantly affects venous capacitance vessels → ↓in venous return, CO, & BP; bradycardia is caused by:
         1. Block of preganglionic cardioaccelerator fibers @ T1-T4
         2. Bezold-Jarisch reflex
         3. Unloading of stretch receptors in SA node
      ii. **Resp**: apnea is usually result of cerebral hypoperfusion **NOT** phrenic nerve paralysis or high concentrations of LA in CSF
      iii. **CNS**: spinal anesthesia reduces sensory input to reticular activating system (RAS) → drowsiness
      iv. **Neuroendocrine**: diminishes stress response d/t reduced circulating levels of catecholamines, renin, angiotensin, glucose, TSH, & growth hormone
      v. **GI**: SNS block → PNS output to gut is unopposed → sphincters relax & ↑peristalsis
      vi. **Kidney & Liver**: unchanged
   j. **Absolute CONTRAINDICATED for regional anesthesia**:

Absolute
- Infection at the site of injection
- Patient refusal
- Coagulopathy or other bleeding diathesis
- Severe hypovolemia
- Increased intracranial pressure
- Severe aortic stenosis
- Severe mitral stenosis

Relative
- Sepsis
- Uncooperative patient
- Preexisting neurological deficits
- Demyelinating lesions
- Stenotic valvular heart lesions
- Left ventricular outflow obstruction (hypertrophic obstructive cardiomyopathy)
- Severe spinal deformity

Controversial
- Prior back surgery at the site of injection
- Complicated surgery
- Prolonged operation
- Major blood loss
- Maneuvers that compromise respiration

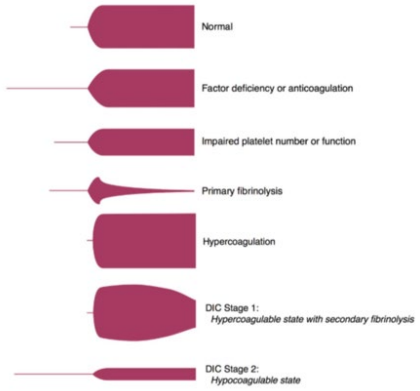
      i. Coagulopathy: PATIENT, aPTT, &/or bleeding time TWICE the normal value
      ii. Hypovolemia → worsening of hypotension d/t sympathectomy
      iii. Severe AS, MS, or HCM
      iv. Full Stomach: hypotension may cause N/V
      v. Multiple Sclerosis: epidural is safe; Spinal may exacerbate s/s
   k. **CSF specific gravity**: 1.002-1.009
   l. **Neuraxial Opioids**: inhibit afferent pain transmission in the substantia gelatinosa in lamina II of dorsal horn by:
      ↓ cAMP
      ↓Ca2+ conductance
      ↑K+ conductance
      i. **SE:** pruritis (MOST COMMON), resp depression (early & late; late only happens w/ hydrophilic drugs like morphine), urinary retention (LC), & N/V
      ii. Morphine: hydrophilic
         1. Spinal: + late resp depression
         2. Epidural: + early depression and + late resp depression

450. **Postintubation Laryngeal Edema (Croup) Tx**: Nebulized racemic epi 0.5 ml of 2.25% solution in 2.5 ml of NS
451. **Peripheral Chemoreceptors**: monitor hypoxemia (PaO2 < 60); don't respond to SaO2 or CaO2

452. **ECG j point**: point at which QRS complex ends & ST segment begins; **elevation could be d/t MI**
453. **Minute ventilation**: Vt X RR
454. **Stress (exercise) ECG**: has 90% specificity (rules in CAD)
455. **Wrong agent in Vaporizer**: High in Low = High; Low in High = Low
456. **LAST Tx**: 20% intralipid; 1.5 mg/kg
457. **2 most important factors for ↑Alveolar Partial Pressure**: 1. Inspired concentration, 2. Blood solubility
458. **Types of drug actions at receptors**:
    a. **Agonist**: affinity & efficacy
    b. **Antagonist**: affinity for receptor but lacks efficacy; can't produce conformational change
        i. **Competitive**: can be overcome by ↑concentrations of agonist
        ii. **Noncompetitive**: can't be overcome by agonist
    c. **Partial Agonist** = **Agonist-Antagonist**: bind w/ receptor & has some efficacy, but can't elicit max tissue response
    d. **Inverse Agonist**: results in opposite reaction of an agonist
459. **Metabolized by CYP450**: Amide LAs, barbs, opioids, antihistamines, tricyclic antidepressants (TCAs)
460. **IVF**:
    a. **LR**: too much = metabolic alkalosis
    b. **NS**: too much = hyperchloremic acidosis
461. **What is it called when someone gives you money**: Kickback = form of Bribery
462. **Cystic fibrosis anesthesia management**: ??
463. **What gets reported to the National Practitioner Data Bank? (from google)**

- Medical malpractice payments
- Adverse clinical privileges actions
- Adverse professional society membership actions
- State licensure and certification actions
- Federal licensure and certification actions
- Negative actions or findings by a peer review organization
- Negative actions or findings by a private accreditation organization
- Exclusions from participation in a Federal or state health care program (including Medicare and Medicaid)
- Other adjudicated actions or decisions
- Health care-related civil judgments and criminal convictions in Federal or state court

464. **TEG**



Normal = Wine glass

465. **Ascending LOH medullary vs cortex**: Where does lasix work? Where does thiazide diuretics work?
466. **CV effects of Cirrhosis (select 2)**

Cardiovascular	
Hyperdynamic circulation	Portosystemic shunt & vasodilator release → • ↓ SVR and BP → ↑ CO • ↑ RAAS activation → ↑ blood volume (pt may behave like he is hypovolemic) • ↑ peripheral blood flow (shunting) → ↑ SvO2 • ↓ Response to vasopressors • Diastolic dysfunction
Portal hypertension	↑ Hepatic vascular resistance → ↑ backpressure to proximal organs Esophageal varices → bleeding Splenomegaly → thrombocytopenia
Ascites	↓ Oncotic pressure ↓ Protein binding ↑ Volume of distribution Drainage → hypotension

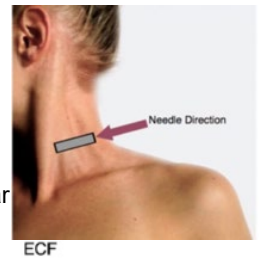
467. Parturient w/ twins on MgSO4 gtt for PreE will most likely need which drug postdelivery? Avoid Methylergonovine d/t intense vasoconstrictive properties. These patients are already hypertensive. Use Carbapost

468. Giving an antisialagogue before an awake intubation does what? ↓effects of sedation/analgesia, 2 options that dealt w/ LAs, ↓SNA response to intubation??

469. How often recalibrate galvanic O2 sensor? 4 hrs, 8 hrs, 16 hrs, 20 hrs "Daily calibration"

470. Give Desmopressin (select 2): vWB Type I>II, Hemophilia A, Hemophilia B, ?

471. Interscalene block: pic of patient lying supine w/ head rotated away. Cl-ick on where to inject.



472. Mapleson circuit that allows lowest FGF to prevent rebreathing in *spontaneously* breathing patient: A

473. During OLV, HPV ↓ BF to ND lung by what %? ????

474. In 2 lung ventilation in lateral decubitus position, HPV % of ND is? Of dependent lung is? ???

475. What nerve is blocked in arm anterolateral toward lateral epicondyle: musculocutaneous, radial, median, ulnar

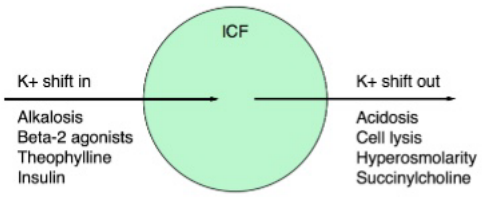
476. What 2 things push K+ out of the cell?

477. Matching of nerve monitoring & Surgery

a. Monitoring: BAEPs, SSEPs, MEPs, EMG

b. Surgery: Crani, Acoustic neuroma, C-spine, Lumbar spine

*I think it is: BAEP = Acoustic neuroma, SSEP = Crani, MEP = C-spine, EMG = Lumbar spine



478. Oxidizing gases (select 2): O2, CO2, N2O, Helium

479. Least common s/s of Pheo: flushing, diaphoresis, HA, AMS

480. Pulse ox can read: reduced Hgb, carboxyHgb, metHgb, sulfoxyHgb

481. 2 devices that prevent tank explosion:

1. Fusible plug→melts at high temperatures→made from Bismuth, Lead, Tin, Cadmium (BLT & Cheese)
2. Valve that opens with elevated pressures
3. Frangible disk that ruptures under pressure

482. Vent mode that supports minimal Vt (select 2): SIMV, PS

483. Vent mode only delivers w/ spontaneous breath: PS

484. Anesthesia management of Aortic Regurg: Full (Maintain preload), Fast (increase with NSR), Forward (decrease Afterload; maintain contractility)

485. What ↓ CPP? Mannitol & HOB elevated 45°

486. _____ stops pain impulse where? Dorso____, 2nd order neuron, ?, ?

487. 2 places where temp doesn't appropriately respond during MH crisis: Distal esophagus, Nasopharyngeal, Rectum, Skin surface

488. 2 risks of immunosuppressive therapy in transplant pts: Malignancy, DM

489. Diagnosis labs of DI:

	Too Much ADH: SIADH	Too Little ADH: DI
Associated Conditions	TBI or SAH (most common) Cancer (non-small cell lung cancer) Noncancerous lung disease Carbamazepine	Pituitary surgery (most common) TBI or SAH
Presentation	Hyponatremia	Polyuria
Plasma *Volume* *Osmolarity* *Sodium*	 Euvolemic (or slightly hypervolemic) Hypotonic (< 275 mOsm/L) Low (<135 mEq/L)	 Euvolemic (or hypovolemic) Hypertonic (>290 mOsm/L) High (145 mEq/L)
Urine *Volume* *Osmolarity* *Sodium*	 Low High (> 100 mOsm/L) High > 30 mEq/L	 High Low (< 200 mOsm/L) Normal
Treatment	Fluid restriction If Na+ 120 mEq/L give hypertonic NaCl (do not correct faster than 1 mEq/L/h) Demeclocycline (↓ ADH responsiveness)	DDAVP or vasopressin Supportive

490. Bone Cement Implantation Syndrome (2 questions)

a. 1 was: 1st priority assessment

Methyl methacrylate (bone cement) is used to bind orthopedic implants to the patient's bone during orthopedic surgery.

- Methyl methacrylate increases intramedullary pressure in the bone (up to 50 mmhg), and this can produce microemboli (fat, bone, marrow, cement) that travel to the lungs. This results in V/Q mismatch (increased dead space), and in extreme cases, it can potentiate right heart failure.
- Residual methyl methacrylate can enter the systemic circulation where it causes bradycardia, dysrhythmias, hypotension (decreased SVR), pulmonary hypertension (increased PVR), hypoxia, and cardiac arrest. Collectively, this constellation of complications is called bone cement implantation syndrome (BCIS).

Hip arthroplasty is associated with the greatest risk of BCIS, but other procedures include knee arthroplasty, vertebroplasty and kyphoplasty.

- In the awake patient under regional anesthesia, the first signs of BCIS are usually dyspnea and altered mental state.
- Under general anesthesia, the first sign of BCIS is usually a decreased EtCO2.

First line treatment: 100% FiO2, IV hydration, and phenylephrine for hypotension.

491. 2 questions on oxygenation & obese positioning

492. ↑IOP is most commonly d/t: venous, arterial, ?, ?

493. MOST COMMON pediatric organ transplant: heart, liver, lungs, kidneys (HLLK)

494. Greatest risk of barotrauma: Using O2 flush valve on inspiration

495. Quality Assurance: typically designed for analysis when performance falls below acceptable levels; at this point, provider and/or systems data are reviewed. Examples of QA programs include peer review systems, or morbidity and mortality reviews.

Root-Cause Analysis:

Continuous Quality Improvement: Continuous quality improvement (CQI) programs identify opportunities for improvement. They recognize that errors will occur, and focus on the processes involved in both successes and failures. These data are analyzed and used to improve the process of care itself, not sanction or penalize individual providers. The CQI process has this key attribute recommended by the 1999 Institute of Medicine report To Err is Human: the implementation of a non punitive system for reporting and analyzing errors.

A few terms to know:

- Outcome = Change in health status after delivery of care
- Process = Planning and coordination of care activities
- Structure = Setting in which care is provided

496. What to do w/ DNR before Surgery: explain risks/benefits of resuscitation, cancel it, CPR if heart stops on induction, ?